Vertex Algebras. IMPA 2023-0

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(with figures and computer calculations contributed by Igor Blatt)

2023-01-09. Lecture 01

1 Introduction

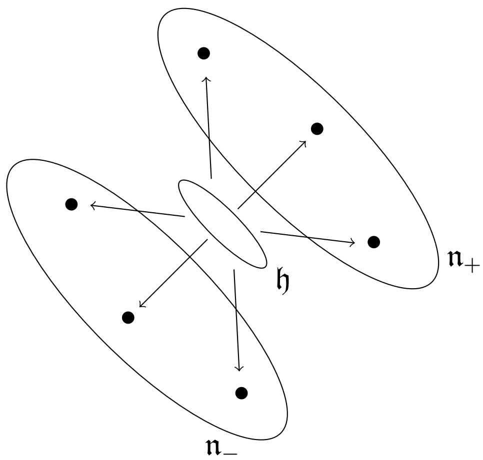
The name of this course is "Vertex Algebras", and that is what it will mostly be about. But I would like to discuss also infinite dimensional Lie algebras, and some other random topics that I find to be relevant.

I would say that having studied Lie algebras is more or less a prerequisite for this course. And in today's introduction I will assume familiarity with Lie algebras. But in the coming lessons I will give a quick review of finite dimensional Lie algebras, so don't be too discouraged if anything I say today seems unfamiliar.

Let $\mathfrak{g}$ be a finite dimensional Lie algebra, with an invariant bilinear form $(,) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ and $U(\mathfrak{g})$ its universal enveloping algebra. The Casimir element Ω is defined to be

$$\Omega = \sum_i e_i e^i \in U(\mathfrak{g}),$$

it is an element of the centre of $U(\mathfrak{g})$, and it is important for studying representations of $\mathfrak{g}$. Simple Lie algebras have positive/negative halves.



At important points in the theory, calculations of the following form appear:

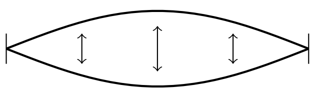
$$\sum_{i\in I} e_i e_{-i} = \sum_{i\in I_+} (e_{-i}e_i + e_i e_{-i}) = 2\sum_{i\in I_+} e_{-i}e_i + \sum_{i\in I_+} [e_i, e_{-i}],$$

where $I = I_+ \cup I_i$ and $I_\pm$ index bases of the positive/negative halves of $\mathfrak{g}$. If $\mathfrak{g}$ is infinite dimensional then how are we supposed to make sense of the infinite sums?

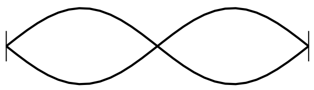
A similar problem appears in quantum field theory, where a particle can be in any of a hierarchy of energy states, transitions are described by "creation/annihilation operators". To illustrate, we consider a vector space (Hilbert space of states) V spanned by vectors

$$\left|1^{N_1}, 2^{N_2}, 3^{N_3}, \dots\right\rangle .$$

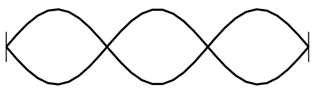
This notation is supposed to indicate N_1 quanta of momentum 1, N_2 quanta of momentum 2, etc. What the different quanta mean is not so important, they might be vibrations of a string for example:



Fundamental: momentum $p_1 = 1$



$p_2 = 2$



$p_3 = 3$

Quantisation of classical relations leads to identities such as

$$H = \frac{1}{2}\sum_{p=1}^{\infty} a_p a_p^\dagger,$$

where $a_p^\dagger$ inserts a quantum of momentum p and a_p removes one. The commutation relations

$$a_p a_p^\dagger - a_p^\dagger a_p = p$$

mean that

$$H\left|0\right\rangle = (1 + 2 + 3 + 4 + \cdots)\left|0\right\rangle .$$

This is solved by physicists using "normal ordering", which is related to renormalisation and which doesn't make much sense mathematically.

Vertex algebra can be viewed as a mathematical theory for this sort of thing.

The main notational device in vertex algebras is the *quantum field*:

$$A(z) = \sum_{n \in \mathbb{Z}} A_n z^n, \qquad A_n \in \mathrm{End}(V)$$

where z is a formal variable. The vector space V above will become our first example of a vertex algebra and the series

$$a(z) = \sum_{n>0} a_n z^{-n-1} + \sum_{n>0} a_n^\dagger z^{n-1},$$

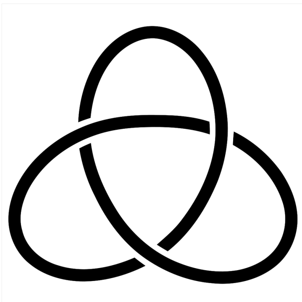
will be one of its quantum fields.

In the 'ementa' of the course I see there is a lot of discussion of algebraic curves and their moduli spaces. This is one of the most interesting applications of vertex algebras, but I'm not sure how much I will discuss it. Let me try to give an indication of this though.

Firstly something that seems unrelated: knot theory. Construction of invariants of knots is a very rich area of mathematics. One famous example is the Jones polynomial

$$\text{Knot } K \longrightarrow \text{Polynomial } J_K(q)$$

For example the Jones polynomials of K_0 the unknot and K_1 the trefoil knot are



are

$$J_{K_0}(q) = 1 \qquad J_{K_1}(q) = -q^{-4} + q^{-3} + q^{-1}.$$

Suppose I had an algebra A (over the base field $\mathbb{C}$) and in the category of A-modules suppose I were able to form tensor products:

$$U, V \mapsto U \otimes V \cong V \otimes U.$$

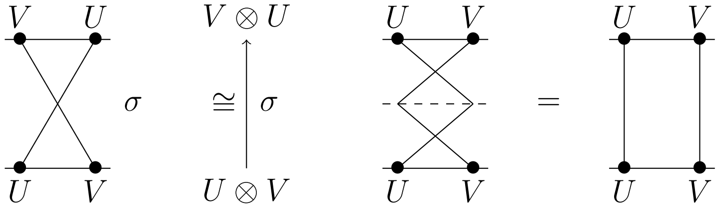
For example we can do this for modules over a Lie algebra $\mathfrak{g}$:

$$x \cdot (u \otimes v) = (x \cdot u) \otimes v + u \otimes (x \cdot v),$$

and the swap

$$\sigma : U \otimes V \to V \otimes U, \qquad u \otimes v \mapsto v \otimes u$$

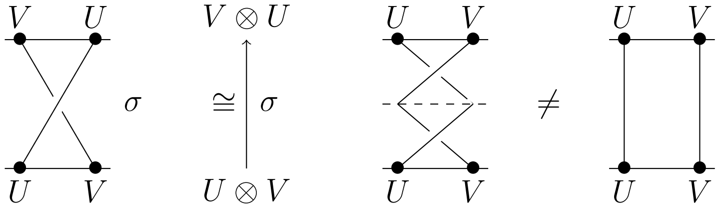
works as an isomorphism. Of course $\sigma^2 = Id$, and we could represent this pictorially as



$\sigma^2 = \mathrm{Id}$

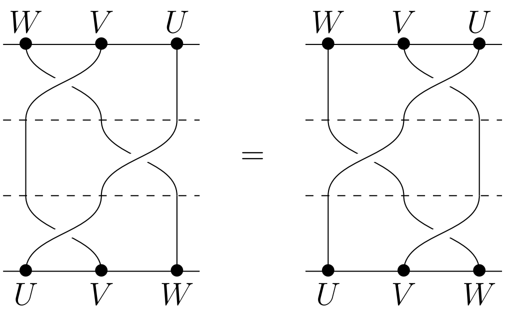
We call such a thing a symmetric tensor category.

We might like to find some more sophisticated sort of algebra A whose modules again have $\otimes, \sigma$ but for which $\sigma^2 \neq Id$, represented pictorially as



$\sigma^2 \neq \mathrm{Id}$

but where, at least,



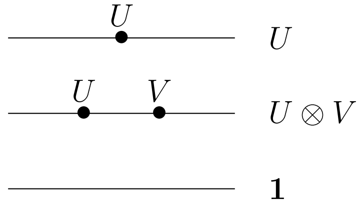
With a few more ingredients we can start assigning morphisms in this category to knots: we need a "neutral object" $\mathbf{1}$ such that

$$\mathrm{Hom}_A(\mathbf{1}, \mathbf{1}) = \mathbb{C}$$

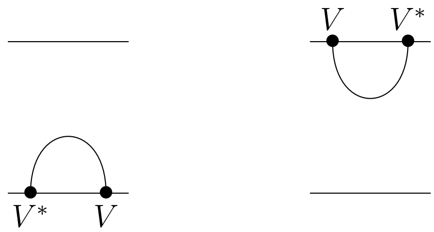
together with unit and counit morphisms

$$V^* \otimes V \to \mathbf{1}, \quad \text{and} \quad \mathbf{1} \to V \otimes V^*.$$

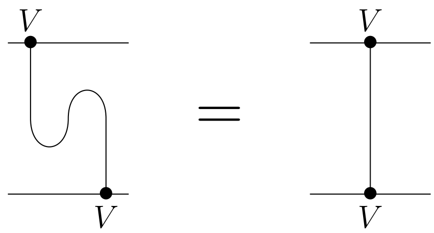
We represented objects and tensor products of objects as sequences of dots on the line, we shall represent the unit object as an absence of dots.



Then the counit and unit morphisms, respectively, will be written as



We wish these morphisms to also satisfy certain axioms, such as:



For example in the category of representations of $\mathfrak{g}$ we can take $\mathbf{1} = \mathbb{C}$ the trivial representation, V^* is the usual dual vector space of V, and

$$V^* \otimes V \to \mathbb{C}, \qquad \lambda \otimes v \mapsto < \lambda, v >= \lambda(v) \in \mathbb{C}$$

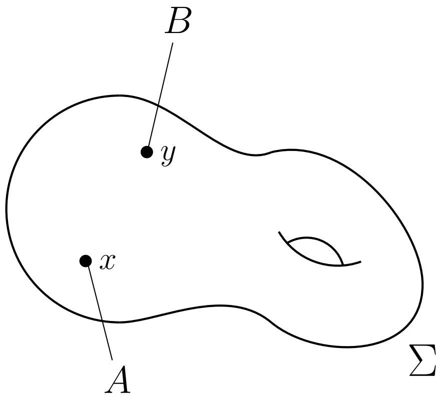
evaluation.

With all these structures in place, we can turn a knot into a morphism $\mathbf{1} \to \mathbf{1}$, that is, a complex number.

As one manipulates a knot in space, its projection changes via series of discrete operations known as Reidemeister moves. Obviously we would like all the Reidemeister moves to correspond to identities in the module category, such as the Yang-Baxter equation above. This translates into a series of structures and conditions on the algebra A.

So what kinds of algebras are good for getting knot invariants? There are two main classes of algebra that do the job: *quantum groups*, and *vertex algebras*.

Let V be a vertex algebra and Σ a Riemann surface. We take a number of marked points on Σ and label each of them with a representation of V. For example



There is a construction which assigns to this data a vector space

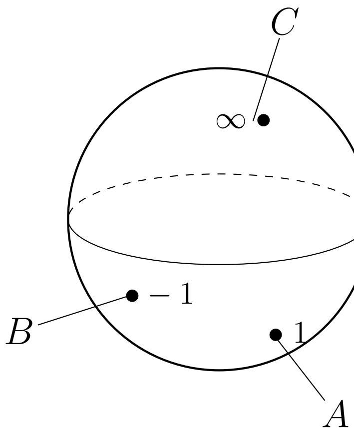
$$\mathcal{C}_\Sigma(A_x, B_y),$$

called *conformal blocks*. In order to carry out this construction it is necessary to interpret the variable z in the quantum field

$$a(z) = \sum_n a_{(n)} z^{-n-1}$$

as a coordinate on the Riemann sphere, which takes quite a lot of preparation.

Anyway, let A, B, C be three representations of V and consider a Riemann sphere S with three points marked:



We have the conformal blocks

$$\mathcal{C}_S(A_1, B_{-1}, C_\infty)$$

for this situation.

We might try to define a tensor product $A \otimes B$ by the condition

$$\mathrm{Hom}(A \otimes B, C) = \mathcal{C}_S(A_1, B_{-1}, C_\infty)$$

for all V-modules C. This is an instance of the philosophy of "representable functors": You characterise an object X in a category by specifying the set $\mathrm{Hom}(X, A)$ for all A in the category. The Yoneda lemma (Xoneda lemma in Argentina) says that this characterises X uniquely if it exists.

More precisely

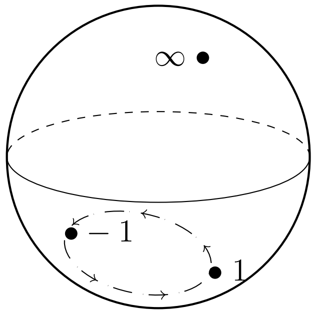
Definition 1. Let $F : \mathcal{C} \to \mathbf{Set}$ be a functor. A representation of F is an object $X \in \mathcal{C}$ and a natural isomorphism

$$\Phi : \mathrm{Hom}_{\mathcal{C}}(X, -) \cong F,$$

i.e., $\Phi_A : \mathrm{Hom}_{\mathcal{C}}(X, A) \cong F(A)$ for all $A \in \mathcal{C}$, compatible with all morphisms $A \to B$ in the obvious way.

Theorem 1. *Any two representations X_1 and X_2 of a functor F are naturally isomorphic.*

In the present case, moving the points 1 and -1 around the unit circle by 180° "deforms" $A \otimes B$ into $B \otimes A$.



So with this sort of logic in mind, it is possible (with a lot of work) to build knot invariants from vertex algebras.

Another famous application of vertex algebras is to monstrous moonshine. Let

$$\mathcal{H} = \{\tau \in \mathbb{C} \mid Im(\tau) > 0\}$$

denote the complex upper half plane. This manifold carries an action of the group $SL_2(\mathbb{Z})$ very important in number theory:

$$\tau \mapsto \frac{a\tau + b}{c\tau + d}.$$

Functions on $\mathcal{H}$ invariant under Γ, that is functions $f : \mathcal{H} \to \mathbb{C}$ such that

$$f(g\tau) = f(\tau), \qquad \text{for all } g \in \Gamma,$$

correspond to functions on the quotient space $\mathcal{H}/SL_2(\mathbb{Z})$. There are some details, but this quotient is (essentially) the Riemann sphere minus some points. If we consider functions that correspond to meromorphic functions on the sphere, then it turns out all such functions are rational functions of one particular function called

$$J(\tau) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + \cdots.$$

In the classification of finite simple groups, there are several infinite families, such as A_n, $n \geq 5$, or $SO_{2n}(\mathbb{F}_{p^k})$. And there are roughly 26 "sporadic" finite simple groups, that don't fit into these classes. The largest and most mysterious of these is the monster group $\mathbb{M}$ with approximately

$$8 \times 10^{53}$$

elements. The irreducible representations of $\mathbb{M}$ have dimensions

$$1, \qquad 196883, \qquad 21296876, \ldots$$

Monstrous moonshine begins with this weird coincidence (which is the first of many).

The (an?) explanation turns out to be that there is a vertex algebra $V^\natural$ which on the one hand has a $\mathbb{Z}_+$-grading

$$V^\natural = V_1^\natural \oplus V_2^\natural \oplus V_3^\natural \oplus \cdots$$

with graded dimension

$$1, \qquad 0, \qquad 196883, \qquad 21296876, \ldots$$

On the other hand $V^\natural$ carries a representation of the group $\mathbb{M}$ preserving the degrees. In particular each graded piece $V^\natural$ must split as a direct sum of irreducible representations of $\mathbb{M}$.

The conceptual part of the explanation is then why we should expect that the graded dimension of a vertex algebra is a modular function. This is related once more to the link with algebraic curves mentioned above. Though it's worth noting that there are some aspects of monstrous moonshine that are still not understood.

Our most important examples will come from certain infinite-dimensional Lie algebras, called Kac-Moody algebras. You should think of these as infinite dimensional Lie algebras that behave in every important respect as finite dimensional simple Lie algebras. So we will spend some time studying Kac-Moody algebras before we get to vertex algebras. And to make sure everybody is on the same page, I will begin with a quick review of the finite dimensional situation.

2023-01-10. Lecture 02

1.1 Review of finite dimensional Lie algebras

Let A be an associative algebra over $\mathbb{C}$, for example $\mathrm{End}(V)$, then A becomes a Lie algebra when equipped with the bracket operation

$$[a, b] = ab = ba.$$

Let R be a commutative algebra over $\mathbb{C}$. A derivation of R is a linear map

$$D : R \to R$$

satisfying

$$D(ab) = D(a)b + aD(b).$$

For example $D = d/dx$ is a derivation of the algebra $R = \mathbb{C}[x]$. In algebraic geometry commutative algebras are thought of as geometric spaces, called "affine schemes", and derivations correspond to vector fields.

The set $Der(R)$ of derivations of R is a vector space. The composition $D_1 \circ D_2$ of derivations is not a derivation, but the bracket is

$$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1.$$

This is another example of a Lie algebra.

Definition 2. A Lie algebra is a vector space $\mathfrak{g}$ together with a bilinear operation $[,] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ such that

$$[y, x] = -[x, y]$$

and

$$[x, [y, z]] = [[x, y], z] + [y, [x, z]].$$

A Lie algebra $\mathfrak{g}$ is called simple if it contains no proper nonzero ideals. A vector subspace $\mathfrak{a} \subset \mathfrak{g}$ is a subalgebra if $[\mathfrak{a}, \mathfrak{a}] \subset \mathfrak{a}$ and an ideal if $[\mathfrak{a}, \mathfrak{g}] \subset \mathfrak{a}$.

Examples of finite-dimensional simple Lie algebras:

- The special linear Lie algebra

$$sl_n = \{A \in Mat_n \mid Tr(A) = 0\},$$

- The special orthogonal Lie algebra

$$so_n = \{A \in Mat_n \mid A + A^T = 0\},$$

- The symplectic Lie algebra

$$sp_{2n} = \{A \in Mat_{2n} \mid \Omega A + A^T \Omega = 0\}, \quad \text{where} \quad \Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

It turns out that these are ALL the finite dimensional simple Lie algebras, with 5 exceptions.

Cartan's criterion.

Let $(,) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ be a bilinear form. We say that $(,)$ is invariant if

$$([x, y], z) = (x, [y, z])$$

for all $x, y, z \in \mathfrak{g}$.

Example 1. For $\mathfrak{g} = sl_n$,

$$(X, Y) = \mathrm{Tr}(XY).$$

Example 2. For any finite dimensional Lie algebra $\mathfrak{g}$ and $x \in \mathfrak{g}$, write $\mathrm{ad}(x) = [x, -]$. The Killing form is

$$\kappa(X, Y) = \mathrm{Tr}_{\mathfrak{g}}\, \mathrm{ad}(X)\, \mathrm{ad}(Y).$$

Exercise 2. *Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.*

Definition: A semisimple Lie algebra is a direct sum of simple ones

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s.$$

Theorem 3 (Cartan)**.** *The finite dimensional Lie algebra $\mathfrak{g}$ is semisimple if and only if its Killing form is nondegenerate.*

The Killing form turns out to be key to determining the structure of simple Lie algebras.

Consider the set of diagonal matrices $\mathfrak{h} \subset \mathfrak{g} = sl_n$, and consider the bracket

$$[h, E_{ij}] = (h_i - h_j)E_{ij}.$$

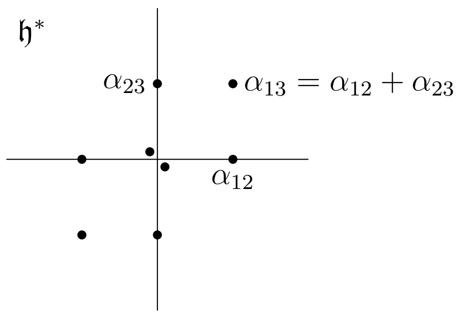
This is saying E_{ij} is an eigenvector for the $\mathrm{ad}(h)$ as $h \in \mathfrak{h}$. The eigenvalues are brought together into a "weight", a linear function

$$\alpha_{ij} : \mathfrak{h} \to \mathbb{C}.$$

The whole of sl_n actually splits into eigenspaces for a finite number of these weights

$$sl_n = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathbb{C}E_{ij}.$$

In fact if we write ε_i for extraction of the ith diagonal entry, then $\alpha_{ij} = \varepsilon_i - \varepsilon_j$. For example for $n = 3$ we have

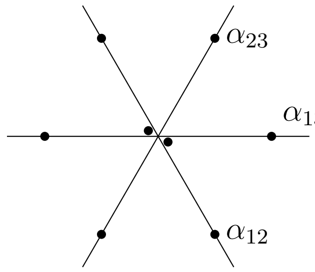


We restrict the Killing form to $(,) : \mathfrak{h} \times \mathfrak{h} \to \mathbb{C}$, and use it to define

$$\nu : \mathfrak{h} \to \mathfrak{h}^*, \quad \text{and} \quad (\alpha, \beta) = (\nu^{-1}(\alpha), \nu^{-1}(\beta)).$$

By the way, we use the notation $\langle \lambda, x\rangle = \langle x, \lambda\rangle = \lambda(x)$ for $x \in \mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$.

If we now redraw the roots with angles dictated by the Killing form, then the figure becomes as follows:



Theorem 4. *Any finite dimensional complex (semi)simple Lie algebra $\mathfrak{g}$ has a decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Delta} \mathbb{C}E_\alpha,$$

where $\mathfrak{h}$ is abelian. The weights $\alpha \in \Delta \subset \mathfrak{h}^*$ are called the "roots" of $\mathfrak{g}$, i.e.,

$$[h, E_\alpha] = \alpha(h)E_\alpha \quad \text{for all } h \in \mathfrak{h}$$

and

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta},$$

(In fact $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$ whenever $\alpha + \beta$ happens to also be a root). The set of roots is invariant under operations

$$r_\alpha : \beta \mapsto \beta - 2\frac{(\alpha, \beta)}{(\alpha, \alpha)}\alpha.$$

The set

$$\{\beta + n\alpha \mid n \in \mathbb{Z}\} \cap (\Delta \cup \{0\})$$

is called the α-string through β.

Key result: for any pair of roots α and β the α-string through β takes the form

$$\{\beta - p\alpha, \dots, \beta, \dots, \beta + q\alpha\}$$

and we have

$$p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)}.$$

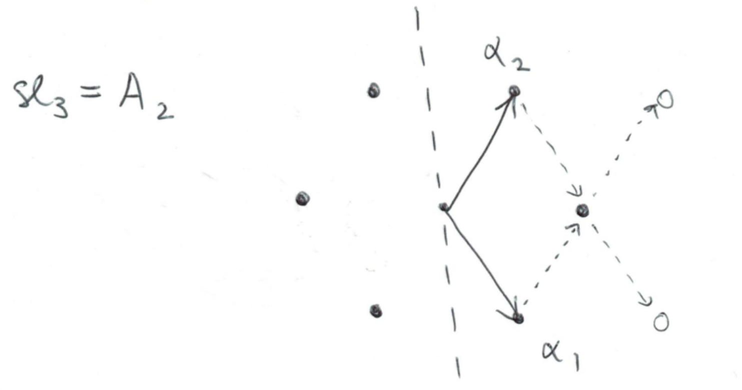
We split Δ as $\Delta_+ \cup \Delta_-$ where $\Delta_- = -\Delta_+$. Then we can define simple roots $\{\alpha_1, \dots, \alpha_r\}$ as those positive roots that cannot be decomposed into a sum of other positive roots.

For simple roots α_i and α_j we have $\alpha_i - \alpha_i \notin \Delta$, therefore

$$a_{ij} = 2\frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} = -q \leq 0$$

represents how many times one may add α_i to α_j before leaving Δ.

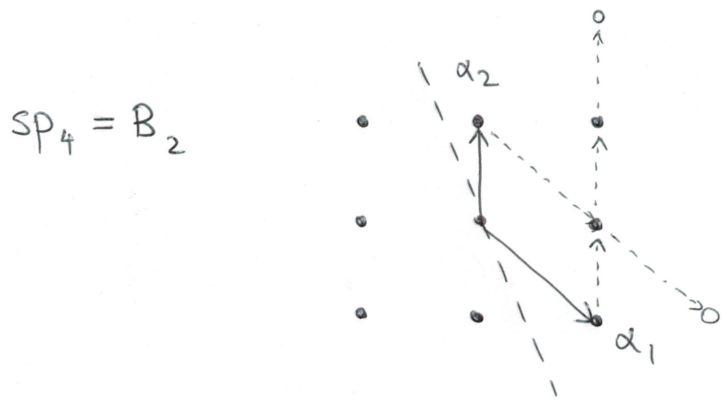
Analysing the Lie algebra sl_3 in this way:



we see that its Cartan matrix must be

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Similarly, analysing sp_4 in this way:



we see (note the importance of getting the order of the roots right) that its Cartan matrix must be

$$\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}.$$

Writing $E_i = E_{\alpha_i}$, we thus we have the relations

$$\mathrm{ad}(E_i)^{1-a_{ij}} E_j = 0,$$

and similar relations for the $F_i = E_{-\alpha_i}$. Also

$$[E_i, F_j] = 0 \quad \text{for } i \neq j.$$

Note also by definition of "weight", we have

$$[h, E_i] =< \alpha_i, h > E_i, \qquad [h, F_i] = - < \alpha_i, h > F_i, \quad \text{for } i = 1 \ldots r,$$

and of course

$$[h, h'] = 0 \quad \text{for all } h, h' \in \mathfrak{h}.$$

Lemma 5. *We have*

$$[E_i, F_i] = c\,\nu^{-1}(\alpha_i)$$

for some nonzero c.

It is standard to normalise E_i and F_i so that

$$[E_i, F_i] = \alpha_i^\vee = \frac{2}{(\alpha_i, \alpha_i)}\nu^{-1}(\alpha_i).$$

WHY? RELATED TO DEFINITION OF P_+ AS WE WILL SEE IN THE NEXT LECTURE.

Theorem 6 (Serre)**.** *The generators E_i, F_i, together with a basis of $\mathfrak{h}$, and the relations presented above, define $\mathfrak{g}$.*

I'll explain this better when we get to Kac-Moody algebras.

1.2 Some infinite dimensional Lie algebras

Let $\mathfrak{g}$ be a finite dimensional Lie algebra, and consider

$$L\mathfrak{g} = \mathfrak{g}[t, t^{-1}] = \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}].$$

If $\mathfrak{g}$ has a basis $\{e_1, \ldots, e_s\}$ then $L\mathfrak{g}$ has a basis

$$\{e_i t^m \mid 1 \le i \le s, m \in \mathbb{Z}\}.$$

There is a natural Lie bracket in $L\mathfrak{g}$:

$$[xt^m, yt^n] = [x, y]t^{m+n}.$$

In fact this is a special case of a very general construction: Let $\mathfrak{g}$ be a Lie algebra (over $\mathbb{C}$) and let R be a commutative $\mathbb{C}$-algebra, then

$$\mathfrak{g} \otimes R, \qquad [x \otimes f, y \otimes g] = [x, y] \otimes fg$$

is a Lie algebra.

Exercise 7. *Prove this statement.*

The particular example of this construction $L\mathfrak{g}$ is known as the "loop algebra" of $\mathfrak{g}$.

Is $L\mathfrak{g}$ simple? Even if $\mathfrak{g}$ is simple, the answer is no in general. Indeed let $I \subset R$ be an ideal, then $\mathfrak{g} \otimes I \subset \mathfrak{g} \otimes R$ is an ideal of Lie algebras. In particular we might consider the kernel of an "evaluation homomorphism":

$$\mathbf{ev}_a : \mathfrak{g}[t, t^{-1}] \to \mathfrak{g}$$

defined for any $a \neq 0$ by

$$\mathbf{ev}_a(xf(t)) = f(a)x.$$

The kernel of an evaluation homomorphism is an ideal in $\mathfrak{g}[t, t^{-1}]$.

Now we come to an interesting construction. Let $(,)$ be an invariant bilinear form on $\mathfrak{g}$. We shall define a Lie algebra structure on the direct sum vector space

$$\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K$$

by declaring K to commute with everyone ($[K, x] = 0$ for all $x \in \hat{\mathfrak{g}}$), and

$$[xt^m, yt^n] = [x, y]t^{m+n} + \delta_{m,-n} m(x, y)K.$$

Exercise 8. *Prove that $\hat{\mathfrak{g}}$ is a Lie algebra.*

The Lie algebra $\hat{\mathfrak{g}}$ is also not simple, indeed it has a nonzero centre. But in a certain precise sense $\hat{\mathfrak{g}}$ is closer to being simple than $L\mathfrak{g}$ is. There is also a sense in which $\hat{\mathfrak{g}}$ is almost a Lie algebra presented by generators and relations just like in Serre's theorem. In particular it has a Cartan matrix. Let's try to see what the Cartan matrix should be in the case $\mathfrak{g} = sl_2$. Recall the usual basis

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

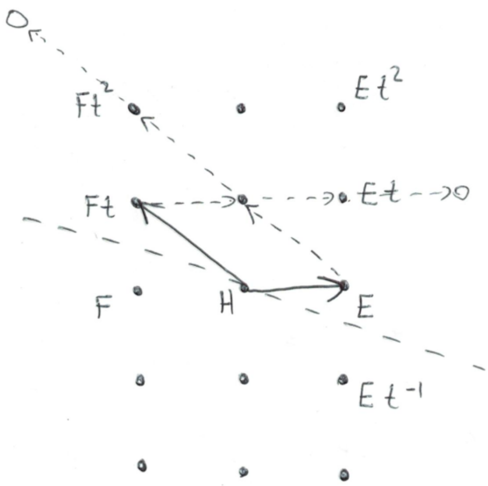
of sl_2.

If x lies in the α weight space of $\mathfrak{g} = sl_2$ then clearly

$$[H, xt^n] = \alpha(H)xt^n,$$

so it seems reasonable to build a Cartan subalgebra starting from H. At the same time it seems unlikely that any at^m for $m \neq 0$ could serve as an element of the Cartan... But $\mathfrak{h} = \mathbb{C}H$ seems too small, since all Et^n still lie in the same weight space, we would like to separate them into distinct weight spaces.

Let's simply draw dots in the plane for the "natural" basis vectors, consistent with their Lie brackets:



Separating the set into positive and negative roots as usual, it seems that

$$\{E, Ft\}$$

should be a set of simple root vectors, and the Cartan matrix ought to be

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

This turns out to be right, though there are a couple of confusing aspects to the story. We know that we need to enlarge $\mathfrak{h}$ in order to separate the xt^n into distinct weight spaces. One might think that the solution is to put $\mathfrak{h} = \mathbb{C}H + \mathbb{C}K$. But this won't work since $[K, \hat{\mathfrak{g}}] = 0$. In fact we must introduce a new element d to do the job, so in the end the Kac-Moody algebra will be

$$\mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d, \quad \text{with} \quad \mathfrak{h} = \mathbb{C}H + \mathbb{C}K + \mathbb{C}d.$$

In particular the number of simple roots is no longer equal to the dimension of the Cartan subalgebra. This is a general feature when the Cartan matrix is nondegenerate.

2023-01-12. Lecture 03

1.3 Review of representations of finite dimensional simple Lie algebras

A representation of a Lie algebra $\mathfrak{g}$ is a vector space V and a homomorphism

$$\rho : \mathfrak{g} \to \mathrm{End}(V)$$

of Lie algebras, i.e.,

$$\rho([x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x).$$

PROBLEM 1: Classify all the representations of a Lie algebra $\mathfrak{g}$.

Impossible in general, so we restrict to "nice", let's say finite dimensional representations. It's still not really possible in general. But we have

Theorem 9 (Weyl)**.** *If $\mathfrak{g}$ is finite dimensional and (semi)simple, then every finite dimensional representation of $\mathfrak{g}$ is isomorphic to a direct sum of irreducible representations.*

Proof uses cohomology, we saw it last semester.

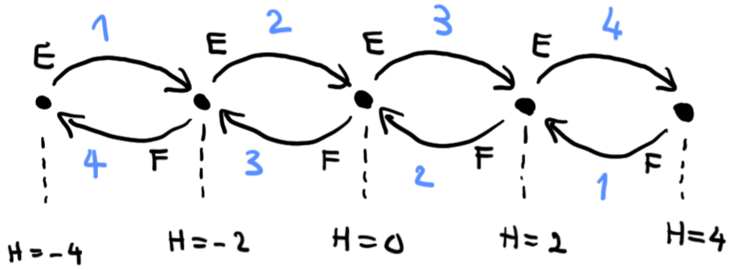
PROBLEM 2: For $\mathfrak{g}$ simple, classify and describe the irreducible finite dimensional representations of $\mathfrak{g}$.

Now this is possible.

Briefly recall the answer for sl_2. Fix the basis $\{E, H, F\}$ of sl_2 as above, the relations are

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

Theorem 10. *For each nonnegative integer $n \in \mathbb{Z}_+$, there is a unique irreducible representation of sl_2 of dimension $n + 1$, denoted V_n. It looks as follows*



Definition 3. Let $\mathfrak{g}$ be semisimple with Cartan subalgebra $\mathfrak{h}$. A weight space decomposition of a representation V of $\mathfrak{g}$ is

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda, \qquad hv = \lambda(h)v \text{ for all } h \in \mathfrak{h} \text{ for all } v \in V_\lambda$$

Any finite dimensional representation of $\mathfrak{g}$ has a weight space decomposition.

EXERCISE: Why?

Clearly

$$\mathfrak{g}_\alpha V_\lambda \subset V_{\lambda+\alpha}.$$

By repeatedly applying elements of $\mathfrak{g}_\alpha$ for $\alpha \in \Delta_+$ it is clear that V has one or more "highest" weights, in the obvious sense. If V is irreducible then the highest weight is unique.

EXERCISE: Why?

The Casimir element. Let $(,)$ be a nondegenerate invariant bilinear form on $\mathfrak{g}$ and let $\{a_i\}$ and $\{b_i\}$ be dual bases of $\mathfrak{g}$. If (V, ρ) is a representation of $\mathfrak{g}$ then we can consider the endomorphism

$$\Omega_V = \sum_i \rho(a_i)\rho(b_i) \in \mathrm{End}(V).$$

Proposition 11. *The operator Ω_V commutes with all $\rho(x)$ for $x \in \mathfrak{g}$. In particular if V is irreducible, Ω_V acts as a constant.*

Let us compute the Casimir in some representations of sl_2. The symmetric bilinear form

$$(E, F) = 1, \quad (H, H) = 2$$

is invariant, and

$$\Omega = EF + FE + \frac{1}{2}H^2 = 2FE + \frac{1}{2}H(H+2).$$

If we evaluate this operator on the highest weight vector (where $E = 0$ and $H = \lambda$) then

$$\Omega = \frac{1}{2}\lambda(\lambda + 2).$$

Now let us think about a general simple Lie algebra $\mathfrak{g}$. Let V be irreducible with highest weight λ and let $v \in V_\lambda$. The three vectors

$$E_i, \quad F_i, \quad \alpha_i^\vee = \frac{2}{(\alpha_i, \alpha_i)}\nu^{-1}(\alpha_i)$$

form an sl_2-triple. We consider

$$v, F_i v, F_i^2 v, \ldots,$$

this sequence terminates, which implies that

$$\alpha_i^\vee v = mv \qquad \text{for some integer } m \geq 0.$$

Therefore

$$< \lambda, \alpha_i^\vee > \in \mathbb{Z}_{\geq 0}.$$

This holds for each $i = 1, \ldots, r$. Denote by $P_+ \subset \mathfrak{h}^*$ this set of weights.

Rather incredibly, the converse is true:

Theorem 12. *If $\lambda \in P_+$ then there is a unique finite dimensional irreducible representation of $\mathfrak{g}$. It is denoted $L(\lambda)$.*

We now describe P_+ a little more concretely. Since the set

$$\Pi = \{\alpha_1, \ldots, \alpha_r\} \subset \mathfrak{h}^*$$

is a basis, the set

$$\Pi^\vee = \{\alpha_1^\vee, \ldots, \alpha_r^\vee\} \subset \mathfrak{h}$$

is a basis. Let

$$\{\varpi_1, \dots, \varpi_r\} \subset \mathfrak{h}^*$$

be the basis of $\mathfrak{h}^*$ naturally dual to $\Pi^\vee$. That is $< \varpi_i, \alpha_j^\vee >= \delta_{ij}$. These weights are called the *fundamental weights* of $\mathfrak{g}$. Clearly

$$P_+ = \{\sum_i k_i \varpi_i \mid k_i \in \mathbb{Z}_+\}$$

the set of nonnegative integer linear combinations of fundamental weights.

There should be some relation between the bases of simple roots and of fundamental weights.

Exercise 13. *Let $\mathfrak{g}$ be a finite dimensional simple Lie algebra, so that we know the nondegenerate invariant form $(,)$ is unique up to rescaling.*

- *Show that the coroots $\alpha_i^\vee$ are well-defined independent of the choice of $(,)$.*
- *Show that the change of basis matrix from $\{\varpi_1, \dots, \varpi_r\}$ to $\{\alpha_1, \dots, \alpha_r\}$ is the Cartan matrix. More precisely, that*

$$\alpha_i = \sum_{j=1}^{r} a_{ji} \varpi_j.$$

1.4 A little bit of quantum mechanics

Let us imagine a particle in three dimensional space. It has a position $\mathbf{x} = (x, y, z)$ relative to some chosen coordinates and a momentum $p = (p_x, p_y, p_z)$. Relative to the chosen origin, it has an angular momentum

$$L = \mathbf{x} \times \mathbf{p}$$

In quantum mechanics the state of the particle (or rather, our knowledge of its state) is described by a "wavefunction" $\psi(x, y, z) \in L^2(\mathbb{R}^3)$, and we interrogate the particle as to its properties by applying linear transformations to it. After measurement the wavefunction changes to an eigenfunction of the observable. The operator corresponding to the x-coordinate of the position is multiplication by x, i.e.,

$$\hat{x} : \psi(\mathbf{x}) \mapsto x\psi(\mathbf{x}),$$

and the operator for momentum p_x is

$$\hat{p}_x : \psi(\mathbf{x}) \mapsto -i\hbar \frac{\partial}{\partial x} \psi(\mathbf{x}).$$

The interpretation of the wavefunction ψ is probabilistic in the sense that if we normalise ψ so that

$$\int_{\mathbb{R}^3} \overline{\psi}(\mathbf{x})\psi(\mathbf{x})\, dx^3 = 1,$$

then the probability of finding the particle in the region $\Omega \subset \mathbb{R}^3$ is

$$\text{Prob(particle is in } \Omega) = \int_\Omega \overline{\psi}(\mathbf{x})\psi(\mathbf{x})\, dx^3 = 1.$$

After measurement of an observable $\hat{A}$, the transition of ψ to an eigenfunction ψ_λ of $\hat{A}$ with eigenvalue λ (i.e., $\hat{A}\psi_\lambda = \lambda\psi_\lambda$) is apparently random, but the probabilities are given by the Born rule

$$\text{Prob}(\psi \to \psi_\lambda) = \left| \int_{\mathbb{R}^3} \overline{\psi}(\mathbf{x})\psi_\lambda(\mathbf{x})\, dx^3 \right|^2.$$

Quantising the three components of angular momentum yields the three operators

$$\begin{aligned} L_x &= yp_z - zp_y \\ L_y &= zp_x - xp_z \\ L_z &= xp_y - yp_x. \end{aligned}$$

One can check that

$$[x\partial_y - y\partial_x, y\partial_z - z\partial_y] = -(z\partial_x - x\partial_z),$$

and cyclic permutations thereof. So

$$[L_x, L_y] = iL_z$$

and cyclic permutations thereof. Now consider $L_\pm = L_x \pm iL_y$, we compute

$$[L_z, L_\pm] = \pm L_\pm, \quad [L_+, L_-] = 2L_z,$$

and so

$$H = L_z, \quad E = \frac{1}{\sqrt{2}}(L_x + iL_y), \quad F = \frac{1}{\sqrt{2}}(L_x - iL_y).$$

is an sl_2-triple.

These operators form a representation of $so(3, \mathbb{C})$ on $L^2(\mathbb{R}^3)$. In spherical coordinates (r, θ, φ) we have

$$x\partial_y - y\partial_x = \partial_\varphi.$$

Time evolution in quantum mechanics is governed by a special operator called the Hamiltonian, according to the Schroedinger equation:

$$i\hbar\frac{\partial}{\partial t}\psi = \hat{H}\psi,$$

and states of well-defined energy E are eigenfunctions of the Hamiltonian:

$$\hat{H}\psi = E\psi.$$

Consider the hydrogen atom: a heavy proton is (essentially) fixed at the origin, and an electron moves around under its influence. The Hamiltonian in this case is

$$\hat{H} = -\frac{p^2}{2m} + V(r)$$

where $V(r)$ is the "potential energy" function

$$V(r) = -\frac{Ze^2}{r}, \qquad r = |\mathbf{x}|$$

The operators L_x, L_y and L_z each commute with $\hat{H}$, but do not commute with each other, so there is no such thing as a simultaneous eigenfunction of the three. So we select L_z and consider simultaneous eigenfunctions of L_z and H. In fact

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

also commutes with both L_z and with $\hat{H}$, so we can take all three at once:

$$\begin{aligned} L_z\psi &= m\psi \\ L^2\psi &= \Omega\psi \\ H\psi &= E\psi. \end{aligned}$$

These are differential equations, which can be written out explicitly. The first one is very simple, it is

$$\frac{\partial}{\partial\varphi}\psi = im\psi.$$

It is possible to completely solve the system using separation of variables. Periodicity in φ and θ impose integrality conditions on m and Ω, and square integrability restricts E to a series of discrete values. This is quantisation.

Explicitly

$$E = -\frac{mZ^2e^4}{2\hbar^2}\cdot\frac{1}{n^2}, \quad \Omega = \ell(\ell+1) \qquad 0 \le |m| \le \ell < n.$$

The first two differential equations involve the radial variable r only trivially, and can be essentially solved by considering the representation of $SO(3)$ on the space of functions on the sphere $L^2(S^2)$.

This is a good excuse to say a few words about representations of compact groups.

Theorem 14 (Peter-Weyl)**.** *Let G be a compact topological group and consider the vector space $L^2(G)$ as a representation of $G \times G$ via*

$$[(g_1, g_2)f](g) = f(g_1^{-1}gg_2).$$

Let $\{V_\lambda\}$ denote the set of all finite dimensional irreducible representations of G. Then

$$L^2(G) \cong \bigoplus_\lambda V_\lambda \otimes V_\lambda^*$$

as representations of $G \times G$.

There are some analytic details to make this precise. But I will not go into that.

Suppose we wish to know $L^2(S^2)$ as a representation of $G = SO(3)$. We can view S^2 as a quotient of G by its (non normal) subgroup $H = SO(2)$. Then

$$L^2(G) \supset L^2(G/H) = \{f \in L^2(G) \mid f(gh) = f(g) \text{ for all } h \in H\},$$

and the Peter-Weyl theorem tells us

$$L^2(G/H) \cong \bigoplus_{\lambda} V_\lambda \otimes (V_\lambda^*)_H$$

where

$$V^H = \{v \in V \mid hv = v \text{ for all } h \in H\}$$

in general denotes the subspace of invariant vectors.

Above we have reviewed the finite dimensional irreducible representations of $sl_2(\mathbb{C})$. The fd irreps of the compact Lie group $SU(2)$ are in bijection with these (in fact one obtains the representation of G on the same underlying vector space by "exponentiating" the representation of $\mathfrak{g}$). The group $SO(3)$ is a quotient of $SU(2)$ by a kernel $\{1, -1\}$ and it happens that HALF of the fd irreps descend to reps of $SO(3)$, namely the sl_2-irreps of EVEN highest weight (i.e., odd dimension) V_0, V_2, V_4, etc.

In this case

$$H = \{\begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} \mid t \in \mathbb{R}\} = \{\exp(tL_z) \mid t \in \mathbb{R}\}$$

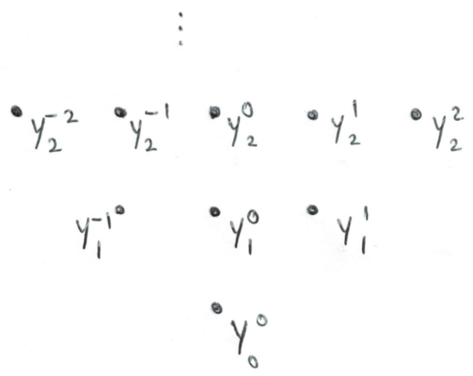
and in any of the representations V_{2k} the invariant subspace for H coincides with the invariant subspace for $\mathfrak{h} = \mathbb{C}L_z$, which is

$$V^{\mathfrak{h}} = \{v \in V \mid xv = 0 \text{ for all } x \in \mathfrak{h}\}$$

in this case just the 0 weight space, which is 1-dimensional. So we conclude

$$L^2(S^2) \cong \bigoplus_{\lambda \in 2\mathbb{Z}} V_\lambda.$$

EXERCISE: Explain (or read about and then explain) why the definitions V^H and $V^{\mathfrak{h}}$ correspond as they do.



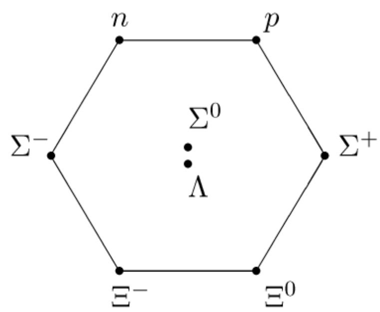
We can index vectors here by eigenvalue of the Casimir Ω and $H = L_z$. Up to a normalisation we see that Ω corresponds to L^2 and λ to ℓ. The weight vectors in the irreps here correspond (up to scaling) to specific functions on the sphere

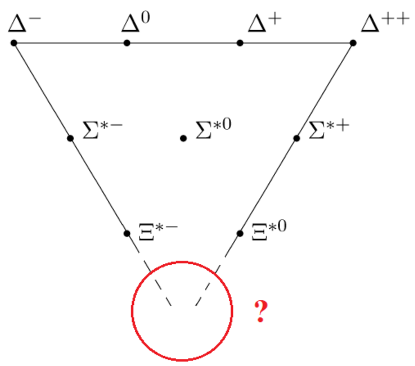
$$Y_\ell^m(\theta, \phi), \qquad 0 \le |m| \le \ell.$$

These are called *spherical harmonics* and are well known in applied mathematics.

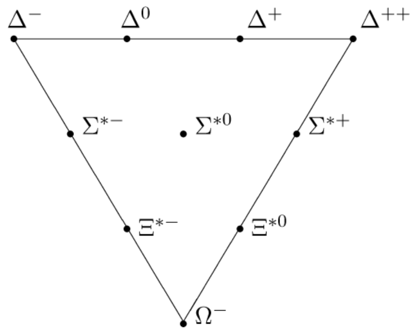
The account above would seem to suggest that there are n^2 independent states with energy $1/n^2$. In fact in a proper relativistic account we find there are $2n^2$ states, and there are slight differences in energy between some of these $2n^2$ states. In fact, in a sense the intrinsic angular momentum of the electron (known as "spin") corresponds to the 2-dimensional representation of sl_2 which does not descend to $SO(3)$.

In the theory of "isospin" the same Lie algebra sl_2 appears in much the same way as it did above as symmetries of a very abstract sort of spin in which "being a proton" and "being a neutron" correspond to the two angular momentum states.





Supposedly what happened was, Gell-Mann (also Neeman) noticed the similarity between these diagrams and the diagrams of the representations of sl_3, and was encouraged to predict the existence of a new particle: the so-called Ω^-.



In these diagrams isospin corresponds to the copy of sl_2 that acts horizontally. A new quantum number called "strangeness" decreases as we go down. The proton and neutron, p and n, have 0 strangeness, the Σ baryons have strangeness -1, etc. Finally the Ω^- has strangeness -3.[1]

2023-01-16. Lecture 04

1.5 Abstract nonsense

What we need to do now is to define universal enveloping algebra and free Lie algebra. We will do this in a way that uses more abstract machinery than is necessary, as an excuse to get familiar with the language of categories.

Definition 4. A category $\mathcal{C}$ consists of a collection of objects, and for each pair X, Y of objects, a set $\mathrm{Hom}(X, Y)$ of "morphisms" or "arrows". Also there are functions

$$\mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \to \mathrm{Hom}(X, Z),$$

of composition of arrows, the composition of $f : X \to Y$ with $g : Y \to Z$ is written $g \circ f$. The composition must be associative, i.e., $(h \circ g) \circ f = h \circ (g \circ f)$, and for all X there is to exist a "unit morphism" $1_X \in \mathrm{Hom}(X, X)$ which is neutral with respect to composition, i.e., if $f : X \to Y$ then $1_Y \circ f = f = f \circ 1_X$.

Definition 5. A functor $F : \mathcal{C} \to \mathcal{D}$ from the category $\mathcal{C}$ to the category $\mathcal{D}$ consists of an object $F(X) \in \mathcal{D}$ for each $X \in \mathcal{C}$, and a morphism

$$F(f) : F(X) \to F(Y)$$

for each $f : X \to Y$ in $\mathcal{C}$, which send units to units and composites to composites.

[1]Why is strangeness not symmetrical around the origin? Each irrep has a corresponding total spin, and by historical convention the strangeness is offset from the more sensible normalisation by this value.

Examples: The category **Set** of sets and functions. The category **Vec** of vector spaces over $\mathbb{C}$ and linear transformations. The category **Ass** of all associative algebras over $\mathbb{C}$ and homomorphisms. The category **Lie** of all Lie algebras over $\mathbb{C}$ and homomorphisms. The category $\mathfrak{g}$**-mod** of representations of a fixed Lie algebra $\mathfrak{g}$ and linear transformations commuting with the action of $\mathfrak{g}$, i.e., for (V,ρ) and (W,σ) representations, linear transformations $f : V \to W$ such that

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\rho(x)"'] & W \arrow[d, "\sigma(x)"] \\ V \arrow[r, "f"] & W \end{tikzcd}$$

Definition 6 (Tensor algebra)**.** Let V be a vector space over $\mathbb{C}$ and consider

$$T(V) = \mathbb{C} \oplus V \oplus (V \otimes V) \oplus (V \otimes V \otimes V) \oplus \cdots,$$

equipped with a product, defined on components

$$V^{\otimes m} \times V^{\otimes n} \to V^{\otimes(m+n)}$$

by

$$(v_1 \otimes v_2) * (w_1 \otimes w_2 \otimes w_3) = v_1 \otimes v_2 \otimes w_1 \otimes w_2 \otimes w_3,$$

for example and similarly in general, and extended in an $\mathbb{F}$-linear fashion to all $T(V)$.

One can check that $V \mapsto T(V)$ defines a functor

$$T : \mathbf{Vec} \to \mathbf{Ass}.$$

There is a functor **Ass** $\to$ **Vec** also, namely the act of forgetting the product.

Definition 7. A pair of functors $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ are said to be adjoint (L being the left adjoint, and R the right adjoint) if there is a bijection

$$\mathrm{Hom}_{\mathcal{C}}(L(X), Y) \to \mathrm{Hom}_{\mathcal{D}}(X, R(Y)),$$

for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, which is "natural" in both X and Y.

What do we mean by "natural"?

Definition 8. A natural transformation from a functor $F : \mathcal{C} \to \mathcal{D}$ to a functor $G : \mathcal{C} \to \mathcal{D}$ consists of a morphism

$$\theta_X : F(X) \to G(X),$$

in $\mathcal{D}$, one for each $X \in \mathcal{C}$ (the θ_X are called the components of the natural transformation θ), such that

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "\theta_X"'] & F(Y) \arrow[d, "\theta_Y"] \\ G(X) \arrow[r, "G(f)"] & G(Y) \end{tikzcd}$$

One writes $\theta : F \Rightarrow G$.

When we say the bijections above are natural, it means that if we fix $X \in \mathcal{C}$ and consider the two functors

$$\mathcal{D} \to \mathbf{Set}, \qquad Y \mapsto \mathrm{Hom}_{\mathcal{D}}(L(X), Y), \quad \text{and} \quad Y \mapsto \mathrm{Hom}_{\mathcal{D}}(X, R(Y)),$$

then there is a natural transformation

$$\theta : \mathrm{Hom}_{\mathcal{D}}(L(X), -) \Rightarrow \mathrm{Hom}_{\mathcal{D}}(X, R(-)).$$

Something similar must happen if we fix Y and vary X.

Exercise 15. *Find* 5 *examples of pairs of adjoint functors on the internet or in books.*

Definition 9 (Tensor algebra again)**.** The tensor algebra of V consists of an associative algebra $T(V)$ and a map of vector spaces $i : V \to T(V)$ such that for all maps $V \to A$ of vector spaces from V to an associative algebra A, there exists a unique homomorphism $T(V) \to A$ such that the following triangle commutes:

$$\begin{array}{ccc} \mathbf{Vec} & & \mathbf{Ass} \end{array}$$

$$\begin{tikzcd} V \arrow[rr, "\phi"] \arrow[dr, "i"'] & & A \\ & T(V) \arrow[ur, "\overline{\phi}"'] & \end{tikzcd}$$

More abstractly, the tensor algebra $T(V)$ of a vector space is obtained by applying the tensor algebra functor T to V. The functor T, in turn, is defined to be the left adjoint of the forgetful functor $\mathbf{Ass} \to \mathbf{Vec}$.

Starting from the notion of adjoint functor, let's see why $T(V)$ satisfies the universal property. Firstly if

$$\mathrm{Hom}_{\mathbf{Ass}}(T(V), A) \cong \mathrm{Hom}_{\mathbf{Vec}}(V, A),$$

then in particular

$$1 \in \mathrm{Hom}_{\mathbf{Ass}}(T(V), T(V)) \cong \mathrm{Hom}_{\mathbf{Vec}}(V, T(V)),$$

corresponds to some canonical linear map

$$i : V \to T(V).$$

Now let $f : V \to A$ be any linear map, and call the corresponding map of associative algebras

$$\overline{f} : T(V) \to A.$$

It's obvious that the unique map g making the diagram

$$\begin{tikzcd} T(V) \arrow[rr, "1"] \arrow[dr, "\overline{f}"] & & T(V) \arrow[dl, "g"] \\ & A & \end{tikzcd}$$

commute, is $g = \overline{f}$. So now in the category **Vec** we have g making

$$\begin{tikzcd} V \arrow[rr, "i"] \arrow[dr, "f"'] & & T(V) \arrow[dl, "g"] \\ & A & \end{tikzcd}$$

commute, and this g comes from a unique algebra morphism, namely $\overline{f}$.

Exercise 16.

- *Explain why an object characterised by a universal property is, if it exists, unique up to isomorphism.*
- *Confirm that the algebra $T(V)$ of the first definition of tensor algebra, satisfies the universal property of the second definition.*

The forgetful functor **Ass** $\to$ **Lie** also has a left adjoint

$$\mathfrak{g} \mapsto U(\mathfrak{g}).$$

This too is easy to define explicitly

$$U(\mathfrak{g}) = \frac{T(\mathfrak{g})}{([x, y] - x \otimes y - y \otimes x \mid x, y \in \mathfrak{g})}$$

(the quotient by the 2-sided ideal generated...).

Question: Does the map **Com** $\to$ **Lie** sending an algebra to its Lie algebra of derivations have an adjoint? Answer: No, in fact the process I just described is not a functor :)

The structure of $U(\mathfrak{g})$ is relatively simple (it looks like a polynomial algebra) but, unlike the case of $T(V)$, it's not very easy to prove it. The precise statement is

Theorem 17 (PBW)**.** *Let $x_1, \ldots, x_n$ be an ordered basis of $\mathfrak{g}$. Then $U(\mathfrak{g})$ has a basis consisting of products*

$$x_{i_1} x_{i_2} \cdots x_{i_s}, \qquad i_1 \leq i_2 \leq \ldots \leq i_s.$$

So $U(\mathfrak{g})$ looks like the polynomial algebra $S(\mathfrak{g})$, but whereas $xy = yx$ in $S(\mathfrak{g})$, we have $xy - yx = [x, y]$ in $U(\mathfrak{g})$.

Remark 18. *Possible project: Study one or more of the proofs of the PBW theorem.*

- *The proof in Humphreys Section 6.3 works by inductively constructing a representation of $\mathfrak{g}$ on $S(\mathfrak{g})$ the symmetric algebra,*
- *The proof in Bergman* The diamond lemma for ring theory *is more general and computer-algorithmic in flavour,*
- *The proof in Braverman, Gaitsgory* The Poincare-Birkhoff-Witt theorem for quadratic algebras of Koszul type *takes the perspective that $U(\mathfrak{g})$ is a "deformation" of $S(\mathfrak{g})$ and uses the link between cohomology and deformation theory.*

1.6 Free Lie algebras

Definition 10. Let V be a vector space. The free Lie algebra on V is the Lie algebra, together with linear map $i : V \to L(V)$, that satisfies the following universal property:

$$\begin{array}{cc} \textbf{Vec} & \textbf{Lie} \end{array}$$

$$\begin{tikzcd} V \arrow[rr, "\phi"] \arrow[dr, "i"'] & & \mathfrak{g} \\ & L(V) \arrow[ur, "\overline{\phi}"'] & \end{tikzcd}$$

As in previous cases, it can also be defined as a left adjoint to the act of forgetting a Lie algebra structure on a vector space.

Definition 11. Let V be a vector space, and define $L(V)$ to be the vector subspace of $T(V)$ spanned by all iterated commutators of elements of $i(V) \subset T(V)$. It is clearly a Lie algebra.

Proposition 19. *The explicit definition of $L(V)$ above is the free Lie algebra on V.*

Proof. We must show that it satisfies the universal property.

Let $\mathfrak{g}$ be any Lie algebra, and $\phi : V \to \mathfrak{g}$ a linear map. We compose this with the canonical map $i : \mathfrak{g} \to U(\mathfrak{g})$. Now by the universal property of $T(V)$ we obtain a unique map $T(V) \to U(\mathfrak{g})$.

Now we compose THIS map with the inclusion of $L(V)$ into $T(V)$. This is a map of Lie algebras. It is immediate that the image of this map is the Lie algebra $i(\mathfrak{g}) \subset U(\mathfrak{g})$. We need to invoke the PBW theorem now: it assures us that $i(\mathfrak{g})$ is an EMBEDDED copy of $\mathfrak{g}$ in $U(\mathfrak{g})$.

So we obtained the required map $L(V) \to \mathfrak{g}$, unique such that blah blah.

$$\begin{array}{ccc} \textbf{Vec} & \textbf{Lie} & \textbf{Ass} \end{array}$$

$$\begin{tikzcd} V \arrow[r, "\phi"] \arrow[d, "i"] \arrow[dr, "i"] & \mathfrak{g} \arrow[r, "i"] & U(\mathfrak{g}) \\ L(V) \arrow[r, "i"] & T(V) \arrow[ur, "\overline{\phi}"'] & \end{tikzcd}$$

□

Proposition 20. *There is an isomorphism of associative algebras*

$$U(L(V)) \cong T(V).$$

Proof. We will show that $U(L(V))$ satisfies the universal property of $T(V)$.

Let A be an associative algebra and $\phi : V \to A$ a linear map. We consider A_- which is A regarded as a Lie algebra via the commutator bracket $[,]$.

By the universal property the map $V \to A_-$ yields a unique map $L(V) \to A_-$. We compose this with the identity map to A. By the universal property the map $L(V) \to A$ yields a unique map $U(L(V)) \to A$.

$$\begin{tikzcd}
\textbf{Vec} & & \textbf{Lie} & \textbf{Ass} \\
V \arrow[rr, "\phi"] \arrow[dr, "i"'] & & A_- \arrow[r, "id"] & A \\
& L(V) \arrow[ur, "\overline{\phi}"] \arrow[urr, "\overline{\phi}"'] \arrow[dr, "i"] & & \\
& & U(L(V)) \arrow[uur, "\overline{\Phi}"'] &
\end{tikzcd}$$

$\square$

We use this proposition, and the PBW theorem, to investigate the structure of $L(V)$.

We give $T(V)$ the structure of a graded associative algebra by setting

$$\deg(1) = 0, \quad \deg(v) = 1 \quad \text{for all } v \in V.$$

This now determines that $\deg(x) = n$ for all $x \in V^{\otimes n}$. The Lie algebra $T(V)_-$ maintains this grading, and is clearly a graded Lie algebra, and the Lie subalgebra $L(V)$ inherits this grading too.

The grading on $L(V)$ induces a grading on $U(L(V))$. Indeed this is a general proposition: if $\mathfrak{g}$ is a graded Lie algebra, then $U(\mathfrak{g})$ carries a unique structure of graded associative algebra in such a way that $i : \mathfrak{g} \to U(\mathfrak{g})$ is a graded homomorphism of Lie algebras (Note: I did not even have to declare that $\deg(1) = 0$ as we have no choice about it – in a graded associative algebra the only possible degree for the unit element is 0).

Note that this grading on $U(L(V))$ is a different thing from the filtration on $U(L(V))$ which is always present on any $U(\mathfrak{g})$.

Lemma 21. *The isomorphism*

$$U(L(V)) \cong T(V)$$

is compatible with gradings.

Proof. Either examine the constructions of the objects and maps involved, or argue via uniqueness of the induced gradings. $\square$

Definition 12. Let U be a $\mathbb{Z}$-graded vector space with finite dimensional graded pieces

$$U = \bigoplus_{n \in \mathbb{Z}} U_n.$$

We define the graded dimension of U to be the formal power series

$$\sum_{n \in \mathbb{Z}} \dim(U_n) q^n.$$

We denote by $Supp(U)$ the support of U the set

$$Supp(U) = \{n \in \mathbb{Z} \mid U_n \neq 0\}.$$

Proposition 22. *Suppose U is a $\mathbb{Z}$-graded vector space with support*

$$Supp(U) \subset \mathbb{Z}_{\geq 1},$$

and $a_n = \dim(U_n) < \infty$ for all n. Then the symmetric algebra $S(U)$ has well-defined graded dimension given by the explicit formula

$$\prod_n (1 - q^n)^{-a_n}.$$

Example: If we take the symmetric algebra on $\mathbb{C}x$ concentrated in degree 1, then we get the polynomial ring $\mathbb{C}[x]$ with graded dimension

$$1 + q + q^2 + q^3 + \cdots = (1 - q)^{-1}.$$

Proof. EXERCISE $\square$

Proposition 23. *Let $d = \dim(v)$, then the graded dimension of $T(V)$ is $(1 - dq)^{-1}$.*

For definiteness suppose $d = \dim(V) = 2$. The free Lie algebra $L(V)$ has some graded dimension

$$a_1 q + a_2 q^2 + \cdots$$

and $U(L(V))$ has the same graded dimension as the symmetric algebra $S(L(V))$ which is

$$(1 - q)^{-a_1}(1 - q^2)^{-a_2}(1 - q^3)^{-a_3} \cdots .$$

When we expand this out, notice that the coefficient of q is a_1 as this and all subsequent terms only contribute to higher powers of q. But the graded dimension of $T(V)$ is

$$1 + 2q + 4q^2 + 8q^3 + 16q^4 + \cdots ,$$

so $a_1 = 2$. So far so good, now we multiply $(1 - 2q)^{-1}$ by $(1 - q)^2$ to cancel this factor, and examine the series expansion

$$1 + q^2 + 2q^3 + 4q^4 + 8q^5 + \cdots ,$$

to deduce $a_2 = 1$. Continuing in this way we determine the a_n recursively. In fact the graded dimension of $L(V)$ is

$$2q + q^2 + 2q^3 + 3q^4 + 6q^5 + 9q^6 + 18q^7 + 30q^8 + 56q^9 + 99q^{10} + 186q^{11} + \cdots$$

Definition 13. A Lyndon word of length n on an alphabet of d letters is a word of length n which is lexicographically first among all of its cyclic permutations.

Apparently the following theorem is true. REFERENCE?

Theorem 24. *Let $d = \dim(V)$. The vector space $L(V)_n$ has a basis indexed by Lyndon words of length n on d letters.*

Definition 14. Let $\{a_n\}$ and $\{b_n\}$ be two sequences of integers related by

$$1 + \sum_{n\geq 1} b_n q^n = \prod_{m\geq 1}(1-q^m)^{-a_m},$$

then $\{b_n\}$ is called the *Euler transform* of $\{a_n\}$.

In light of this definition, the graded dimension of $L(V)$ is given by the inverse Euler transform of the sequence $\{d^n\}$.

Theorem 25. *The inverse Euler transform is given "explicitly" by*

$$a_n = \frac{1}{n}\sum_{d|n}\mu(n/d)c_d$$

where c_n defined by

$$c_n = nb_n - \sum_{k=1}^{n-1} c_k b_{n-k},$$

Here $\mu(n)$ is the Moebius function: $\mu(n)$ is 0 if n is divisible by a square, and otherwise is $(-1)^N$ where N is the number of prime factors of n.

The Euler transform has quite a few applications in combinatorics.

Definition 15. A partition of $N \in \mathbb{Z}_+$ is a list

$$(\lambda_1, \lambda_2, \ldots \lambda_s), \quad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_s \geq 1$$

such that $\sum \lambda_i = N$. A coloured partition of N is a partition of N in which each copy of n is assigned one of a set of a_n "colours". We do not distinguish the order of the colours.

For example $(4, 2, 1, 1)$ is a partition of $N = 8$. The set of colours might be the same for each n, or it might be different. If there are two colours, $\{+, -\}$ say, for each n then $a_n = 2$ for all n and for example

$$(4_+, 2_+, 1_+, 1_-) = (4_+, 2_+, 1_-, 1_+)$$

is a two-coloured partition of $N = 8$. If we take the sequence

$$1, 0, 1, 0, 1, 0, 1, 0, 1, \ldots,$$

then even λ are not allowed in the partition, and an $\{a_n\}$-partition is the same thing as a partition into odd parts.

Theorem 26. *The number b_N of $\{a_n\}$-parititions of N is given by the Euler transform of $\{a_n\}$.*

In particular the number $p(N)$ of honest partitions of N is given by the generating function

$$1 + \sum_{n \geq 1} p(n) q^n = \prod_{m \geq 1} (1 - q^m)^{-1}.$$

You might be very familiar with this fact. If not, I strongly suggest thinking about it for a few hours to make sure you really understand why it is true. It will be useful in this course.

Remark 27. *I noticed that*

$$\dim L(V)_n = \frac{1}{n} \sum_{d|n} \mu(n/d) 2^d,$$

so apparently the transform $\{b_n\} \to \{c_n\}$ in the theorem sends $\{2^n\}$ to itself. How general is this, is it a coincidence? Can you prove it?

Remark 28. *If we consider $\mathcal{C} = \mathcal{D} = \mathfrak{g}$-**mod** and the functors $F = G = Id$, then a natural transformation is a $\theta_V \in \operatorname{End} V$ for each (V, ρ) such that*

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ {\scriptstyle \theta_V}\downarrow & & \downarrow{\scriptstyle \theta_W} \\ V & \xrightarrow{f} & W \end{array}$$

But now just grab $x \in \mathfrak{g}$, it is clear that setting $\theta_V = \rho(x)$ does exactly what we need. So there is a natural transformation from the identity functor to itself, for each $x \in \mathfrak{g}$. But that's not all, we can compose natural transformations $Id \Rightarrow Id$ component-by-components, so clearly in this case they form an associative algebra containing $\mathfrak{g}$. Hmm, I wonder what this algebra could be?

2023-01-17. Lecture 05

2 Kac-Moody algebras

2.1 Kac-Moody algebras

In the finite dimensional case we had roots and coroots

$$\alpha_i \in \mathfrak{h}^*, \quad \alpha_i^\vee \in \mathfrak{h},$$

and the Cartan matrix

$$a_{ij} = < \alpha_i^\vee, \alpha_j > = 2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}.$$

Clearly $a_{ii} = 2$. Although it may happen that $a_{ij} \neq a_{ji}$, certainly

$$a_{ij} = 0 \Leftrightarrow a_{ji} = 0.$$

We now define a *generalised Cartan matrix* to be an $n \times n$ integer matrix $A = (a_{ij})$, such that

- $a_{ii} = 2$.
- $a_{ij} \leq 0$ if $i \neq j$.
- $a_{ij} = 0$ iff $a_{ji} = 0$.

There is one more condition we might like to impose. Returning to the classical case, let D be the diagonal matrix

$$\epsilon_{ii} = 2/(\alpha_i, \alpha_i).$$

Then

$$A = DB$$

where $B_{ij} = (\alpha_i, \alpha_j)$ is the symmetric Gram matrix of the invariant bilinear form.

So we say that the generalised Cartan matrix is *symmetrisable*, if there exist diagonal D and symmetric B such that $A = DB$. We shall fix a choice D with diagonal entries $\epsilon_1, \ldots \epsilon_n$ (all nonzero).

Definition 16. Let A be a generalised Cartan matrix. A complex vector space $\mathfrak{h}$ and linearly independent subsets $\Pi^\vee = \{\alpha_1^\vee, \ldots, \alpha_n^\vee\} \subset \mathfrak{h}$ and $\Pi = \{\alpha_1, \ldots, \alpha_n\} \subset \mathfrak{h}^*$ such that

$$< \alpha_i^\vee, \alpha_j >= a_{ij}, \quad \text{for all } i, j$$

is called a "realisation" of A.

EXERCISE: Show that a realisation $\mathfrak{h}$ of A must satisfy

$$\dim \mathfrak{h} \geq 2n - \operatorname{Rank} A$$

Let $\tilde{\mathfrak{n}}_+$ (resp. $\tilde{\mathfrak{n}}_-$) be the free Lie algebra with the generators $e_1, \ldots e_n$ (resp. $f_1, \ldots f_n$). We put $\tilde{\mathfrak{g}}(A) = \mathfrak{h} \oplus \tilde{\mathfrak{n}}_+ \oplus \tilde{\mathfrak{n}}_-$ and make it into a Lie algebra by setting

- $[\mathfrak{h}, \mathfrak{h}] = 0$.
- For all $h \in \mathfrak{h}$, $[h, e_i] = \langle \alpha_i, h \rangle e_i$ and $[h, f_i] = - \langle \alpha_i, h \rangle f_i$.
- $[e_i, f_j] = \delta_{ij} \alpha_i^\vee$.

$$Q = \mathbb{Z}\alpha_1 + \cdots \mathbb{Z}\alpha_n \subseteq \mathfrak{h}^*$$
$$Q^\vee = \mathbb{Z}\alpha_1^\vee + \cdots \mathbb{Z}\alpha_n^\vee \subseteq \mathfrak{h},$$

Also write Q_+ and $Q_+^\vee$ where we take $\mathbb{Z}_+$ instead of $\mathbb{Z}$ in each case.

If $\beta = \sum_i k_i\alpha_i$ then we put

$$\tilde{\mathfrak{g}}(A)_\beta = \{\text{spans of nested commutators containing } k_i \text{ copies of } e_i\}$$

similarly for $\tilde{\mathfrak{n}}_-$ and $\tilde{\mathfrak{g}}(A)_0 = \mathfrak{h}$. Thus $\tilde{\mathfrak{g}}(A)$ is a Q-graded Lie algebra.

Proposition 29. *The space $\tilde{\mathfrak{g}}(A)$ with the brackets given* is *a Lie algebra.*

This was proved in my Lie algebras class 2022-2, and is Kac IDLA Theorem 1.2 (a).

Proof. The idea is to first consider the free Lie algebra F with generators

$$e_1, \dots e_n, f_1, \dots, f_n, h_1, \dots h_N,$$

where $\{h_1, \dots h_N\}$ is a basis of a realisation $\mathfrak{h}$ of A. Then quotient this free Lie algebra by the ideal generated by the relations above.

Introduce a Q-grading in F, this descends to the quotient, since the ideal is compatible with the Q-grading. This will be $\tilde{\mathfrak{g}}(A)$.

The rest of the proof consists in showing (using the defining relations) that an arbitrary nested commutator of elements e, h, f can be written as a commutator purely of one type of element: e, h or f. $\square$

Theorem 30 (Gabber-Kac)**.** *Denote by $I \subset \tilde{\mathfrak{g}}(A)$ the maximal ideal such that $I \cap \mathfrak{h} = 0$. Then I is generated by the Serre relations:*

$$(\operatorname{ad} e_i)^{1-a_{ij}} e_j \quad \text{and} \quad (\operatorname{ad} f_i)^{1-a_{ij}} f_j \quad \text{for } i \neq j.$$

PROJECT: Study the proof of this theorem. It appears in IDLA as Theorem 9.11. The preliminary result is to prove that generators of the maximal ideal lie in degrees $\beta \in Q$ that satisfy

$$(\beta, \beta) = 2(\rho, \beta).$$

Definition 17. The Kac-Moody algebra associated with A is the quotient $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)/I$.

Additional notation:

Write $\lambda \leq \mu$ whenever $\mu - \lambda \in Q_+$. This makes Q (all of $\mathfrak{h}^*$ in fact) into a partially ordered set.

For $\alpha = \sum_i k_i \alpha_i \in Q_+$, write $\mathrm{ht}(\alpha) = \sum_i k_i$ the height of αt.

We write

$$P = \{\lambda \in \mathfrak{h}^* | \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z} \text{ for each } i\} \subset \mathfrak{h}^*,$$
$$P_+ = \{\lambda \in \mathfrak{h}^* | \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_+ \text{ for each } i\} \subset \mathfrak{h}^*.$$

We call P the *weight lattice*, though this is a slight misnomer as P is only discrete if A is invertible. The elements of P_+ are called *dominant integral weights*.

We write

$$\mathrm{mult}\, \beta = \dim \mathfrak{g}(A)_\beta = \dim \tilde{\mathfrak{g}}(A)_\beta - \dim I_\beta.$$

For some $\beta \in Q$ we have $\mathrm{mult}\, \beta = 0$, and we define the root system of $\mathfrak{g}(A)$ to be

$$\Delta = \{\alpha \in Q \backslash 0 \mid \mathrm{mult}\, \alpha > 0\}.$$

By construction we have $\Delta = \Delta_+ \cup \Delta_-$ where $\Delta_\pm = \Delta \cap \pm Q_+$.

2.2 Some examples

(1) Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

This is the finite dimensional simple Lie algebra sl_3. Here are the root spaces:

$$\begin{array}{ccc} & 1 & 1 \\ 1 & \bullet & 1 \\ 1 & 1 & \end{array}$$

(2) Cartan matrix

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

We will later prove that this Kac-Moody algebra is isomorphic to the central extension of the loop algebra

$$\tilde{sl}_2 = sl_2[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d.$$

Here are (some of) the root spaces:

$$\begin{array}{ccccccccccc}
 & & & & & & & & 1 & 1 & 1 \\
 & & & & & & & 1 & 1 & 1 & \\
 & & & & & & 1 & 1 & 1 & & \\
 & & & & & 1 & 1 & 1 & & & \\
 & & & & 1 & 1 & 1 & & & & \\
 & & & 1 & 1 & 1 & & & & & \\
 & & 1 & \bullet & 1 & & & & & & \\
 & 1 & 1 & 1 & & & & & & & \\
1 & 1 & 1 & & & & & & & & \\
1 & 1 & & & & & & & & &
\end{array}$$

(3) Cartan matrix

$$A = \begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix}.$$

This is an example of a *hyperbolic* Kac-Moody algebra. There is no known way to describe this Lie algebra other than through its definition as a Kac-Moody algebra.

						1	6	27	73	162	288	449	600
						2	9	27	60	107	162	211	240
						3	9	23	39	60	73	80	73
					1	4	9	16	23	27	27	23	16
					1	4	6	9	9	9	6	4	1
				1	2	3	4	4	3	2	1		
				1	1	2	1	1					
			1	1	1	1							
		1	•	1									
1	1	1	1										
2	1	1											
3	2	1											

(Thanks to Igor Blatt for calculating these root multiplicities.)

2.3 The Killing form and the Weyl group

Construction of a realisation: Let

$$\mathfrak{h} = \mathbb{C}\alpha_1^\vee + \cdots + \mathbb{C}\alpha_n^\vee + \mathbb{C}d_1 + \cdots \mathbb{C}d_?,$$

Let $\Lambda_1, \ldots, \Lambda_n, \delta_1, \ldots \delta_?$ be the dual basis, and define

$$\alpha_j = \sum_i a_{ij}\Lambda_i + ?,$$

this guarantees $< \alpha_i, \alpha_j^\vee >= a_{ij}$ as required. The extra terms ? are added to make the α_i linearly independent.

Example: For the Cartan matrix $\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$ we take

$$\mathfrak{h} = \mathbb{C}\alpha_0^\vee + \mathbb{C}\alpha_1^\vee + \mathbb{C}d, \qquad \mathfrak{h}^* = \mathbb{C}\Lambda_0 + \mathbb{C}\Lambda_1 + \mathbb{C}\delta,$$

and

$$\alpha_0 = 2\Lambda_0 - 2\Lambda_1 + \delta, \qquad \alpha_1 = -2\Lambda_0 + 2\Lambda_1.$$

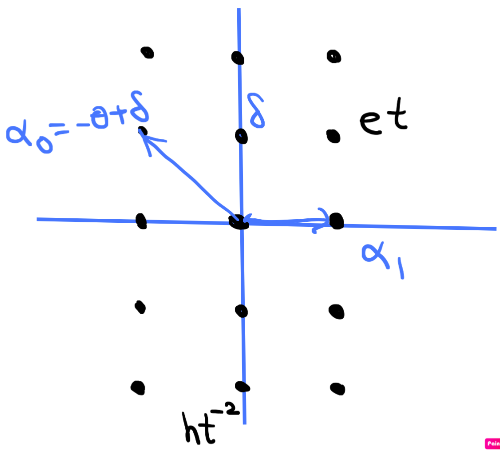
A symmetric bilinear form on $\mathfrak{h}$ can be reconstructed as follows (denote $\mathfrak{h}_c$ the span of the d's):

$$(\alpha_i^\vee, h) = (h, \alpha_i^\vee) = \epsilon_i < \alpha_i, h > \text{ for all } h \in \mathfrak{h}, \quad (\mathfrak{h}_c, \mathfrak{h}_c) = 0.$$

For the $\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$ case, a more convenient basis is:

$$\mathfrak{h}^* = \mathbb{C}\alpha_1 + \mathbb{C}\delta + \mathbb{C}\Lambda_0,$$

Then $(\alpha_1, \alpha_1) = 2$, $(\Lambda, \delta) = 1$, all other pairsings are zero. In particular this form (working over $\mathbb{R}$) has signature $(2, 1)$.



Extending $(,)$ to all of $\mathfrak{g}$:

First extend $(\cdot, \cdot)$ to a bilinear form

$$\tilde{\mathfrak{g}} \times \tilde{\mathfrak{g}} \to \mathbb{C}$$

by putting $(\mathfrak{h}, \mathfrak{n}_\pm) = 0$ and

$$(e_i, f_j) = \delta_{ij} \frac{2}{(\alpha_i, \alpha_i)}$$

(which is forced on us), and then extending to all combinations by requiring *symmetry* and *invariance*, i.e.,

$$(x, [y, z]) = ([x, y], z).$$

One may verify that this form on $\tilde{\mathfrak{g}}$ factors to an invariant symmetric bilinear form on $\mathfrak{g}$. Furthermore, the restriction is nondegenerate. It is called the Killing form.

Definition 18. The reflection in the simple root α_i is the linear transformation $r_i : \mathfrak{h}^* \to \mathfrak{h}^*$ defined by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \, \alpha_i.$$

One may verify that $(r_i\lambda, r_i\mu) = (\lambda, \mu)$ for all $\lambda, \mu \in \mathfrak{h}^*$.

Definition 19. The Weyl group of $\mathfrak{g}(A)$ is the subgroup

$$W \subset O(\mathfrak{h}^*, (,))$$

generated by the simple reflections r_i.

Example: affine sl_2 as usual.

2023-01-19. Lecture 06

2.4 Highest weight modules and the BGG category $\mathcal{O}$

Definition 20. Let $\Lambda \in \mathfrak{h}^*$. The Verma module $M(\Lambda)$ of highest weight Λ is by definition

$$M(\Lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{h}\oplus\mathfrak{n}_+)} \mathbb{C}_\Lambda,$$

where $\mathbb{C}_\Lambda = \mathbb{C}v_\Lambda$ is the 1-dimensional $(\mathfrak{h} \oplus \mathfrak{n}_+)$-module defined by $\mathfrak{n}_+ v_\Lambda = 0$ and $hv_\Lambda = \langle\Lambda, h\rangle\, v_\Lambda$ for all $h \in \mathfrak{h}$.

Recall in general if A is an associative algebra over $\mathbb{C}$ and M and N are right and left A-modules, then we have the vector space

$$M \otimes_A N = \frac{M \otimes_{\mathbb{C}} N}{(m \cdot a \otimes n = m \otimes a \cdot n)},$$

with variations for bimodules, etc. By the PBW theorem, as a vector space $M(\Lambda)$ is isomorphic to $U(\mathfrak{n}_-)$, and has a basis

$$f_{i_1} f_{i_2} \cdots f_{i_s} v_\Lambda, \qquad i_1 \leq \ldots \leq i_s,$$

where $f_1, f_2, \ldots$ is any fixed ordered basis of $\mathfrak{n}_-$.

Definition 21. Let M be a representation of $\mathfrak{g}$, and let $v \in M_\lambda$ be a vector of weight λ. Then v is called a singular vector if $\mathfrak{n}_+ v = 0$, and a highest weight vector if $M_{\lambda+\alpha} = 0$ for all $\alpha \geq 0$. A highest weight module is one generated by a highest weight vector.

A singular vector v in a representation V determines a unique morphism

$$M(\Lambda) \to V$$

by $v_\Lambda \mapsto v$. If v is a highest weight vector, then the morphism is surjective.

A representation of highest weight Λ is graded by $\Lambda - Q_+$.

Lemma 31. *If $I \subseteq M(\Lambda)$ is a submodule then $I = \oplus_\lambda (I \cap M(\Lambda))_\lambda$.*

Proof. This was proved in the Lie algebras course 2022-2. □

The main use of this lemma to guarantee that the sum of proper submodules of $M(\Lambda)$ is proper, so that there exists a maximal proper submodule $J \subset M(\Lambda)$. We denote the quotient (which is irreducible)

$$L(\Lambda) = M(\Lambda)/J.$$

Definition 22. The BGG Category $\mathcal{O}$ is the full subcategory of the category of all $\mathfrak{g}$-modules whose objects M satisfy the following conditions

- M is finitely generated as a $U(\mathfrak{g})$-module,
- M is a weight representation, i.e.,

$$M = \bigoplus_{\lambda \in \mathfrak{h}^*} M_\lambda,$$

- the action of $\mathfrak{n}_+$ on M is locally finite, i.e., for any $v \in M$ we have

$$\dim U(\mathfrak{n}_+)v < \infty.$$

From the definition it follows (See Humphreys "BGG" book) that

- The weights of M lie in a finite union of sets of the form $\Lambda - Q_+$.
- The weight spaces with respect to $\mathfrak{h}$ are finite dimensional.

EXERCISE: Prove these properties.

Definition 23. The formal character of $V \in \mathcal{O}$ is the sum

$$\mathrm{Ch}_V = \sum_{\lambda \in \mathfrak{h}^*} \dim V_\lambda e^\lambda.$$

Example: For the finite dimensional representation

$$L(3\varpi) = M(3\varpi)/J$$

of sl_2, we have write $x = e^\varpi$

$$\mathrm{Ch}_{M(3\varpi)} = x^3(1 + x^{-2} + x^{-4} + \cdots) = \frac{x^3}{1 - x^{-2}}, \qquad \mathrm{Ch}_{L(3\varpi)} = x^3 + x + x^{-1} + x^{-3}.$$

In general the character of the Verma module $M(\Lambda)$ is easily seen to be e^Λ/R where

$$R = \prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})^{-\operatorname{mult}(\alpha)}.$$

Modules in $\mathcal{O}$ are NOT completely reducible in general (even if $\mathfrak{g}$ is finite type). Just look at the case above where $J \subset M(3\varpi)$ is a submodule, but there is NO complementary subrepresentation isomorphic to $L(3\varpi)$.

This is similar to something that happens for the universal enveloping algebra $U(\mathfrak{g})$.

Definition 24. An increasing filtration F on an associative algebra A is a sequence

$$0 = F_{-1} \subset F_0 \subset F_1 \subset F_2 \subset \ldots \subset A$$

such that $F_i F_j \subset F_{i+j}$. The associated graded of the filtration is the vector space

$$\mathrm{gr}(A) = \bigoplus_i \mathrm{gr}_i(A), \qquad \mathrm{gr}_i(A) = F_i/F_{i-1}.$$

It carries a product $\mathrm{gr}_p(A) \times \mathrm{gr}_q(A) \to \mathrm{gr}_{p+q}(A)$ defined by

$$(x + F_{p-1}) \cdot (y + F_{q-1}) = xy + F_{p+q-1}.$$

Example: Let $F_p \subset U(\mathfrak{g})$ be the span of all products of elements of $\mathfrak{g}$ of length at most p. Then the associated graded is commutative (this is easy to see), and in fact

$$\mathrm{gr}_F\, U(\mathfrak{g}) \cong S(\mathfrak{g}).$$

(This statement is actually equivalent to the PBW theorem.)

Definition 25. Let V be a representation of a Lie algebra $\mathfrak{g}$. A decreasing filtration on V is a sequence

$$V = V_0 \supset V_1 \supset V_2 \supset \ldots \supseteq V_n \supseteq \ldots$$

of submodules V_n. A composition series of V is a finite filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \ldots \supseteq V_n = 0$$

by submodules, such that V_j/V_{j+1} is irreducible for each j.

Definition 26. Let V be a representation of $\mathfrak{g}$ and $\lambda \in \mathfrak{h}^*$. A weak (LOCAL?) composition series of V is a filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \ldots \supseteq V_n = 0$$

such that for each $j = 0, \ldots n$ one of the following holds:

- $(V_j/V_{j+1})_\mu = 0$ for all $\mu \in \lambda + Q_+$,
- $V_j/V_{j+1} \cong L(\lambda_j)$ for some λ_j, and in particular is irreducible.

Lemma 32 (Kac, Lemma 9.6)**.** *Every $V \in \mathcal{O}$ has local composition series for every $\lambda \in \mathfrak{h}^*$.*

Proof. The proof is by induction on $a(V, \lambda) = \sum_{\mu \geq \lambda} \dim V_\mu$. Choose $v_\mu \in V_\mu$ a singular vector with $\mu \geq \lambda$. Let $U = U(\mathfrak{g})v_\mu \subseteq V$. U is a highest weight module and so when we divide U by its maximal proper submodule $\overline{U}$ we get $U/\overline{U} \cong L(\mu)$. Now consider the chain $V \supseteq U \supseteq \overline{U} \supseteq 0$. The middle quotient is isomorphic to $L(\mu)$ and we can apply the inductive assumption to say that the strings before and after split, so the claim follows. □

The number of times $L(\mu)$ occurs in the 'composition series' is well-defined and is the same for all choices of $\lambda \leq \mu$. So we define $[V : L(\mu)]$ to be this number.

We claim that for $V \in \mathcal{O}$,

$$\operatorname{Ch} V = \sum_{\lambda \in \mathfrak{h}^*} [V : L(\lambda)] \operatorname{Ch} L(\lambda). \tag{1}$$

In fact if we fix λ and a local composition series with respect to λ, then

$$\operatorname{Ch} V - \sum_{\mu \geq \lambda} [V : L(\mu)] \operatorname{Ch} L(\mu)$$

is a linear combination of e^{ν} with $\nu \not\geq \lambda$. By exhaustion, the formula stated holds.

2023-01-23. Lecture 07

2.5 The Casimir Operator and the weights of singular vectors

Definition 27. A Weyl vector is a vector $\rho \in \mathfrak{h}^*$ such that

$$\langle \rho, \alpha_i^\vee \rangle = 1$$

for each simple coroot $\alpha_i^\vee$. (Equivalently $(\rho, \alpha_i) = (\alpha_i, \alpha_i)/2$ for each i.)

The Weyl vector is not necessarily unique, though if $\mathfrak{g}$ is of finite type then the Weyl vector is unique and is

$$\rho = \sum_{i=1}^{n} \varpi_i.$$

For each root space $\mathfrak{g}_\alpha$ choose a basis $\{e_\alpha^{(1)}, \dots e_\alpha^{(k)}\}$ in such a way that

$$\{e_{-\alpha}^{(1)}, \dots e_{-\alpha}^{(k)}\}$$

is the basis of $\mathfrak{g}_{-\alpha}$ dual with respect to $(\cdot, \cdot)$. Also choose dual bases $\{u^i\}$ and $\{u_i\}$ of $\mathfrak{h}$. Now

Definition 28. The Casimir operator is

$$\Omega = 2 \sum_{\alpha \in \Delta_+} \sum_i e_{-\alpha}^{(i)} e_\alpha^{(i)} + 2\nu^{-1}(\rho) + \sum_i u_i u^i.$$

In the general case Ω is not an element of $U(\mathfrak{g})$. But if $V \in \mathcal{O}$ then any $v \in V$ is annihilated by $\mathfrak{g}_\alpha$ for all but finitely many $\alpha \in \Delta_+$, and so $\Omega|_V$ is a finite sum.

Thus what we have is $\Omega|_V \in \operatorname{End}(V)$, well defined for each $V \in \mathcal{O}$.

Proposition 33. *The endomorphism $\Omega|_V$ commutes with the action of x for every $x \in \mathfrak{g}$.*

Proposition 34. *Let $v \in V_\lambda \subseteq V$ be a singular vector. Then*

$$\Omega v = 2(\rho, \lambda)v + [\sum_i \lambda(u_i)\lambda(u^i)]v = (\lambda + 2\rho, \lambda)v.$$

Corollary 35. *In a highest weight representation V with highest weight Λ, any singular weight λ must satisfy*

$$(\lambda + 2\rho, \lambda) = (\Lambda + 2\rho, \Lambda)$$

2.6 Integrable modules

One might ask: why should we care specifically about finite dimensional representations of $\mathfrak{g}$? For some applications, the most relevant representations are those which "exponentiate" or "integrate" to representations of the corresponding Lie group G.

Definition 29. A $\mathfrak{g}$-module V is said to be *integrable* if

- it is a weight module, i.e., $V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda$ with finite dimensional weight spaces V_λ,
- All e_i and f_i act locally nilpotently on V (i.e., for each $v \in V$, $e_i^N v = f_i^N v = 0$ for some $N \in \mathbb{Z}_+$ depending on v).

For general Kac-Moody algebras, there are few finite dimensional representations. But there are still many interesting integrable representations.

Three facts

- The Verma module $M(\Lambda)$ is never integrable.
- The adjoint representation of $\mathfrak{g}$ is integrable.
- $L(\Lambda)$ is integrable when the highest weight Λ is dominant integral, $\Lambda \in P_+$.

These are straightforward to prove, but certainly not trivial.

That $L(\Lambda)$ is integrable. Let $\mathfrak{a}_i \subset \mathfrak{g}$ be the copy of sl_2 spanned by the sl_2-triple $\{E_i, \alpha_i^\vee, F_i\}$ and consider

$$V = U(\mathfrak{a}_i)v_\Lambda \subset M(\Lambda).$$

Since $\Lambda \in P_+$ we have $< \Lambda, \alpha_i^\vee >= m \in \mathbb{Z}_+$ by definition, and we easily verify (by representation theory of sl_2) that $F_i^{m+1}v_\Lambda$ is a singular vector. Hence $F_i^{m+1}v_\Lambda = 0$ in $L(\Lambda)$, which means F_i acts nilpotently on v_Λ. Clearly E_i does too.

By an induction (and using the Serre relations), one can show that F_i and E_i act nilpotently on any vector

$$F_{i_1}^{k_1} \cdot F_{i_s}^{k_s} v_\Lambda,$$

so integrability follows by induction. □

For the rest of this section let V be an integrable $\mathfrak{g}$-module. We claim that V decomposes into a direct sum of finite-dimensional $\mathfrak{a}_i$-modules, and on each one $\mathfrak{h}$ acts semisimply.

Pick $v \in V$ and let $U = \sum F_i^m E_i^n v \subseteq V$, it is easy to confirm that U is $\mathfrak{a}_i$-invariant and $\mathfrak{h}$-diagonal. Local nilpotence of E_i and F_i implies U is finite dimensional. By the Weyl complete reducibility theorem, U decomposes into a direct sum of irreducibles. We have shown each $v \in V$ lies in a finite direct sum of irreducible $\mathfrak{a}_i$-modules, hence V is a direct sum of irreducible $\mathfrak{a}_i$-modules.

EXERCISE: Check carefully that this argument is valid.

Now pick any nonzero $V_\lambda \subseteq V$ and consider the set of weights

$$\{\lambda + n\alpha_i \mid n \in \mathbb{Z}\} \cap \{\text{weights of } V\}.$$

By $\mathfrak{sl}_2$-theory it is an interval, a priori it could be infinite. However, by the considerations above, since the weight space dimensions are all finite, the string has to be finite.

Suppose, therefore, that

$$\{\lambda + n\alpha_i \mid n \in \mathbb{Z}\} \cap \{\text{weights of } V\} = \{\lambda + n\alpha_i \mid -p \leq n \leq q\}.$$

The weight multiplicities are symmetrical in a certain sense. In particular, the highest and lowest weights in our string must be negatives of each other (upon evaluation on $\alpha_i^\vee$), i.e.,

$$\langle \lambda + q\alpha_i, \alpha_i^\vee \rangle = - \langle \lambda - p\alpha_i, \alpha_i^\vee \rangle .$$

This implies

$$p - q = \langle \lambda, \alpha_i^\vee \rangle . \tag{2}$$

The symmetry of the multiplicities about the mid-point can be expressed as

$$\dim V_\lambda = \dim V_{r_i\lambda}, \quad \text{where} \quad r_i\lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i.$$

Thus we have

Theorem 36. *The character*

$$\mathrm{Ch}_V = \sum_{\mu \in \mathfrak{h}^*} \dim(V_\mu) e^\mu$$

of an integrable $\mathfrak{g}$-module V is invariant under the Weyl group action.

2.7 The Kac-Weyl character formula for dominant integral highest weight modules

We will prove a formula for the character of $L(\Lambda)$ for $\Lambda \in P_+$.

Recall that we have

$$\mathrm{Ch}_{M(\Lambda)} = \sum_{\lambda \leq \Lambda} [M(\Lambda) : L(\lambda)] \, \mathrm{Ch}_{L(\lambda)} \, .$$

Let us choose a linear order on P consistent with the partial order $\leq$. The set of coefficients

$$b_{\Lambda,\lambda} = [M(\Lambda) : L(\lambda)]$$

can be viewed as a $\mathbb{Z} \times \mathbb{Z}$ matrix. This matrix is upper triangular with ones on the diagonal. Hence the matrix can be inverted and we may write

$$\mathrm{Ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} \, \mathrm{Ch}_{M(\lambda)} \, . \tag{3}$$

We have $m_{\Lambda,\Lambda} = 1$ and $m_{\Lambda,\lambda} \in \mathbb{Z}$.

Combining the formula for the character of the Verma with equation (3) and multiplying through by e^{ρ} yields

$$R \, \mathrm{Ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda}. \tag{4}$$

For reasons that will become clear soon, we will multiply through to obtain:

$$e^{\rho} R \, \mathrm{Ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda+\rho}. \tag{5}$$

First we will show that $e^{\rho} R$ is W-skew invariant by checking it on the generating reflections r_i. We need the following two facts:

- For all $\alpha \in \Delta$, $\operatorname{mult} r_i(\alpha) = \operatorname{mult} \alpha$. Indeed Δ is W-invariant.
- The set $\Delta_+ \backslash \{\alpha_i\}$ is invariant under r_i.

The first fact follows from the W-invariance of characters of integrable modules, since the adjoint module $\mathfrak{g}$ is integrable.

As for the second fact: write $\beta \in \Delta_+ \backslash \{\alpha_i\}$ as a $\mathbb{Z}_+$-linear combination of the α_k with a positive coefficient for some simple root $\alpha_j \neq \alpha_i$. Then $r_i(\beta)$ also has a positive coefficient of α_j and hence lies in $\Delta_+ \backslash \{\alpha_i\}$, because every root is either positive or negative.

Now we calculate

$$r_i(e^{\rho} R) = e^{\rho - \alpha_i}(1 - e^{\alpha_i}) \prod_{\alpha \in \Delta_+ \backslash \{\alpha_i\}} (1 - e^{-\alpha})^{\operatorname{mult} \alpha} = -e^{\rho} R,$$

which is the W-skew invariance asserted above.

We have assumed that $\Lambda \in P_+$, which implies that $L(\Lambda)$ is integrable, which in turn implies that $\mathrm{Ch}_{L(\Lambda)}$ is W-invariant. So

$$\sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda+\rho}$$

must be W-skew invariant.

Fix a weight λ such that $m_{\Lambda,\lambda} \neq 0$. If $w(\lambda+\rho) = \mu+\rho$ then $m_{\Lambda,\lambda} = \epsilon(w) m_{\Lambda,\mu}$ because of skew invariance. Because $m_{\Lambda,\lambda} \neq 0$ only for $\lambda \leq \Lambda$ we deduce that $w(\lambda+\rho) \leq \Lambda+\rho$ for all $w \in W$.

Next we will show that the range of summation in (5) is $\lambda \in W \circ \Lambda$, where $\circ$ is the *shifted action* of the Weyl group $w \circ \lambda = w(\lambda+\rho) - \rho$.

Proof. First we claim that

$$[M(\Lambda) : L(\lambda)] \neq 0$$

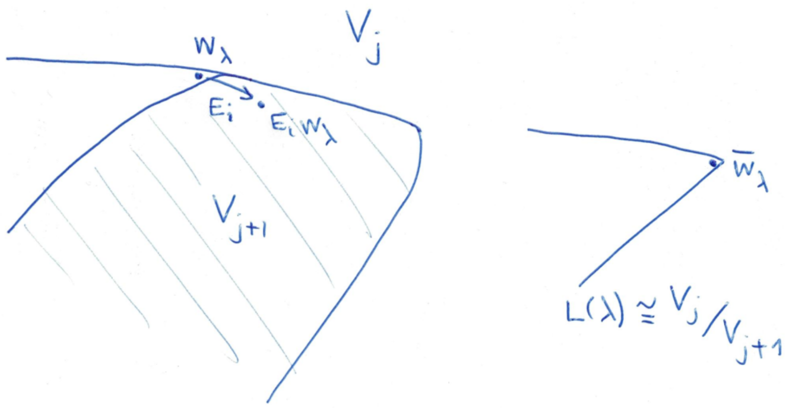
only if

$$|\lambda+\rho|^2 = |\Lambda+\rho|^2.$$

This is not as obvious as it might look. Certainly Ω acts as the scalar $|\Lambda+\rho|^2 - |\rho|^2$ on all vectors in $M(\Lambda)$. From the nonzero multiplicity, we infer that there exists a local composition series

$$V = V_0 \supset V_1 \supset \ldots \supset V_n = 0,$$

with $V_j/V_{j+1} \cong L(\lambda)$ for some j. It is tempting to claim that V_j must have a singular vector of weight λ and we would be done. However we need to be slightly more careful: The following could happen



Nevertheless we know that Ω descends to the same scalar on the quotient of V_j which, being isomorphic to $L(\lambda)$, yields the desired equality.

Since the coefficients $m_{\Lambda,\lambda}$ arise from the inverse matrix, it follows that $m_{\Lambda,\lambda} \neq 0$ only for Λ and λ having the same relation $|\lambda+\rho|^2 = |\Lambda+\rho|^2$ as above.

Now let $\mu = w \circ \lambda$ be such that $\mathrm{ht}(\Lambda - \mu)$ takes its minimal value as w ranges over W. We will prove that $\mu = \Lambda$. First we will prove that $\mu \in P_+$. Indeed if $\langle \mu + \rho, \alpha_i^\vee \rangle < 0$ for some i then

$$\Lambda - r_i \circ \mu = \Lambda + \rho - r_i(\mu + \rho) = \Lambda - \mu + \langle \mu + \rho, \alpha_i^\vee \rangle \, \alpha_i$$

has strictly lower height than $\Lambda - \mu$. Hence $\langle \mu + \rho, \alpha_i^\vee \rangle \geq 0$ for each i, which is to say

$$\mu + \rho \in P_+.$$

Now, $\Lambda + \rho$ and $\mu + \rho$ are in P_+ and $(\Lambda + \rho) - (\mu + \rho) = \sum_i k_i \alpha_i$ where all $k_i \geq 0$. Also $|\Lambda + \rho|^2 = |\mu + \rho|^2$, which implies

$$0 = |\Lambda + \rho|^2 - |\mu + \rho|^2 = (\Lambda + \mu + 2\rho, \Lambda - \mu) = \sum_i k_i \, \langle \Lambda + \mu + 2\rho, \alpha_i^\vee \rangle \, (\alpha_i, \alpha_i).$$

The terms in the summation on the right are all positive, so each $k_i = 0$ in fact, hence $\mu = \Lambda$.

It follows that $\lambda \in W \circ \Lambda$, as claimed. $\square$

We have shown that

$$R \, \mathrm{Ch} \, L(\Lambda) = \sum_{\lambda \in W \circ \Lambda} m_{\Lambda, \lambda} e^{\lambda}.$$

There is one more step to the final version:

Theorem 37 (Kac)**.** *Let $\Lambda \in P_+$, so that $L(\lambda)$ is integrable and lies in $\mathcal{O}$. Then*

$$R \, \mathrm{Ch} \, L(\Lambda) = \sum_{w \in W} \epsilon(w) e^{w \circ \Lambda}. \tag{6}$$

To complete the proof we just need to show that the stabiliser of Λ in $(W, \circ)$ is $\{1\}$.

2.8 The dominant integral weights form a fundamental domain for W

It remains to prove the following fact: Let $\Lambda \in P_+$, then the stabiliser of Λ under the dot action of W is trivial. The proof relies on some annoyingly complicated lemmas about the geometry of the W-action on $\mathfrak{h}^*$. These are given here for completeness, but I intend to skip this section in class.

Now, A is an integer matrix, so we could have defined $\mathfrak{h}_{\mathbb{R}}$ and $\mathfrak{h}_{\mathbb{R}}^*$, using $\mathbb{R}$ in place of $\mathbb{C}$. We can embed $\mathfrak{h}_{\mathbb{R}} \hookrightarrow \mathfrak{h}$. The action of W may be defined on $\mathfrak{h}_{\mathbb{R}}^*$ since the definition makes no reference to the base field.

The *positive Weyl chamber* is the set

$$C = \{h \in \mathfrak{h}_{\mathbb{R}} | \langle \alpha_i, h \rangle \geq 0 \text{ for each } i\}.$$

We claim that C is a fundamental domain for W, i.e., for all $h \in C$, $w(h) = h$ if an only if $w = 1$. The claim above about P_{++} will follow.

The basic combinatorial lemma is the following:

Lemma 38. *If $r_{i_1} \cdots r_{i_t}(\alpha_i) < 0$ then there exists s $(1 \leq s \leq t)$ such that*

$$r_{i_1} \cdots r_{i_s} \cdots r_{i_t} r_i = r_{i_1} \cdots r_{i_{s-1}} r_{i_{s+1}} \cdots r_{i_t}.$$

Proof. For each $k < t$ let $\beta_k = r_{i_{k+1}} \cdots r_{i_t}(\alpha_i)$, and let $\beta_t = \alpha_i$. By hypothesis $\beta_0 < 0$, and $\beta_t > 0$.

Let $\beta_{s-1} < 0$, $\beta_s > 0$, we have $\beta_{s-1} = r_{i_s}\beta_s$. Now, if $\beta \in \Delta_+$, then $(\beta + \mathbb{Z}\alpha_k) \cap \Delta \subseteq \Delta_+$ unless $\beta = \alpha_i$ simply because every root is either positive or negative. Therefore if $\beta \in \Delta_+$ and $r_i(\beta) < 0$ then $\beta = \alpha_i$. Applying this to the situation at hand shows that $\beta_s = \alpha_{i_s}$. Also $\alpha_{i_s} = w(\alpha_i)$ where $w = r_{i_{s+1}} \cdots r_{i_t}$.

The subspaces $\mathbb{R}\alpha_i$ and $\alpha_i^\perp$ are complementary, and r_i is the unique linear transformation that sends α_i to $-\alpha_i$ and restricts to the identity on $\alpha_i^\perp$. The same goes for α_{i_s}. Because w is an orthogonal transformation sending α_i to α_{i_s}, it sends $\alpha_i^\perp$ to $\alpha_{i_s}^\perp$. Therefore $r_{i_s} = w r_i w^{-1}$.

Substituting this for r_{i_s} into $r_{i_1} \cdots r_{i_s} \cdots r_{i_t} r_i$ yields $r_{i_1} \cdots r_{i_{s-1}} r_{i_{s+1}} \cdots r_{i_t}$ as desired. □

Suppose now that $h \in C$ and $h' = w(h) \in C$ also. Let $w = r_{i_1} \cdots r_{i_t}$ be a *reduced expression* for w, meaning that $t \in \mathbb{Z}_+$ is minimal among all possible such representations of w. We have $\langle \alpha_{i_t}, h \rangle \geq 0$ by hypothesis, therefore $\langle w(\alpha_{i_t}), h' \rangle \geq 0$.

On the other hand, $w(\alpha_{i_t}) < 0$. For assume $w(\alpha_{i_t}) > 0$, then $r_{i_1} \cdots r_{i_{t-1}}(\alpha_{i_t}) < 0$. Lemma 38 then shows that our expression for w was not reduced, a contradiction.

Therefore $\langle w(\alpha_{i_t}), h' \rangle \leq 0$, hence $\langle w(\alpha_{i_t}), h' \rangle = 0$, hence $\langle \alpha_{i_t}, h \rangle = 0$, hence $r_{i_t} h = h$.

An induction on the length t of the reduced expression for w shows that $w(h) \in C$ implies $w = 1$.

2023-01-24. Lecture 08

2.9 An example of the Weyl-Kac character formula for $\mathfrak{g} = \mathfrak{sl}_3$

I worked out in detail for

$$L(2\varpi_1 + \varpi_2),$$

with `Mathematica` code.

....

2.10 Affine Lie algebras are Kac-Moody algebras

We had a class of examples

$$\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K,$$

which, as we saw, appear very close to being Kac-Moody algebras. We wanted to make the subspaces $\mathfrak{g}_\alpha t^m$ ($\alpha \in \Delta$ the root system of $\mathfrak{g}$, and for $m > 0$) into positive root spaces, and we wanted these to be linearly independent from the $\alpha \in \Delta$ themselves. This led us to extend $\hat{\mathfrak{g}}$ to $\widetilde{\mathfrak{g}}$:

$$\tilde{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d,$$

where d is a new element in the Cartan which "measures the power of t" in a root vector, i.e.,

$$[d, et^m] = met^m.$$

Explicitly then $\mathrm{ad}(d) = td/dt$.

So our Cartan subalgebra should be

$$\tilde{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}K \oplus \mathbb{C}d.$$

Let us write the dual Cartan as

$$\tilde{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta,$$

here Λ_0 is the vector dual to K, i.e., $\Lambda_0(K) = 1$ and $\Lambda_0(d) = \Lambda_0(\mathfrak{h}) = 0$, and $\delta(d) = 1$ and $\delta(K) = \delta(\mathfrak{h}) = 0$.

Besides the simple roots of $\mathfrak{g}$ there ought to be one more simple root α_0, and its root vector ought to be $F_\theta t$ (where $\theta \in \Delta_+$ is the highest root of $\mathfrak{g}$ and we take $\{E_\theta, \theta^\vee, F_\theta\}$ and sl_2-triple in $\mathfrak{g}$ with $E_\theta \in \mathfrak{g}_\theta$). By definition of δ, then we have

$$\alpha_0 = \delta - \theta.$$

We confirm that $\Pi = \{\alpha_0, \alpha_1, \ldots, \alpha_n\}$ is linearly independent.

What can we say about the coroot $\alpha_0^\vee$? Well in a Kac-Moody algebra we ought to have

$$[E_i, F_i] = \alpha_i^\vee.$$

But for $i = 0$ and $E_0 = F_\theta t$ we should take $F_0 = e_\theta t^{-1}$, and then we find

$$\alpha_0^\vee = [F_\theta t, E_\theta t^{-1}] = K - \theta^\vee.$$

We have the following recognition theorem

"Anything that looks like a Kac-Moody algebra, is one."

Theorem 39 (IDLA, Section 1.4)**.** *Let $\mathfrak{g}$ be a Lie algebra generated by $e_1, \dots, e_n$ together with $f_1, \dots, f_n$ and an abelian subalgebra $\mathfrak{h} \subset \mathfrak{g}$. Suppose, further, that there are linearly independent sets*

$$\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\} \subset \mathfrak{h}, \quad \Pi = \{\alpha_1, \dots, \alpha_n\} \subset \mathfrak{h}^\vee,$$

such that

- *The relations*

$$[e_i, f_j] = \delta_{ij}\alpha_i^\vee$$
$$[h, e_i] = < \alpha_i, h > e_i, \quad [h, f_i] = - < \alpha_i, h > f_i,$$

  hold in $\mathfrak{g}$,

- *The only ideal in $\mathfrak{g}$ with zero intersection with $\mathfrak{h}$ is 0.*

- *$\mathfrak{h}$ is a realisation of the matrix $A = (a_{ij})$ defined by $a_{ij} = < \alpha_i^\vee, \alpha_j >$.*

Then $\mathfrak{g}$ is isomorphic to the Kac-Moody algebra associated with A.

Proof. Basically trivial. We immediately get a surjection $\tilde{\mathfrak{g}}(A) \to \mathfrak{g}$ since the former is so close to being free, then this must factor to $\mathfrak{g}(A) \to \mathfrak{g}$. □

Theorem 40 (IDLA Theorem 7.4)**.** *The Lie algebra $\tilde{\mathfrak{g}}$ with roots and coroots as above, is a Kac-Moody algebra.*

Proof. We have constructed α_0 and $\alpha_0^\vee$ in such a way that the relations of Theorem 39 hold.

We shall check that $\tilde{\mathfrak{g}}$ has no ideals meeting $\tilde{\mathfrak{h}}$ trivially. If it had such an ideal $\mathfrak{a}$ then we would have

$$\mathfrak{a} \cap \tilde{\mathfrak{g}}_\beta \neq 0$$

for $\beta \in Q$, i.e., $xt^n \in \mathfrak{a}$ for some vector $x \in \mathfrak{g}$ (lying either in $\mathfrak{h} = \mathfrak{g}_0$ or $\mathfrak{g}_\alpha$ for some root α). Now choose $y \in \mathfrak{g}_{-\alpha}$ such that $(x, y) \neq 0$ and consider

$$[xt^n, yt^{-n}] = [x, y] + n(x, y)K.$$

In order to guarantee the K component vanishes, we must have $n = 0$. But then we have $\alpha \neq 0$, and so

$$[x, y] = (x, y)\nu^{-1}(\alpha) \neq 0$$

nonzero element of $\mathfrak{h}$. So the condition on ideals is satisfied.

It remains to check that our choice of $\tilde{\mathfrak{h}}, \Pi, \Pi^\vee$ is a realisation of the Cartan matrix. This is true, essentially by definition. □

Example: Let's determine the Cartan matrices of $\tilde{sl}_3$ and $\tilde{G}_2$. We obtain

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}.$$

2.11 Affine Weyl group

We work out the bilinear form on $\tilde{\mathfrak{h}}$ as per the general recipe, then transfer it to

$$\tilde{\mathfrak{h}}^* = \mathfrak{h}^* + \mathbb{C}\delta + \mathbb{C}\Lambda_0.$$

We find

$$(\mathfrak{h}, \mathbb{C}\delta + \mathbb{C}\Lambda_0) = 0, \qquad (\Lambda_0, \delta) = 1, \qquad |\Lambda_0| = |\delta| = 0.$$

Typically we write weights in $\tilde{\mathfrak{h}}^*$ as

$$\tilde{\lambda} = \lambda + n\delta + k\Lambda_0, \qquad \text{where } \lambda \in \mathfrak{h}^*.$$

The simple reflections $r_i = r_{\alpha_i}$ for $i = 1, \dots, n$ are easy to describe:

$$r_i(\tilde{\lambda}) = \tilde{\lambda} - 2\frac{(\lambda, \alpha_i)}{(\alpha_i, \alpha_i)}\alpha_i$$

leaves the δ and Λ_0 components of $\tilde{\lambda}$ unchanged, and acts on the component in $\mathfrak{h}$ as the usual finite reflection. On the other hand, recall

$$\alpha_0 = -\theta + \delta, \qquad \text{so } (\alpha_0, \alpha_0) = (\theta, \theta) = |\theta|^2,$$

and

$$r_0(\tilde{\lambda}) = \tilde{\lambda} - \frac{2}{|\theta|^2}\left[k - (\lambda, \theta)\right](-\theta + \delta).$$

Observe that the coefficient k of Λ_0 is fixed by r_0 and hence by ALL elements of the affine Weyl group. Thus we may imagine $\tilde{\mathfrak{h}}^*$ as a collection of "slices" of the form

$$k\Lambda_0 + (\mathfrak{h}^* + \mathbb{R}\delta)$$

each slice preserved by the action of the group.

Let us investigate the action of the group more closely now. We compute explicitly

$$r_0 r_\theta(\hat{\lambda}) = \hat{\lambda} + k\nu(\theta^\vee) - \left[< \lambda, \theta^\vee > + k|\theta^\vee|^2/2\right]\delta$$

Let us imitate this formula to get a general definition:

Definition 30. For any $\alpha \in \mathfrak{h}^*$ let

$$t_\alpha(\hat{\lambda}) = \hat{\lambda} + k\alpha - \left[(\lambda, \alpha) + k|\alpha|^2/2\right]\delta$$

The composition computed above is t_α for $\alpha = \nu(\theta^\vee)$.

Lemma 41. *For $w \in W$ the finite Weyl group, and for all $\alpha, \beta \in \mathfrak{h}^*$ we have*

$$t_{\alpha+\beta} = t_\alpha t_\beta,$$

and

$$t_{w(\alpha)} = w t_\alpha w^{-1}.$$

Proposition 42. *The Weyl group $\hat{W}$ of $\tilde{\mathfrak{g}}$ contains W as a subgroup and*

$$T = \{t_\alpha \mid \alpha \in M\}$$

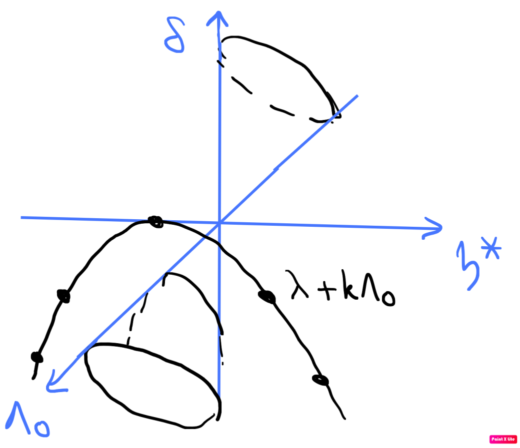
as a normal subgroup. Here $M \subset \mathfrak{h}^*$ is the lattice

$$M = \nu(\mathbb{Z}W\theta^\vee).$$

In other words

$$\hat{W} \cong W \ltimes T$$

Remark: if $\mathfrak{g}$ is simply laced (such as sl_n), then $M = Q$ the root lattice.



2.12 Integrable highest weight representations of affine Lie algebras

We will reserve the symbol P_+ for the set of dominant integral weights of $\mathfrak{g}$, spanned over $\mathbb{Z}_+$ by the basis

$$\{\varpi_1, \ldots, \varpi_n\} \subset \mathfrak{h}^*$$

of fundamental weights for $\mathfrak{g}$.

Now let $\tilde{\lambda} = \lambda + n\delta + k\Lambda_0$ as before. Since $\alpha_0^\vee = -\theta^\vee + K$, the condition of being dominant integral says

$$\lambda \in P_+, \qquad \text{and } k- < \lambda, \theta^\vee > \in \mathbb{Z}_+.$$

Well, let us define

$$\theta^\vee = \sum_{i=1}^{n} a_i^\vee \alpha_i^\vee.$$

Writing

$$\lambda = \sum_i m_i \varpi_i, \quad m_i \in \mathbb{Z}_+,$$

then the condition becomes

$$\sum_i a_i^\vee m_i \leq k.$$

Definition 31. The set P_+^k of dominant integral weights of level $k \in \mathbb{Z}_+$ is

$$P_+^k = \{\sum m_i \varpi_i + k\Lambda_0 \mid \sum a_i^\vee m_i \leq k\}.$$

(technically dominant integral weights are those of the form $\hat{\lambda} + n\delta$ for some $\hat{\lambda} \in P_+^k$ for some $k \in \mathbb{Z}_+$, and some $n \in \mathbb{C}$. But we tend to "normalise" by putting $n = 0$).

For example for $\tilde{sl}_2$ we have

$$\begin{aligned} P_+^0 &= \{0\} \\ P_+^1 &= \{\Lambda_0, \Lambda_0 + \varpi\} \\ P_+^2 &= \{2\Lambda_0, 2\Lambda_0 + \varpi, 2\Lambda_0 + 2\varpi\}, \end{aligned}$$

etc.

Let's use the Weyl character formula to compute the character of $L(\Lambda_0)$.

As we know in general

$$R\,\mathrm{Ch}\,L(\Lambda) = \sum_{w \in W} \epsilon(w) e^{w \circ \Lambda}.$$

For an affine Lie algebra we have

$$R = \prod_{n=1}^{\infty} \left[(1 - e^{-n\delta})^{rank} \prod_{\alpha \in \Delta_+} (1 - e^{\alpha - n\delta})(1 - e^{-\alpha-(n-1)\delta}) \right]$$

In this particular case let us put for convenience

$$q = e^{-\delta}, \qquad y = e^{\alpha/2},$$

$$R = \prod_{n=1}^{\infty} (1 - y^{-1} q^{n-1})(1 - q^n)(1 - yq^n)$$

We need a Weyl vector $\hat{\rho}$, some vector satisfying

$$< \rho, \alpha_i^\vee >= 1, \quad \text{for } i = 0, \ldots, n.$$

We take the vector

$$\hat{\rho} = c\Lambda_0 + \rho, \quad \rho = \sum_{i=1}^{n} \varpi_i$$

(which already satisfies the conditions for $i = 1, \ldots, n$) and determine c by

$$< c\Lambda_0 + \sum_i \varpi_i, K - \theta^\vee >= c - \sum a_i^\vee.$$

Definition 32. The dual Coxeter number of $\mathfrak{g}$ is the integer

$$h^\vee = 1 + \sum a_i^\vee$$

So

$$\hat{\rho} = h^\vee \Lambda_0 + \rho.$$

For example for sl_2 is happens that $h^\vee = 2$ (and for E_8 it happens that $h^\vee = 30$, etc.), so

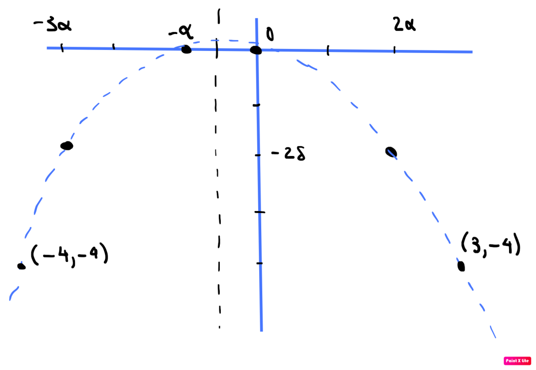
$$\hat{\rho} = 2\Lambda_0 + \varpi.$$

By hand we can compute a few of the translations $t_{m\alpha} \circ \Lambda_0$. We get

$$\ldots, \quad \Lambda_0 - 3\alpha - 2\delta \quad \Lambda_0, \quad \Lambda_0 + 3\alpha - 4\delta, \ldots$$

We have $r_1 \circ \Lambda_0 = \Lambda_0 - \alpha$, and a few of its translations are:

$$\ldots, \quad \Lambda_0 - 4\alpha - 4\delta \quad \Lambda_0 - \alpha, \quad \Lambda_0 + 2\alpha - 2\delta, \ldots$$



It is not hard to find an explicit formula for all these images. Finally we obtain

$$\mathrm{Ch}_{L(\Lambda_0)} = \frac{\sum_{n\in\mathbb{Z}} y^{3n} q^{3n^2+n} - \sum_{n\in\mathbb{Z}} y^{3n-1} q^{3n^2-n}}{\prod_{n=1}^{\infty}(1-y^{-1}q^{n-1})(1-q^n)(1-yq^n)}$$

Computing a few terms explicitly, we obtain

		1		
	1	1	1	
	1	2	1	
	2	3	2	
1	3	5	3	1
1	5	7	5	1
2	7	11	7	2

Interestingly, there is an alternative formula for this character:

$$\mathrm{Ch}_{L(\Lambda_0)} = \prod_{n=1}^{\infty}(1-q^n)^{-1}\cdot\sum_{m\in\mathbb{Z}} y^m q^{m^2}.$$

The identity between the two formulas was apparently known to Jacobi in 1820.

For now it is not really clear why (other than pure coincidence) the character should have an alternate description of this sort, notice that all the columns of numbers "look the same" (each corresponds to an individual summand $y^m q^{m^2}\prod_{n=1}^{\infty}(1-q^n)^{-1}$). After we become familiar with vertex algebras, an explanation will become clear.

Very importantly, the function $\mathrm{Ch}_{L(\Lambda_0)}$ is an example of a modular function.

2.13 Theta functions

If we set $y=1$ in the formula for $\mathrm{Ch}_{L(\Lambda_0)}(q,y)$ above we get

$$\begin{aligned}\mathrm{Ch}_{L(\Lambda_0)}(q) &= \prod_{n=1}^{\infty}(1-q^n)^{-1}\cdot\sum_{m\in\mathbb{Z}} q^{m^2}\\ &= 1+3q+4q^2+7q^3+13q^4+19q^5+29q^6+43q^7+62q^8+\cdots\end{aligned}$$

We could do all the same calculations again, this time for $L(\Lambda_0+\varpi)$, and this time we would obtain

$$\begin{aligned}\mathrm{Ch}_{L(\Lambda_0+\varpi)}(q) &= \prod_{n=1}^{\infty}(1-q^n)^{-1}\cdot\sum_{m\in\mathbb{Z}} q^{(m+1/2)^2-1/4}\\ &= 2+2q+6q^2+8q^3+14q^4+20q^5+34q^6+46q^7+70q^8+\cdots\end{aligned}$$

Functions like $\sum q^{m^2}$ are called theta functions. More precisely we denote

$$\mathcal{H} = \{\tau\in\mathbb{C} \mid Im\tau > 0\}$$

We also usually write $q=e^{2\pi i\tau}$, thus $\tau\in\mathcal{H}$ if and only if

$$|q|<1.$$

Definition 33. The theta function is defined as

$$\theta(\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau}.$$

By the comparison test this series is absolutely convergent in $\mathcal{H}$.

Consider in general the Fourier transform

$$\hat{g}(y) = \int_{-\infty}^{\infty} g(x)e^{2\pi i x y}\,dx.$$

and

Proposition 43 (Poisson summation formula)**.**

$$\sum_{n\in\mathbb{Z}} g(n) = \sum_{n\in\mathbb{Z}} \hat{g}(n).$$

Now let

$$g(x,t) = e^{-\pi t x^2}$$

so that

$$\theta(it) = \sum_{n\in\mathbb{Z}} g(n,t).$$

In this case

$$\hat{g}(y) = \sqrt{t} e^{-\pi y^2/t},$$

from which it follows that

$$\theta(-1/\tau) = \sqrt{\tau/i}\ \theta(\tau).$$

We call such a transformation rule "modular invariance".

Theorem 44. *The Dedekind eta function*

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1-q^n)$$

is modular invariant in the sense that

$$\eta(-1/\tau) = \sqrt{\tau/i}\ \eta(\tau).$$

Examining carefully the character of $L(\Lambda_0)$, we see that it reduces to

$$\chi_0(\tau) = q^{-1/24} \operatorname{Ch}_{L(\Lambda_0)} = \frac{\theta(2\tau)}{\eta(\tau)}$$

Two observations:

- firstly we have to normalise the character, multiplying by a specific power of q,
- even then, we do not get modular invariance, because 2τ is going to turn into $\tau/2$.

But now the other module saves the day:

$$\chi_1(\tau) = q^{1/4-1/24} \operatorname{Ch}_{L(\Lambda_0)} = \frac{1}{\eta(\tau)} \cdot \sum_{m\in\mathbb{Z}} q^{(m+1/2)^2},$$

is such that the vector space

$$\mathbb{C}\chi_0 + \mathbb{C}\chi_1$$

is modular invariant. In fact we can calculate that

$$\chi_i(-1/\tau) = \sum_{j=0}^{1} S_{ij}\chi_j(\tau),$$

where $\{S_{ij}\}$ is the matrix

$$S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Since

$$-\frac{1}{-1/\tau} = \tau,$$

we should expect $S^2 = I$, and indeed this is true.

PROJECT: The set of normalised characters

$$\chi_\lambda(\tau) = (\text{factor})\, \mathrm{Ch}_{L(\lambda+k\Lambda_0)}(q)$$

spans a finite dimensional modular invariant vector space. The S-matrix is known explicitly. It is related to the tensor product of modules by something called the Verlinde formula. Investigate the details and tell this story.

2.14 Peterson formula

D. Peterson found an algorithm to compute the multiplicities of root spaces of a symmetrisable Kac-Moody algebra. His proof uses the Weyl-Kac character formula. Here is the argorithm. For $\beta \in Q_+$ we define a number c_β recursively as follows: $c_\beta = 1$ for $\beta \in \Pi$ a simple root, and in general

$$c_\beta = \frac{1}{(\beta, \beta - 2\rho)} \sum_{\gamma_1+\gamma_2=\beta} (\gamma_1, \gamma_2) c_{\gamma_1} c_{\gamma_2}$$

where the sum is over $\gamma_1, \gamma_2 \in Q_+$. Then the multiplicities are recovered from $\{c_\beta\}$ by the formula

$$c_\beta = \sum_{n\in Z_{\geq 1}} \frac{\mathrm{mult}(\beta/n)}{n}.$$

PROJECT: In the programming language of your choice (but I suggest `SAGE`), implement Peterson's algorithm and compute root space multiplicities for a few Kac-Moody algebras.

2023-01-26. Lecture 09

3 Vertex Algebras

Main references: Kac-Raina-Rozhkovskaya [KRR] Chapters 13-18, Kac [KVA]

3.1 Quantum Fields / Heisenberg vertex algebra

As a starting point, I wish to discuss two Lie algebras, and how the relation between them illustrates the intuition for vertex algebras.

(1) The first is a sort of baby affine Kac-Moody algebra, known as the oscillator Lie algebra:

$$\hat{\mathfrak{a}} = \mathbb{C}K \oplus \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n,$$

with Lie brackets

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0,$$

i.e., K is central.

(2) The second is the Witt Lie algebra

$$W = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}D_n$$

with

$$[D_m, D_n] = (m-n)D_{m+n}.$$

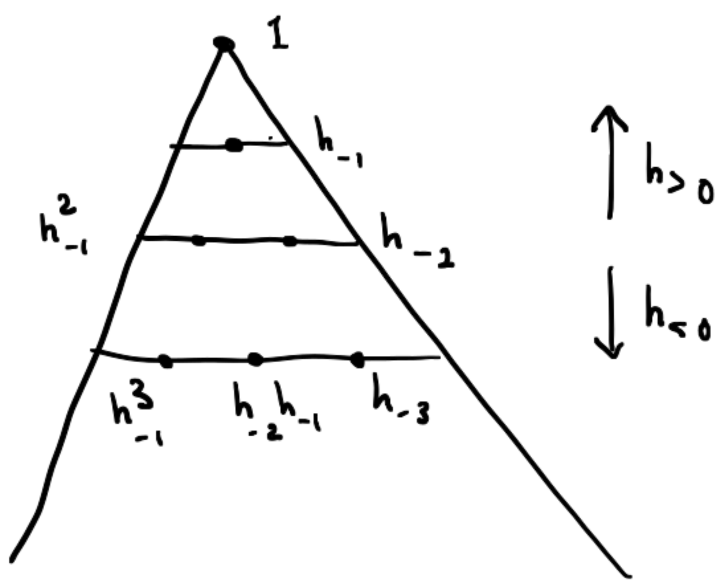
The oscillator Lie algebra has a well-known representation, which we shall call Fock space. Explicitly,

$$H = \mathbb{C}[x_{-1}, x_{-2}, x_{-3}, \ldots],$$

the space of polynomials. For $n > 0$ the action of h_{-n} is multiplication by x_{-n}, and the action of h_n is

$$h_n = n\frac{\partial}{\partial x_{-n}}.$$

Finally K acts by I_H. And $h_0 = 0$.



Exercise 45. *Check that this defines a representation of $\hat{\mathfrak{a}}$.*

Actually the condition $h_0 = 0$ can be relaxed to $h_0 = \mu I_H$ and we obtain a family of representations which we denote H^μ, namely

$$H^\mu = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}\left|\mu\right\rangle,$$

where $\hat{\mathfrak{a}}_+ \subset \hat{\mathfrak{a}}$ is the subalgebra $\mathbb{C}K \oplus \bigoplus_{n\geq 0} \mathbb{C}h_n$, and $\mathbb{C}\left|\mu\right\rangle$ is the $\hat{\mathfrak{a}}_+$-module in which $h_0 = \mu I_{H^\mu}$ and $h_{>0} = 0$. This is a lot like a Verma module of a Kac-Moody algebra.

More generally let $\mathfrak{h}$ be a finite dimensional vector space, with a symmetric bilinear form $(\cdot,\cdot)$, and define

$$\hat{\mathfrak{h}} = \mathbb{C}K \oplus \mathfrak{h} \otimes \mathbb{C}[t, t^{-1}]$$

with Lie brackets

$$[at^m, bt^n] = m\delta_{m,-n}(a,b)K, \qquad [K, \hat{\mathfrak{h}}] = 0,$$

This is a Lie algebra, and for any $\mu \in \mathfrak{h}$ we can define a representation

$$H(\mathfrak{h})^\mu = U(\hat{\mathfrak{h}}) \otimes_{U(\mathbb{C}K + \mathfrak{h}[t])} \mathbb{C}\left|\mu\right\rangle,$$

where $h_0\left|\mu\right\rangle = (\mu, h)\left|\mu\right\rangle$ and $h_{>0}\left|\mu\right\rangle = 0$.

Remark 46.

(1) The Fock space H should be thought of as a quantum mechanical state space. *The operators $h_{<0}$ are creation operators, and the $h_{>0}$ are annihilation operators.*

(2) Vertex algebras can be motivated as a way of systematising calculations that physicists perform in quantum field theory.

OK now let's do something interesting. Introduce some new operators acting on H:

$$L_m = \frac{1}{2}\sum_{k\in\mathbb{Z}} h_k h_{m-k}.$$

Why might we be interested in doing this?

- The Lie algebra $\mathfrak{a}$ can be given an invariant bilinear form:

  $$(xt^m, yt^n) = (x,y)\delta_{m,-n}.$$

  The corresponding Casimir operator on H is essentially L_0.

- These infinite sums arise in calculations in quantum field theory.

For $m \neq 0$ the operator L_m is well-defined. Indeed for any $v \in H$ there exists $N > 0$ such that $h_n v = 0$ for all $n > N$. Since h_k and h_{m-k} commute,

$$h_k h_{m-k} v = h_{m-k} h_k v$$

and in all but finitely many terms either $k > N$ or else $m - k > N$. So $L_m v$ is a finite sum.

Exercise 47. *If $m, n, m+n \neq 0$ then*

$$[L_m, L_n] = (m-n)L_{m+n}.$$

So we get something approaching a representation of W in the space H, by sending $W_n \mapsto L_n$.

The problem with L_0 is that the operators h_n and h_{-n} do not commute and so we get

$$L_0 \mathbf{1} = \sum_{n>0} n \frac{\partial}{\partial x_{-n}} x_{-n} = 1 + 2 + 3 + 4 + \cdots .$$

A trick, which is called normal ordering because of physics, is to simply ignore the fact that h_n and h_{-n} do not commute, and see what happens. In other words we redefine

$$L_0 = \frac{1}{2} \sum_{k \in \mathbb{Z}} : h_k h_{-k} :$$

where

$$: h_k h_{-k} := \begin{cases} h_k h_{-k} & \text{if } k < 0 \\ h_{-k} h_k & \text{if } k \geq 0 \end{cases}$$

Remark 48. *This is precisely the kind of thing we did in the case of the Casimir operator of a Kac-Moody algebra.*

This suggests we gather our operators into generating functions[2] like:

$$h(z) = \sum_{n \in \mathbb{Z}} h_n z^{-n-1}$$

Definition 34.

- Let V be a vector space. A formal distribution on V is a series

  $$a(z) = \sum a_{(n)} z^{-n-1}$$

  where $a_{(n)} \in V$. The vector space of all of them is denoted $V[[z, z^{-1}]]$.

- A *quantum field* on V is a formal distribution $a(z)$ on $\mathrm{End}(V)$ such that for any $v \in V$ one has $a_{(n)} v = 0$ for all $n > N$ for some N (which is allowed to depend on v).

[2] There are physical reasons for doing this too, but it's quite complicated.

Example 1. $h(z)$ is a quantum field on H.

Now

Definition 35. Let $a(z)$, $b(z)$ be quantum fields. Their normally ordered product is

$$: a(z)b(z) := a(z)_+ b(z) + b(z)a(z)_-,$$

where

$$a(z)_+ = \sum_{n<0} a_{(n)} z^{-n-1}, \qquad a(z)_- = \sum_{n\geq 0} a_{(n)} z^{-n-1}.$$

Proposition 49. *If $a(z)$ and $b(z)$ are quantum fields then $: a(z)b(z) :$ is a well-defined quantum field.*

Proof. We shall show that, for each fixed $v \in V$, each coefficient of each part

$$a(z)_+ b(z) \quad \text{and} \quad b(z)a(z)_-$$

is a finite sum. Let us examine the z^{-N-1} coefficient of

$$a(z)_+ b(z)v = \sum_{m<0} a_{(m)} z^{-m-1} \sum_n b_{(n)} z^{-n-1} v = \sum_{m<0} \sum_n a_{(m)} b_{(n)} v z^{-(m+n+1)-1}.$$

It is

$$\sum_{m<0} \sum_n a_{(m)} b_{(N-m-1)} v. \tag{7}$$

Having fixed N here, and letting $m \to -\infty$, the subscript on $b_{(-)}$ goes to $+\infty$. But beyond some fixed point, all subsequent terms vanish. So the sum is finite.

The treatment of the other product is similar, even more straightforward in fact. The coefficient of z^N is now

$$\sum_{m\geq 0} \sum_n b_{(N-m-1)} a_{(m)} v, \tag{8}$$

which is finite since $a_{(m)} v = 0$ for sufficiently large m.

It remains to show that, again for fixed $v \in V$, both sums above vanish for sufficiently large positive N. This is clear in the case of (7). In the case of (8) we first look at the (finite) set of nonzero $a_{(m)} v$, $m \geq 0$, then among them take the supremum of the M such that $b_{(M)} a_{(m)} v$, and use that value for N. $\square$

The first key idea of vertex algebras is that we want to work with quantum fields as our basic objects, multiplying them, etc. The second key idea of vertex algebras is to study commutation relations by introducing a second formal variable:

$$\begin{aligned} h(z)h(w) &= \sum_{m,n} h_m h_n z^{-m-1} w^{-n-1} \\ &= \sum_{m<0,n} h_m h_n z^{-m-1} w^{-n-1} + \sum_{m\geq 0,n} (h_n h_m + m\delta_{m,-n}) z^{-m-1} w^{-n-1} \\ &=: h(z)h(w) : + \sum_{m\geq 0} m z^{-m-1} w^{m-1} \\ &=: h(z)h(w) : + \iota_{z,w} \frac{1}{(z-w)^2}. \end{aligned} \tag{9}$$

Here the symbol $\iota_{z,w}$ denotes Taylor series expansion in positive powers of w. Similarly

$$h(w)h(z) =: h(z)h(w) : + \iota_{w,z} \frac{1}{(z-w)^2}. \tag{10}$$

Thus we may subtract the two expressions and obtain

$$[h(w), h(z)] = \iota_{z,w} \frac{1}{(z-w)^2} - \iota_{w,z} \frac{1}{(z-w)^2}. \tag{11}$$

Explicitly all this is saying is that

$$[h(w), h(z)] = \sum_{m\in\mathbb{Z}} m z^{-m-1} w^{m-1}$$

which is straightforward enough. But analytically this is senseless of course, so we had better be careful with this "series expansion" notation.

Some terminology.

- $\mathbb{C}[z]$ the ring of polynomials in one variable,
- $\mathbb{C}[[z]]$ the ring of Taylor series in one variable,
- we can allow negative powers of z and get $\mathbb{C}[z, z^{-1}]$ the ring of Laurent polynomials in one variable, but if we relax too much we get...
- $\mathbb{C}[[z, z^{-1}]]$ the *vector space* of formal distributions in z, note that this is not a ring – the obvious multiplication is not well-defined! Though if we consider those vectors with a limited (i.e., finite) number of negative powers of z then we get...
- $\mathbb{C}((z))$ the field of Laurent series in one variable. Yes, it's a field! More generally if $\mathbb{F}$ is a field then $\mathbb{F}((z))$ is again a field, so we have...
- $\mathbb{C}((z))((w))$ and $\mathbb{C}((w))((z))$ two fields. Of course they are isomorphic in an obvious way, but on the other hand they both embed into...

- $\mathbb{C}[[z^{\pm 1}, w^{\pm 1}]]$ the vector space of formal distributions in z and w, and their images in there are distinct.
- We introduce the linear maps $i_{z,w}$ and $i_{w,z}$ as being maps from $\mathbb{C}((z,w))$ the field of fractions of the ring (which is an integral domain) $\mathbb{C}[[z,w]]$, to the fields $\mathbb{C}((z))((w))$ and $\mathbb{C}((w))((z))$.

The point is, the diagram

$$\begin{tikzcd} \mathbb{C}((z,w)) \arrow[r, "i_{z,w}"] \arrow[d, "i_{w,z}"'] & \mathbb{C}((z))((w)) \arrow[d] \\ \mathbb{C}((w))((z)) \arrow[r] & \mathbb{C}[[z^{\pm 1}, w^{\pm 1}]] \end{tikzcd}$$

actually does NOT commute!

Why do we call $\mathbb{C}[[z,z^{-1}]]$ the space of formal distributions? Consider $\mathbb{C}[z,z^{-1}]$ as a ring of functions on some space (it's the ring of algebraic functions on $\mathbb{C}\backslash\{0\}$ in fact). The functions z^n form a basis indexed by $\mathbb{Z}$, and the vector space dual of $\mathbb{C}[z,z^{-1}]$ can be identified with $\mathbb{C}[[z,z^{-1}]]$. A suggestive way to do it is to identify the vector ε_n dual to the basis vector z^n as $\varepsilon_n \leftrightarrow z^{-n-1}$, so the pairing is

$$< \sum a_n z^{-n-1}, \sum f_n z^n >= \sum_n a_n f_n.$$

Here the sum in the left hand argument is infinite, while in the right hand argument it is finite. It is customary to write this pairing $<,>$ instead as

$$\int a(z)f(z)dz, \qquad \operatorname{Res}_z a(z)f(z).$$

Here, and in general:

Definition 36. The residue map

$$\operatorname{Res}_z : U[[z,z^{-1}]] \to U$$

takes a formal distribution to its z^{-1} coefficient.

Depending on your point of view, the fact that distributions cannot be multiplied is the essential obstruction to making sense of quantum field theory.

2023-01-30. Lecture 10

Definition 37. The formal Dirac delta function is by definition

$$\delta(z,w) = \sum_{n\in\mathbb{Z}} z^{-m-1}w^m = \iota_{z,w}\frac{1}{z-w} - \iota_{w,z}\frac{1}{z-w}.$$

So our calculation above reveals that

$$[h(w), h(z)] = \partial_w \delta(z, w). \tag{12}$$

Properties of the delta function.

Lemma 50. *The delta function satisfies*

$$(z - w)\delta(z, w) = 0. \tag{13}$$

Let us write ∂_w for $\partial/\partial w$, etc., and $\partial^{(j)} = \frac{1}{j!}\partial^j$. Then

$$i_{z,w}\frac{1}{(z-w)^{j+1}} - i_{w,z}\frac{1}{(z-w)^{j+1}} = \partial_w^{(j)}\delta(z, w)$$

is annihilated by $(z - w)^{j+1}$, recovering the prior statement as the case $j = 0$. Finally

$$(z - w)^m \partial_w^{(j)}\delta(z, w) = \partial_w^{(j-m)}\delta(z, w) \tag{14}$$

whenever $m \leq j$.

Proof. The first claim follows either by direct calculation, or by observing that

$$(z - w)\delta(z, w) = (z - w)\iota_{z,w}\frac{1}{z-w} - (z - w)\iota_{w,z}\frac{1}{z-w} = i_{z,w}1 - i_{w,z}1 = 1 - 1 = 0.$$

The equality in the next formula is easy to check, and the vanishing follows as in the first claim. The final claim follows from the equaity in the second claim. □

Properties of the residue:

Lemma 51. *The residue operation satisfies*

$$\operatorname{Res}_z f(z)\delta(z, w) = f(w) \tag{15}$$

and

$$\operatorname{Res}_z f(z)\partial_z g(z) = -\operatorname{Res}_z(\partial_z f(z))g(z)$$

Proof. The first claim is easy to check directly. The second claim follows from the Leibniz relation

$$\partial_z(f(z)g(z)) = \partial_z(f(z))g(z) + f(z)\partial_z(g(z)),$$

and the fact that

$$\operatorname{Res}_z \circ\, \partial_z = 0.$$

□

So consequently, in our oscillator algebra example, we have

$$(z-w)^2[h(z), h(w)] = 0.$$

Now we come to:

Definition 38. Let $a(z)$ and $b(z)$ be two quantum fields on a vector space V. We say they form a *local pair* if

$$(z-w)^N[a(z), b(w)] = 0, \qquad \text{for some } N \geq 0. \tag{16}$$

More generally, we say that a formal distribution $D \in U[[z^{\pm 1}, w^{\pm 1}]]$ on a vector space U is local if

$$(z-w)^N D = 0$$

for some $N \geq 0$.

Remark 52. *It is easy to see that locality of quantum fields is a symmetric relation. But it is neither reflexive nor transitive.*

The connection with (9) appears in the following result.

Proposition 53. *Let $a(z)$ and $b(z)$ be a local pair as in Definition 38. Then*

$$\begin{aligned} a(z)b(w) &= :a(z)b(w): + \sum_{j=0}^{N-1} c_j(w) i_{z,w} \frac{1}{(z-w)^{j+1}}, \\ \textit{and} \quad b(w)a(z) &= :a(z)b(w): + \sum_{j=0}^{N-1} c_j(w) i_{w,z} \frac{1}{(z-w)^{j+1}}, \end{aligned} \tag{17}$$

for some finite set of quantum fields $c_j(w)$. The coefficients $c_j(w)$ in the expansion (17) *are uniquely determined by $a(z)$ and $b(z)$ and are given by*

$$c_j(w) = \mathrm{Res}_z (z-w)^j [a(z), b(w)]. \tag{18}$$

Formula (18) motivates another definition (or notation, really)

Definition 39. Let $a(w)$ and $b(w)$ be two quantum fields, then for $j \in \mathbb{Z}_+$ we denote by $a(w)_{(j)} b(w)$ the quantum field

$$\mathrm{Res}_z (z-w)^j [a(z), b(w)].$$

Example 2. In our space H we have $h(w)_{(0)} h(w) = 0$ and $h(w)_{(1)} h(w) = \mathbf{1}(w)$ where by definition $\mathbf{1}(w) = I_H$.

Proof. Let V be a vector space and

$$D(z,w) = \sum_{i,j} D_{i,j} z^i w^j \in U = V[[z^{\pm 1}, w^{\pm 1}]],$$

be a formal distribution in two variables. We will show that

$$(z-w)^N D(z,w) = 0 \tag{19}$$

is equivalent to

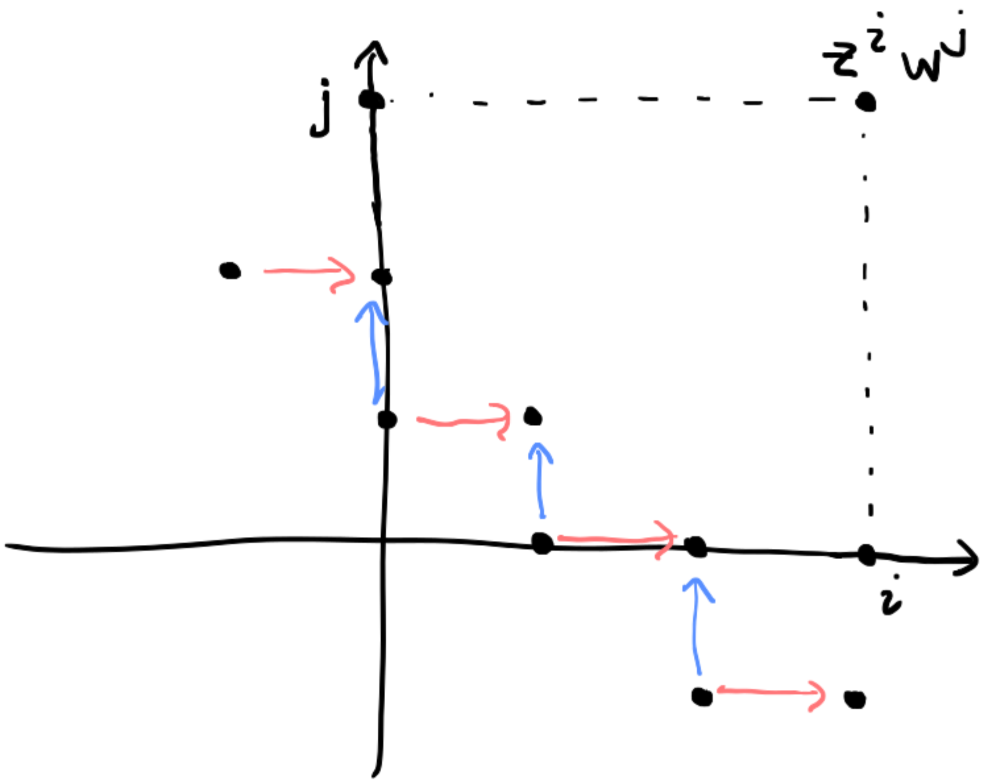
$$D(z,w) = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z,w) \tag{20}$$

for some collection of formal distributions $c_j(w) \in V[[z, z^{-1}]]$.

One direction follows immediately from the properties of the Dirac delta function discussed above.

To prove the other direction we proceed by induction. Assume first that

$$(z-w)D(z,w) = 0.$$



We easily see that $D_{i,j} = D_{i+1,j-1}$, i.e., the coefficients of $D(z,w)$ are constant along diagonals, and we have $D(z,w) = c(w)\delta(z,w)$ for some formal distribution $c(w) \in V[[w,w^{-1}]]$. As for the formula for $c(w)$, we have

$$\mathrm{Res}_z\, D(z,w) = \mathrm{Res}_z\, c(w)\delta(z,w) = c(w).$$

Now suppose that the implication from (19) to (20) has been established and assume

$$(z-w)^{N+1} D(z,w) = 0.$$

Then since $(z-w)(z-w)^N D(z,w) = 0$ we have

$$(z-w)^N D(z,w) = c_N(w)\delta(z,w), \qquad \text{and} \qquad c_N(w) = \mathrm{Res}_z (z-w)^N D(z,w).$$

Now put

$$D'(z,w) = D(z,w) - c_N(w)\partial_w^{(N)}\delta(z,w)$$

and note that

$$\begin{aligned}(z-w)^N D'(z,w) &= (z-w)^N D(z,w) - c_N(w)(z-w)^N \partial_w^{(N)}\delta(z,w) \\ &= c_N(w)\delta(z,w) - c_N(w)\delta(z,w) \\ &= 0.\end{aligned}$$

So we can write $D'(z,w)$ in the form (20) and consequently

$$D(z,w) = \sum_{j=0}^{N} c_j(w)\partial_w^{(j)}\delta(z,w),$$

where

$$c_j(w) = \mathrm{Res}_z (z-w)^j D(z,w).$$

The uniqueness of the $c_j(w)$ is clear since we have an explicit formula.

To connect these considerations with normal ordering and prove (17) we consider

$$D(z,w)_+ = \sum_{m \geq 0} D_{m,n} z^m w^n, \qquad D(z,w)_- = \sum_{m<0} D_{m,n} z^m w^n$$

for a formal distribution in general. It is easy to see that

$$\partial_w^{(j)}\delta(z,w)_- = i_{z,w}\frac{1}{(z-w)^{j+1}} \quad \text{and} \quad \partial_w^{(j)}\delta(z,w)_+ = -i_{w,z}\frac{1}{(z-w)^{j+1}}.$$

So if we apply the operation $(-)_-$ to (20) we get

$$D(z,w)_- = \sum_{j=0}^{N-1} c_j(w) i_{z,w}\frac{1}{(z-w)^{j+1}}.$$

In particular if $a(z)$ and $b(z)$ form a local pair and we take $D(z,w) = [a(z), b(w)]$ then we obtain

$$[a(z)_-, b(w)] = \sum_{j=0}^{N} c_j(w) i_{z,w} \frac{1}{(z-w)^{j+1}}.$$

But

$$a(z)b(w) = [a(z)_-, b(w)] + :a(z)b(w):,$$

from which we deduce the first relation of (17). The other is deduced in the same way.

□

For arbitrary $n \in \mathbb{Z}$, let us define

$$a(w)_{(n)}b(w) = \mathrm{Res}_z \left(a(z)b(w) i_{z,w}(z-w)^n - b(w)a(z) i_{w,z}(z-w)^n \right).$$

Exercise 54. *Verify that*

$$a(w)_{(-1)}b(w) = a(w)_+ b(w) + b(w)a(w)_- =: a(w)b(w): .$$

(So the definition of $a(w)_{(n)}b(w)$ unifies the OPE and the normally ordered product.)

Verify that

$$a(w)_{(-n-1)}b(w) =: [\partial_w^{(n)} a(w)]b(w):$$

for all $n \geq 0$.

In Proposition 53 we claimed that the $c_j(w)$ are quantum fields, but we did not prove it yet.

EXERCISE: Prove that $a(w)_{(n)}b(w)$ is a quantum field. You can consider separately the terms

$$\mathrm{Res}_z\, a(z)b(w) i_{z,w}(z-w)^n$$

(which is pretty easy, and in which the residue is irrelevant) and

$$\mathrm{Res}_z\, b(w)a(z) i_{w,z}(z-w)^n$$

(which is a little more complicated, and in which the residue plays a role).

3.2 Vertex algebras and state-field correspondence

Now we come to the main definition.

Definition 40. A *vertex algebra* is a vector space V, a vector $|0\rangle \in V$, an operator $T : V \to V$, and a set $\mathcal{F}$ of quantum fields on V such that:

- $T|0\rangle = 0$ and $[T, a(z)] = \partial_z a(z)$ for all $a(z) \in \mathcal{F}$,

- V is spanned by
$$a^1_{(n_1)}a^2_{(n_2)}\dots a^k_{(n_k)}\left|0\right\rangle$$
where $a^i(z) \in \mathcal{F}$ (if not just take the subspace of V spanned by these terms),
- all fields from $\mathcal{F}$ are mutually local.

We claim that
$$\begin{aligned} V &= H, \\ \left|0\right\rangle &= 1, \\ T &= L_{-1} \\ \mathcal{F} &= \{h(z)\}. \end{aligned}$$
is a vertex algebra.

We have already proved locality of the pair $h(z)$ with itself. The second axiom, of spanning, is clear from the construction of H. The operator $T = L_{-1}$ has come out of no where to some extent, so let us dwell on it a little.

Firstly, if we had not been given the formula $T = L_{-1}$, we could have constructed T from the properties it must satisfy:
$$[T, h(z)] = \partial_z h(z) \quad \Leftrightarrow \quad [T, h_{(n)}] = -nh_{(n-1)}.$$
Thus
$$Th_{(n)}v = -nh_{(n-1)}Tv, \quad \text{and} \quad T\left|0\right\rangle = 0$$
characterises the action of T on H uniquely. by induction on length of monomials. One might question whether this definition of T really is well-defined. In fact
$$T(x_{-n_1}\cdots x_{-n_s}) = -\sum_{j=1}^{s} n_j x_{-n_1}\cdots x_{-n_j-1}\cdots x_{-n_s}$$
We're done as far as checking the vertex algebra structure. But let's just confirm that $T = L_{-1}$. By definition
$$L_{-1} = \frac{1}{2}\sum_{n\in\mathbb{Z}} h_{-1-n}h_n = \sum_{n\geq 1} nx_{-n-1}\frac{\partial}{\partial x_n},$$
which clearly coincides with T.

EXERCISE: Let V be a representation of a Lie algebra $\mathfrak{g}$. Consider the general problem of constructing $T : V \to V$ compatible with the action of a derivation d of $\mathfrak{g}$. What can be said?

2023-01-31. Lecture 11

Swap notation for $\overline{\mathcal{F}}$ and $\mathcal{F}'$. Usually A' denotes a commutant

Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra and let us define

$\tilde{\mathcal{F}} = \{\text{quantum fields } a(z) \text{ on } V \mid [T, a(z)] = \partial_z a(z)\}$

$\overline{\mathcal{F}} = \{a(z) \in \tilde{\mathcal{F}} \mid a(z) \text{ forms a local pair with every } b(z) \in \mathcal{F}\}$

$\mathcal{F}' =$ "minimal subspace of $\tilde{\mathcal{F}}$ containing I_V and $\mathcal{F}$ and closed under ∂_z and all n^{th} products".

Lemma 55 (Dong's Lemma)**.** *If $a(z)$, $b(z)$ and $c(z)$ are mutually local, then so are $a(z)_{(n)}b(z)$ and $c(z)$ for all $n \in \mathbb{Z}$.*

Proof. By hypothesis we have

$$
\begin{aligned}
(z-w)^r[a(z), b(w)] &= 0\\
(x-w)^r[b(w), c(x)] &= 0\\
(x-z)^r[a(z), c(x)] &= 0
\end{aligned}
$$

for some $r \geq 0$. Let us assume $r > |n|$ too.

By definition

$$a(w)_{(n)}b(w) = \mathrm{Res}_z \left[a(z)b(w)i_{z,w} - b(w)a(z)i_{w,z}\right](z-w)^n$$

we wish to show that

$$(x-w)^N[c(x), a(w)_{(n)}b(w)] = 0$$

for some $N \geq 0$. We will prove an apparently stronger result, that the vanishing holds even without the residue.

Schematically we are trying to prove vanishing of

$$c(x)a(z)b(w) - c(x)b(w)a(z) - a(z)b(w)c(x) + b(w)a(z)c(x)$$

after multiplying by

$$(x-w)^N = (x-w)^r \sum_s \binom{N-r}{s}(x-z)^{N-r-s}(z-w)^s,$$

(binomial expansion). We consider in turn each of these summands (indexed by s) multiplied by the combination of four terms above. There are two cases.

We shall choose $N > 4r$.

The first case is that $s \geq 2r$. In this case the total power of $z-w$ is at least r, this is positive so the $i_{z,w}$ and $i_{w,z}$ can be removed, and furthermore by locality of a and b the order of $a(z)$ and $b(w)$ can be exchanged anywhere. There is thus cancellation between terms (1) and (2) and between (3) and (4).

The second case is that $s < 2r$, in which case $N - r - s > r$. In each such term we can exchange a and c and we can exchange b and c. Thus we obtain cancellation between terms (1) and (3) and between terms (2) and (4).

□

We have

$$\mathcal{F} \subset \mathcal{F}' \subset \overline{\mathcal{F}} \subset \tilde{\mathcal{F}}.$$

The first and last containment follow from the definitions. The middle containment is a direct consequence of Dong's lemma. Actually, we should check that the definition of $\mathcal{F}'$ makes sense, i.e., that if $a(w), b(w) \in \tilde{\mathcal{F}}$ then $\partial_w a(w)$ and $a(w)_{(n)}b(w)$ are too. The first of these claims is immediate. The second is easy too

$$\begin{aligned}\partial_w \operatorname{Res}_z a(z)b(w)i_{z,w}(z-w)^n &= \operatorname{Res}_z a(z)(\partial_w b(w))i_{z,w}(z-w)^n + \operatorname{Res}_z a(z)b(w)i_{z,w}\partial_w(z-w)^n \\ &= \operatorname{Res}_z a(z)(\partial_w b(w))i_{z,w}(z-w)^n - \operatorname{Res}_z a(z)b(w)i_{z,w}\partial_z(z-w)^n \\ &= \operatorname{Res}_z a(z)(\partial_w b(w))i_{z,w}(z-w)^n - \operatorname{Res}_z(\partial_z a(z))b(w)i_{z,w}(z-w)^n\end{aligned}$$

and similarly for the other summand. Thus

$$\partial_w(a(w)_{(j)}b(w)) = \partial_w(a(w))_{(j)}b(w) + a(w)_{(j)}\partial_w b(w).$$

Therefore

$$[T, a(w)_{(n)}b(w)] = [T, a(w)]_{(n)}b(w) + a(w)_{(n)}[T, b(w)] = \cdots = \partial_w(a(w)_{(n)}b(w))$$

(note that if $a(z)$ is a quantum field then so is $[T, a(z)]$).

The space $\mathcal{F}'$ is thus, concretely, the span of all iterated nth products

$$(a^1)_{(n_1)}((a^2_{(n_2)}a^3)_{(n_3)}a^4)$$

of fields $a^i \in \mathcal{F} \cup \{I_V\}$ and their derivatives.

Theorem 56.

1. *For any $a(z) \in \tilde{\mathcal{F}}$ we have*

   $$a(z)\,|0\rangle = v + (Tv)z + \frac{1}{2}(T^2 v)z^2 + \cdots = e^{zT}v$$

   for some $v \in V$. Thus we define $s : \tilde{\mathcal{F}} \to V$ by

   $$s(a(z)) = v.$$

2. $s(a(z)_{(n)}b(z)) = a_{(n)}s(b)$.

3. *Let $\mathcal{G} \subset \tilde{\mathcal{F}}$ such that $s(\mathcal{G}) = V$ and suppose $a(z)$ is local with each $b(z) \in \mathcal{G}$. If $s(a(z)) = 0$ then $a(z) = 0$.*

Proof.

(1) Let

$$a(z)\,|0\rangle = v(z) = \sum_{n\in\mathbb{Z}} v_n z^n.$$

Since $\partial_z v(z) = Tv(z)$ we have

$$v_n = \frac{1}{n} T v_{n-1},$$

But since $v_n = 0$ for all $n < N$ for some $N \in \mathbb{Z}$, we have $v_n = 0$ for all $n < 0$ in fact. It also follows that $v(z) = e^{zT} v$.

(2) We consider

$$\begin{aligned} a(w)_{(n)} b(w) |0\rangle &= \mathrm{Res}_z \left(a(z) b(w) |0\rangle\, i_{z,w} (z-w)^n - b(w) a(z) |0\rangle\, i_{w,z} (z-w)^n \right) \\ &= \mathrm{Res}_z\, a(z) b(w) |0\rangle\, i_{z,w} (z-w)^n \end{aligned}$$

(the second term on the right hand side vanished because $a(z)|0\rangle \in V[[z]]$ and $i_{w,z}(z-w)^n$ both contain only positive powers of z).

By definition then

$$\begin{aligned} s(a(w)_{(n)} b(w)) &= \mathrm{Res}_z\, a(z) b(w) |0\rangle\, i_{z,w} (z-w)^n |_{w=0} \\ &= \mathrm{Res}_z\, z^n a(z) b(w) |0\rangle\, |_{w=0} \\ &= \mathrm{Res}_z\, z^n a(z) s(b(w)) \\ &= a_{(n)} s(b(w)). \end{aligned}$$

(3) By locality we have

$$\begin{aligned} (z-w)^N a(z) b(w) |0\rangle &= (z-w)^N b(w) a(z) |0\rangle \\ &= (z-w)^N b(w) e^{zT} s(a(z)) \\ &= 0. \end{aligned}$$

Put $w = 0$ to obtain

$$z^N a(z) s(b(w)) = 0.$$

Since this holds for all $b(z) \in \mathcal{G}$, we have $a(z)V = 0$, i.e., $a(z) = 0$. □

Corollary 57.

1. *The restriction $s|_{\mathcal{F}'}$ is surjective.*
2. *The restriction $s|_{\overline{\mathcal{F}}}$ is injective.*
3. *The restriction $s|_{\overline{\mathcal{F}}} : \overline{\mathcal{F}} \to V$ is an isomorphism. We denote the inverse map by*

$$a \mapsto Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}.$$

This is the state-field correspondence. *Now $(V, |0\rangle, T, \overline{\mathcal{F}})$ is a vertex algebra, and*

$$Y(a_{(n)} b, z) = Y(a, z)_{(n)} Y(b, z)$$

for all $a, b \in V$ and all $n \in \mathbb{Z}$.

Proof. By iterated application of

$$a_{(n)}s(b) = s(a(z)_{(n)}b(z))$$

we obtain

$$a^1_{(n_1)}a^2_{(n_2)}s(b) = a^1_{(n_1)}s(a^2(z)_{(n_2)}b(z)) = s(a^1(z)_{(n_1)}a^2(z)_{(n_2)}b)$$

and, in general, that all the monomials

$$a^1_{(n_1)}a^2_{(n_2)}\dots a^k_{(n_k)}\left|0\right\rangle$$

are in the image of $s|_{\mathcal{F}'}$. But by definition these span V, so indeed the map is surjective.

If $s|_{\overline{\mathcal{F}}}$ were not injective we would have $b(z) \in \overline{\mathcal{F}}$ with $s(b(z)) = 0$. Dong's lemma implies that $b(z)$ is local with $\mathcal{F}'$, so applying Goddard's lemma with $\mathcal{G} = \mathcal{F}'$ yields $b(z) = 0$, proving injectivity.

The two previous parts together imply that $\mathcal{F}' = \overline{\mathcal{F}}$, and $s|_{\overline{\mathcal{F}}}$ is an isomorphism. So we write Y for the inverse as in the theorem statement.

Now let's apply part (2) of Theorem 56 to $a(z) = Y(a,z)$ and $b(z) = Y(b,z)$. Apply Y to both sides of

$$s(Y(a,z)_{(n)}Y(b,z)) = a_{(n)}b$$

to obtain

$$Y(a_{(n)}b,z) = Y(a,z)_{(n)}Y(b,z).$$

□

Proposition 58 (Commutator formula)**.** *Let $(V, \left|0\right\rangle, T, \mathcal{F})$ be a vertex algebra, and $a, b \in V$. Then*

$$[Y(a,z), Y(b,w)] = \sum_{j\in\mathbb{Z}_+} Y(a_{(j)}b,w)\partial_w^{(j)}\delta(z,w).$$

Proof. Since the fields are mutually local, we have

$$[Y(a,z), Y(b,w)] = \sum_{j\in\mathbb{Z}_+} c_j(w)\partial_w^{(j)}\delta(z,w)$$

where

$$c_j(w) = \mathrm{Res}_z(z-w)^j[Y(a,z),Y(b,w)] = Y(a,w)_{(j)}Y(b,w).$$

But by the corollary above, this coincides with

$$Y(a_{(j)}b,w)$$

so we are done. □

Just for fun, let us expand out the commutator formula. We apply $\operatorname{Res}_z \operatorname{Res}_w z^m w^n$ to both sides to obtain

$$a_{(m)}b_{(n)}c - b_{(n)}a_{(m)}c = \sum_{j\in\mathbb{Z}_+} \binom{m}{j} (a_{(j)}b)_{(m+n-j)}c.$$

This holds for all $a, b, c \in v$ and all $m, n \in \mathbb{Z}$. Suppose we take the simplest possible special case: $m = n = 0$. Then

$$a_{(0)}b_{(0)}c - b_{(0)}a_{(0)}c = (a_{(0)}b)_{(0)}c.$$

This is just the Jacobi identity for the operation

$$V \times V \to V, \quad (a, b) \mapsto a_{(0)}b.$$

We do not get a Lie algebra as the skew-symmetry does not hold. But something close to this does hold. We will discuss this later.

Exercise 59. *Prove the Borcherds identity*

$$(Y(a,z)Y(b,w)i_{z,w} - Y(b,w)Y(a,z)i_{w,z})(z-w)^n = \sum_{j\in\mathbb{Z}_+} Y(a_{(n+j)}b, w)\partial_w^{(j)}\delta(z,w). \tag{21}$$

Hint: The left hand side is local (why?), and so has an expansion involving derivatives of the delta function as we saw in the first lecture. Now use the state-field correspondence.

2023-02-02. Lecture 12

3.3 Calculating in vertex algebras

We begin with notation:

Definition 41. Let V be a vertex algebra and $a, b \in V$. The λ-bracket (or *operator product expansion* or *OPE*) of a and b is:

$$[a_\lambda b] = \sum_{j\in\mathbb{Z}_+} \lambda^{(j)} a_{(j)}b = \operatorname{Res}_z e^{\lambda z} a(z)b.$$

The normally ordered product of a with b is

$$:ab := a_{(-1)}b.$$

For example in the Heisenberg vertex algebra H we have

$$[h_\lambda h] = \lambda \left|0\right\rangle.$$

In any vertex algebra V and for any $a \in v$ we have

$$[a_\lambda \left|0\right\rangle] = [\left|0\right\rangle_\lambda a] = 0.$$

We will tend to write $|0\rangle$ as 1 and omit it from products in the usual way, whenever this does not lead to confusion.

The following identities are easy to prove.

$$
\begin{aligned}
[(\partial a)_\lambda b] &= -\lambda[a_\lambda b] \\
[a_\lambda (\partial b)] &= (\partial + \lambda)[a_\lambda b] \\
\partial :ab: &= :(\partial a)b: + :a(\partial b): .
\end{aligned}
$$

EXERCISE: Prove them.

More difficult, but useful, identities are the following *skew-symmetry*

$$[b_\lambda a] = -[a_{-\lambda-\partial} b]$$

and *noncommutative Wick formula*:

$$[a_\lambda :bc:] = :[a_\lambda b]c: + :b[a_\lambda c]: + \int_0^\lambda [[a_\lambda b]_\mu c] d\mu,$$

valid for a, b and c mutually local fields.

Exercise 60. *Apply* $\operatorname{Res}_z e^{\lambda z}(-)$ *to both sides of the commutator formula (proved last time), obtaining*

$$[a_\lambda Y(b,w)c] = Y(b,w)[a_\lambda c] + e^{\lambda w} Y([a_\lambda b], w)c.$$

Deduce the noncommutative Wick formula.

OK now let's do some calculations. As we noted above, in the λ-bracket notation, we have

$$[h_\lambda h] = \lambda,$$

Let us consider

$$L = \frac{1}{2} :hh: .$$

so that

$$L(z) = \frac{1}{2} :h(z)h(z):$$

is the quantum field on H that started this whole adventure. We confirm (from the rules above) that

$$[L_\lambda h] = \partial h + \lambda h.$$

Now we use the noncommutative Wick formula to prove that

$$[L_\lambda L] = \partial L + 2\lambda L + \frac{\lambda^3}{12}.$$

WORK THIS OUT IN CLASS.

Now let's return to the quantum field notation. We see that

$$[L(z), L(w)] = \partial_w L(w)\delta(z,w) + 2L(w)\partial_w\delta(z,w) + \frac{1}{2}\partial_w^{(3)}\delta(z,w). \tag{22}$$

Now let us expand even further. Since we had defined

$$L_m = \frac{1}{2}\sum_{k\in\mathbb{Z}} : h_k h_{n-k} :\Leftrightarrow L(z) = \sum L_m z^{-m-2},$$

we take $\operatorname{Res}_z \operatorname{Res}_w z^{m+1}w^{n+1}$ applied to (22), obtaining

$$[L_m, L_n] = (m-n)L_{m+n} + \delta_{m,-n}\frac{m^3-m}{12}I_H.$$

Summarising what we have achieved, without the vertex algebra language:

Corollary 61. *The assignment*

$$\begin{aligned} L_m &= \frac{1}{2}\sum_{k\in\mathbb{Z}} h_k h_{n-k}, \quad m \neq 0, \\ L_0 &= \sum_{k\in\mathbb{Z}_+} h_{-k}h_k \end{aligned}$$

defines a representation of the Virasoro algebra in H in which $C = 1$.

We could have proved this by direct computation, but in the time it would have taken we developed a theory which allows similar and much more complicated calculations to be done easily.

More poetically, fixing the infinities in the naive definition of the action of W, resulted in the appearance of a nontrivial central extension.

Exercise 62. *Let β be a constant and define*

$$B = L + \beta\partial x.$$

Use the noncommutative Wick formula to prove that

$$[B_\lambda B] = \partial B + 2\lambda B + \frac{\lambda^3}{12}(1-12\beta^2).$$

3.4 The Virasoro vertex algebras

What can we say about this representation of Vir in H afforded by $L =: hh : /2$? Not very much. If we make the following definition:

Definition 42. A highest weight representation of Vir is a representation of Vir containing a vector v such that

- $V = U(Vir)v$,
- $L_n v = 0$ for all $n > 0$,
- $L_0 v = hv$ and $Cv = cv$ for some constants h and c.

Then $V_0 = U(Vir)\left|0\right\rangle \subset H$ is a highest weight representation, with $h = 0$ and $c = 1$. we see this containment is strict because $h_{-1}\left|0\right\rangle \notin V_0$. Indeed by the PBW theorem, and a grading argument, any element of V_0 with a chance of being $h_{-1}\left|0\right\rangle$ is $L_{-1}\left|0\right\rangle$. But

$$L_{-1}\left|0\right\rangle = \frac{1}{2}\sum_{k\in\mathbb{Z}} h_k h_{n-k}\left|0\right\rangle .$$

It is easy to see that every term on the right hand side annihilates $\left|0\right\rangle$.

Now let us consider V_0 with $\left|0\right\rangle$ as in V and the restriction of T from V to V_0. Since we checked already that $T = L_{-1}$ we already known that $T : V_0 \to V_0$. Finally put

$$\mathcal{F} = \{L(z)\}.$$

We claim this is a vertex algebra. The locality of $L(z)$ with itself follows from (22), indeed

$$(z-w)^4[L(z), L(w)] = 0.$$

The translation invariance for this field already follows from $\mathcal{F}' \subset \tilde{\mathcal{F}}$, proved in the previous lectures. It remains only to verify that V_0 is generated from $\left|0\right\rangle$ by the coefficients $\{L_m\}$, but we have defined V_0 so that this is the case.

Inspired by this example, we define a class of vertex algebras.

Definition 43. Denote by Vir^c the highest weight Vir-module

$$U(Vir) \otimes_{U(Vir_{\geq -1})} \mathbb{C}\left|0\right\rangle$$

where $C\left|0\right\rangle = c$, and $L_m\left|0\right\rangle = 0$ for all $m \geq -1$. By the PBW theorem, this module has a basis consisting of monomials

$$L_{-m_1}L_{-m_2}\cdots L_{-m_s}\left|0\right\rangle , \quad m_1 \geq \ldots \geq m_s \geq 2.$$

We now equip Vir^c with $T = L_{-1}$ and the set of quantum fields $\{L(z)\}$, where

$$L(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}.$$

This is a vertex algebra, known as the *universal Virasoro vertex algebra of central charge c.*

The fact that $L(z)$ is a quantum field is seen in the same way that $h(z)$ is a quantum field on H. Locality of $L(z)$ with itself needs to be proved from its definition. But by now we know that the defining relations of Virasoro are equivalent to

$$[L(z), L(w)] = \partial_w L(w)\delta(z, w) + 2L(w)\partial_w \delta(z, w) + \frac{c}{2}\partial_w^{(3)}\delta(z, w). \tag{23}$$

We also need to check that

$$[L_{-1}, L(z)] = \partial_z L(z),$$

this is checked from the Virasoro relations. Finally we require $L_{-1} |0\rangle = 0$, which we have imposed by fiat.

Rmk: any variation on $L_0 |0\rangle = 0$ would violate the axioms.

EXERCISE: Check that such-and-such vector is a singular vector in $Vir^{-22/5}$.

2023-02-06. Lecture 13

3.5 Skew-symmetry

Last time we proved all the identities we needed to do our computations except one:

$$[b_\lambda a] = -[a_{-\lambda-T} b].$$

Now we will prove this. Firstly we translate back into the quantum field notation.

Lemma 63 (FBZ, 3.2.3)**.**

$$e^{wT} Y(a, z) e^{-wT} = i_{z,w} Y(a, z + w).$$

Proof. By a standard sort of calculation

$$e^{wT} X e^{-wT} = e^{\mathrm{ad}(wT)} X.$$

Thus, since $[T, a(z)] = \partial_z a(z)$, we have

$$e^{wT} Y(a, z) e^{-wT} = e^{\mathrm{ad}(wT)} Y(a, z) = e^{w\partial_z} Y(a, z).$$

Now for any $n \in \mathbb{Z}$ we have

$$e^{w\partial_z} z^n = \sum_{k=0}^{\infty} \frac{w^k}{k!} \partial_z^k (z^n) = \sum_{k=0}^{\infty} \frac{w^k}{k!} n(n-1)\cdots(n-k+1) z^{n-k} = \sum_{k=0}^{\infty} \binom{n}{k} w^k z^{n-k} = i_{z,w}(z+w)^n.$$

And we are done. $\square$

Proposition 64 (Skew-symmetry)**.** *Let V be a vertex algebra, and $a, b \in V$. Then*

$$Y(b, z) a = -e^{zT} Y(a, -z) b.$$

Proof. We start from locality relation, both sides of which we apply to $|0\rangle$:

$$(z-w)^N Y(a,z)Y(b,w)\,|0\rangle = (z-w)^N Y(b,w)Y(a,z)\,|0\rangle\,,$$

that is,

$$(z-w)^N Y(a,z)e^{wT}b = (z-w)^N Y(b,w)e^{zT}a.$$

Now by the translation lemma we rewrite the RHS:

$$(z-w)^N Y(a,z)e^{wT}b = (z-w)^N e^{zT}Y(b,w-z)a.$$

In the expression $Y(b,w-z)a$ there is some limit to how negative a power of $w-z$ occurs, so if we take N sufficiently large, only non-negative powers of w appear on both sides.

So do this, and put $w=0$, thus obtaining

$$z^N Y(a,z)b = z^N e^{zT}Y(b,-z)a,$$

from which we can cancel z^N . □

By expanding out the skew-symmetry identity in powers of z and equating coefficients, we obtain the equivalent form

$$b_{(n)}a = -\sum_{j\in\mathbb{Z}_{\geq 0}} ()$$

3.6 A second definition of vertex algebra

If we take the Borcherds identity and extract the coefficient of $z^{-m-1}w^{-k-1}$ we obtain:

$$\sum_{j\in\mathbb{Z}_+}\binom{m}{j}(a_{(n+j)}b)_{(m+k-j)}c = \sum_{j\in\mathbb{Z}_+}(-1)^j\binom{n}{j}\left(a_{(m+n-j)}b_{(k+j)}c - (-1)^n b_{(n+k-j)}a_{(m+j)}c\right). \tag{24}$$

This is valid in any vertex algebra V, for all $a,b,c\in V$ and all $m,k,n\in\mathbb{Z}$.

If we put $m=0$ in (24) we obtain a formula

$$(a_{(n)}b)_{(k)}c = \text{stuff}.$$

If we multiply the LHS by w^{-k-1} and sum on k we get a formula

$$Y(a_{(n)}b,w)c = \text{stuff}.$$

If you do the computation in detail, you will find that the formula obtained is

$$Y(a_{(n)}b,w)c = Y(a,w)_{(n)}Y(b,w)c, \tag{25}$$

which we have already seen. Likewise we can deduce (24) if we know (25) for all n, so these identities are equivalent.

Definition 44 (KVA Proposition 4.8 (b))**.** A vertex algebra is a vector space V, a nonzero vector $|0\rangle \in V$, and a collection of bilinear products

$$V \times V \to V, \qquad (a, b) \mapsto a_{(n)}b$$

satisfying:

- For each $a, b \in V$ there exists N such that $a_{(n)}b = 0$ for all $n \geq N$,
- For all $a \in V$ we have $|0\rangle_{(n)} a = \delta_{n,-1}a$
- For all $a \in V$ we have $a_{(-1)} |0\rangle = a$ and $a_{(\geq 0)} |0\rangle = 0$,
- Equation (24) holds for all $a, b, c \in V$ and $m, k, n \in \mathbb{Z}$.

If we write

$$Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$$

then the first axiom expresses the fact that these formal distributions are required to be quantum fields. The second axiom states that $Y(|0\rangle, z) = I_V$. The third axiom states that

$$Y(a, z) |0\rangle \in V[[z]], \quad \text{and} \quad s(Y(a, z)) = a.$$

Locality can be deduced from the Borcherds identity (most easily when it is put back into the form (21)).

We remark that $T : V \to T$ is recovered from this definition by setting

$$Ta = a_{(-2)} |0\rangle .$$

This is not so surprising since, we recall, in the first definition of vertex algebra

$$a(z) |0\rangle = e^{zT} a,$$

so

$$a_{(-1)} |0\rangle + a_{(-2)} |0\rangle + a_{(-3)} |0\rangle + \cdots = a + (Ta)z + \frac{1}{2}(T^2 a)z^2 + \cdots .$$

Theorem 65. *These two definitions of vertex algebra are equivalent.*

The nice thing about the second definition of vertex algebra is that it makes the definition of homomorphism of vertex algebras become obvious.

Definition 45. A homomorphism of vertex algebras is a linear map $\varphi : V_1 \to V_2$ satisfying

$$\varphi(|0\rangle_1) = |0\rangle_2$$

and

$$\varphi(a_{(n)}b) = \varphi(a)_{(n)}\varphi(b)$$

for all $a, b \in V_1$ and $n \in \mathbb{Z}$.

Proposition 66. *There is a (unique) homomorphism of vertex algebras*

$$\varphi : Vir^{c=1} \to H$$

such that $\varphi(L) = \frac{1}{2} : hh :$.

It is important to recognise that we have not yet proved this fact, although we have good reason to believe it. Indeed we have confirmed already that

$$[L_\lambda L] = TL + 2\lambda L + \frac{1}{2}\lambda^{(3)},$$

and the same λ-bracket relation is satisfied by $\varphi(L)$. But how can we be sure that, for example, $\varphi(: LL :) =: \varphi(L)\varphi(L) :$? Is it automatic, or is it something we need to check? If so, how?

We will return to this point in future. For now rest assured that φ is indeed a homomorphism.

The notions of isomorphism and automorphism of vertex algebras now also make sense.

Lemma 67. *Let V be a vertex algebra and G a group of automorphisms of V. Then the fixed-point subspace*

$$V^G = \{v \in V \mid g(v) = v \text{ for all } g \in G\}$$

is a vertex algebra.

Proof. □

So for example we might consider the sign representation of $\mathbb{Z}/2\mathbb{Z} = \{1, -1\}$ on the 1-dimensional vector space $\mathfrak{h} = \mathbb{C}x$, and use it to induce a representation of $\mathbb{Z}/2\mathbb{Z}$ on the Heisenberg vertex algebra

$$H = S(t^{-1}\mathfrak{h}[t^{-1}]) = \mathbb{C}[x_{-1}, x_{-2}, \ldots],$$

more explicitly

$$(-1) \cdot x_{-i_1}x_{-i_2}\cdots x_{-i_s} = (-1)^s x_{-i_1}x_{-i_2}\cdots x_{-i_s}.$$

It is straightforward to check that this is an automorphism of order 2 of the vertex algebra. The fixed point subalgebra is the span of the monomials of even length. We may compute the graded dimension of this vertex algebra as follows: the graded dimension of H itself is given by the partition counting function

$$\mathrm{Ch}_H = \prod_{m=1}^{\infty}(1 - q^m)^{-1}.$$

If we consider instead

$$\prod_{m=1}^{\infty}(1 + q^m)^{-1}$$

then we obtain the count of partitions with partitions of odd length counted with $-$ sign. If we want to count only the partitions of even length, then we should take the sum

$$\prod_{m=1}^{\infty}(1-q^m)^{-1}+\prod_{m=1}^{\infty}(1+q^m)^{-1}$$

to cancel the contributions of odd-length partitions. Now we have overcounted the even-length partitions, so we divide by 2 to correct this:

$$\begin{aligned}\mathrm{Ch}_{H^+}&=\frac{1}{2}\left(\prod_{m=1}^{\infty}(1-q^m)^{-1}+\prod_{m=1}^{\infty}(1+q^m)^{-1}\right)\\&=1+q^2+q^3+3q^4+3q^5+6q^6+7q^7+12q^8+14q^9+22q^{10}+\cdots\end{aligned}$$

Here $H^+=H^{\mathbb{Z}/2\mathbb{Z}}$ denotes the $+1$ eigenspace of $(-1)\in\mathbb{Z}/2\mathbb{Z}$. We may similarly compute the graded dimension of the other eigenspace:

$$\begin{aligned}\mathrm{Ch}_{H^-}&=\frac{1}{2}\left(\prod_{m=1}^{\infty}(1-q^m)^{-1}-\prod_{m=1}^{\infty}(1+q^m)^{-1}\right)\\&=q+q^2+2q^3+2q^4+4q^5+5q^6+8q^7+10q^8+16q^9+20q^{10}+\cdots\end{aligned}$$

Let us consider again the homomorphism $\varphi:Vir^1\to H$. It is already clear that this morphism is not surjective, as H contains the nonzero vector $h=x_{-1}$ of degree 1 while Vir^c contains nothing of this degree. We observe now that $\varphi(L)=(1/2):hh:$ lands in H^+, and so since L generates Vir^1 we must have $\varphi(Vir^1)\subset H^+$. Could this be an isomorphism? Since the graded dimension of Vir^c in general is

$$\begin{aligned}\mathrm{Ch}_{Vir^c}&=\prod_{m=2}^{\infty}(1-q^m)^{-1}\\&=1+q^2+q^3+2q^4+2q^5+4q^6+4q^7+7q^8+8q^9+12q^{10}+\cdots,\end{aligned}$$

we see that no, this is not an isomorphism.

In general the kernel of a homomorphism should be an ideal. So let us define ideal in a vertex algebra.

Definition 46. An ideal I of a vertex algebra V is a vector subspace such that

$$a_{(n)}i\in I\quad\text{and}\quad i_{(n)}a\in I$$

for all $i\in I$ and $a\in V$.

Remark 68. *In general $Tx=x_{(-2)}\,|0\rangle$ for all $x\in V$. Therefore ideals, as defined above, are stable under T, that is $TI\subset I$.*

Comment on alternative equivalent definition of ideal

3.7 Vertex algebra modules

Now that we have the Borcherds-style definition of vertex algebra, it also becomes clear how to define vertex algebra module. Well, not so clear, as indeed we now give two potential definitions of V-module. And they are not (at least, not obviously) equivalent!

Definition 47 (1)**.** A module over a vertex algebra V is a vector space M and, for every $a \in V$ a quantum field

$$Y^M(a,z)$$

on M such that

$$\begin{aligned} Y^M(\left|0\right\rangle, z) &= I_M, \\ Y^M(a_{(n)}b, z) &= Y^M(a,z)_{(n)}Y^M(b,z). \end{aligned}$$

This is the definition given in KRR, Chapter 18.

Exercise 69. *From the first definition, deduce the relation* $Y^M(Ta,z) = \frac{\partial}{\partial z} Y^M(a,z)$*. Hint: First prove that* $Ta = a_{(-2)}\left|0\right\rangle$*. The relation*

$$\operatorname{Res}_z\left(\partial_z f(z) g(z)\right) = -\operatorname{Res}_z\left(f(z)\partial_z g(z)\right)$$

is also useful.

Definition 48 (2)**.** A module over a vertex algebra V is a vector space M, an operator $T^M : M \to M$ and, for every $a \in V$ a quantum field

$$Y^M(a,z)$$

on M such that

$$\begin{aligned} Y^M(\left|0\right\rangle, z) &= I_M, \\ [T^M, Y^M(a,z)] &= \partial_z Y^M(a,z) \\ Y^M(a_{(n)}b, z) &= Y^M(a,z)_{(n)}Y^M(b,z). \end{aligned}$$

Of course a module of the second kind yields a module of the first kind by forgetting T^M. In spite of the exercise above, it is not possible to reconstruct the operator T^M of the second definition from the first definition.

In much of the literature on vertex algebras, a compatible action of the Virasoro algebra is assumed as part of the structure, and modules are assumed to have this action too. For example see FHL, Definition 4.1.1.

We already have a few examples of modules.

Example 3. Let $(V, \left|0\right\rangle, T, Y)$ be any vertex algebra. Then (V, T, Y) is a V-module in the second sense.

If we consider $V = H$ and $M = H^\mu$, then M becomes a V-module upon putting

$$Y^M(h,z) = \sum_{n\in\mathbb{Z}} h_n^M z^{-n-1} = h(z) + \mu z^{-1}.$$

Remark 70. *Why this defines a module structure is not so easy to answer convincingly, see the third lecture.*

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3.8 Lie conformal algebras

Just a few brief remarks on this. There is a point of view according to which vertex algebras are like associative algebras, and that there is another kind of object analogous to Lie algebras. I.e., there is a dictionary

$$\text{Associative algebras} \longleftrightarrow \text{Lie algebras}$$

$$\text{Vertex algebras} \longleftrightarrow \text{Lie conformal algebras}$$

More precisely there is a pair of adjoint functors

$$\mathbf{Ass} \xrightarrow{(-)_-} \mathbf{Lie}$$

$$\mathbf{Ass} \xleftarrow[U(-)]{} \mathbf{Lie}$$

where the right adjoint functor $A \mapsto A_-$ means to turn an associative algebra A into a Lie algebra by equipping it with the commutator bracket $[a,b] = ab - ba$, and the left adjoint functor is the universal enveloping algebra $\mathfrak{g} \mapsto U(\mathfrak{g})$.

Similarly there is a way to take a vertex algebra, "forget some of its structure", obtaining a Lie conformal algebra, and there is an adjoint functor which sends a Lie conformal algebra R to its *universal enveloping vertex algebra* $V(R)$.

So what is a Lie conformal algebra?

Definition 49. A Lie conformal algebra is a vector space R, together with a linear endomorphism $T : R \to R$, and a λ-*bracket*

$$R \otimes R \to R[\lambda].$$

These structures are to satisfy

- (skew-symmetry) $[b_\lambda a] = -[a_{-\lambda-T} b]$

- (Jacobi) $[a_\lambda[b_\mu c]] - [b_\mu[a_\lambda c]] = [[a_\lambda b]_{\lambda+\mu} c]$
- () $[(Ta)_\lambda b] = -\lambda[a_\lambda b]$ and $[a_\lambda(Tb)] = (\lambda + T)[a_\lambda b]$

(In fact one of the conditions in the third axiom is redundant; using skew-symmetry each of the two conditions can be derived from the other.)

What the definition of a morphism of Lie conformal algebras ought to be is an easy exercise left to the student.

Every vertex algebra V, equipped with T, and with the λ-bracket as defined in a previous lecture, is an example of a Lie conformal algebra. In fact axioms (1) and (3) we have already checked. As for axiom (2), it can be verified as well, using for example the commutator formula. Since the λ-bracket $[a_\lambda b]$ contains all $a_{(j)}b$ for $j \geq 0$, a Lie conformal algebra is like a vertex algebra which has forgotten its normally ordered product $: ab := a_{(-1)}b$. The universal enveloping vertex algebra construction takes a Lie conformal algebra R and adjoins a normally ordered product to it in the "freest possible way".

I will not give the construction here, but the universal property that $V(R)$ should satisfy is already clear: There should be a canonical map $i : R \to V(R)$ of Lie conformal algebras, such that given any morphism $\phi : R \to V$ of Lie conformal algebras, there exists a unique morphism of vertex algebras $\overline{\phi} : V(R) \to V$ such that

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$$\begin{tikzcd} R \arrow[rr, "\phi"] \arrow[dr, "i"'] & & V \\ & V(R) \arrow[ur, "\overline{\phi}"'] & \end{tikzcd}$$

Example 4. Let $R = \mathbb{C}[T]h \oplus \mathbb{C}\mathbf{1}$, that is to say R has a $\mathbb{C}$-basis

$$\{h, Th, T^2h, \ldots\} \cup \{\mathbf{1}\},$$

and let $T\mathbf{1} = 0$. The λ-bracket is uniquely defined by

$$[h_\lambda h] = \lambda \mathbf{1}$$

and

$$[\mathbf{1}_\lambda a] = 0 \qquad \text{for all } a \in R.$$

The third axiom ensures uniqueness.

Some calculations must be performed to check that the λ-bracket is well-defined and satisfies the axioms. Another approach might be to consider the Heisenberg vertex algebra H and its vector subspace $R' = \mathbb{C}\left|0\right\rangle + \mathbb{C}h + \mathbb{C}Th + \cdots$. If you can confirm this space is closed under $[-_\lambda -]$ then this will prove well-definedness.

Example 5. Let $c \in \mathbb{C}$ and let $R^c = \mathbb{C}[T]L \oplus \mathbb{C}\mathbf{1}$ with $T\mathbf{1} = 0$. The λ-bracket is uniquely defined by

$$[L_\lambda L] = TL + 2\lambda L + \lambda^{(3)}\frac{c}{2}\mathbf{1}$$

and

$$[\mathbf{1}_\lambda a] = 0 \qquad \text{for all } a \in R^c.$$

The origin of the following example will become clearer in the next section.

Example 6. Let $\mathfrak{g}$ be a Lie algebra and $(,) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ an invariant bilinear form. Let $R^{\mathfrak{g},(,)} = \mathbb{C}[T] \otimes \mathfrak{g} \oplus \mathbb{C}\mathbf{1}$ with $T\mathbf{1} = 0$. The λ-bracket is uniquely defined by

$$[a_\lambda b] = [a, b] + \lambda(a, b)\mathbf{1}$$

and

$$[\mathbf{1}_\lambda a] = 0 \qquad \text{for all } a \in R^{\mathfrak{g},(,)}.$$

What we will (essentially) have is that $H = U(R)$, $Vir^c = U(R^c)$ and $V^{(,)}(\mathfrak{g}) = U(R^{\mathfrak{g},(,)})$. Here $V^{(,)}(\mathfrak{g})$ is the universal affine vertex algebra. It will be defined in the next section, and we will not use the notation $V^{(,)}(\mathfrak{g})$ but rather $V^k(\mathfrak{g})$.

3.9 Commutative vertex algebras

Definition 50. Let $(V, \mathcal{F}, |0\rangle, T)$ be a vertex algebra. We say the vertex algebra is commutative if

$$[a(z), b(w)] = 0$$

for all $a, b \in \mathcal{F}$.

Remark 71. *Clearly this property is preserved upon passage from $\mathcal{F}$ to $\mathcal{F}'$, so the definition makes sense.*

It is quite easy to construct examples of commutative vertex algebras. Let A be a commutative algebra over $\mathbb{C}$ with unit element 1. Furthermore let $D : A \to A$ be a derivation of A. We should like to construct a vertex algebra structure in which $|0\rangle = 1$ and $T = D$. We know already by general principles that

$$Y(a, z)\,|0\rangle = e^{zT}a = (e^{zD}a)1,$$

we shall define $Y(a, z)$ by

$$Y(a, z)b = (e^{zD}a)b.$$

This definition satisfies the first two axioms. We need only check locality. But since A is commutative we have

$$[e^{zD}a, e^{zD}b] = 0.$$

So we have locality, and that the vertex algebra is in fact commutative.

Theorem 72. *Any commutative vertex algebra is obtained by the construction above.*

Proof. Let V be a commutative vertex algebra, and let $a, b \in V$. By definition we know that

$$Y(a,z)b \in V((z)).$$

But

$$Y(a,z)b = Y(a,z)Y(b,w)\left|0\right\rangle|_{w=0} = Y(b,w)Y(a,z)\left|0\right\rangle|_{w=0}$$

by commutativity. Since $Y(a,z)\left|0\right\rangle \in V[[z]]$ we conclude that

$$Y(a,z)b \in V[[z]].$$

Let us now define

$$ab = Y(a,z)b|_{z=0}.$$

Since the vertex algebra is commutative we have the identity

$$a(bc) = b(ac) \quad \text{for all } a, b, c \in A.$$

By the following lemma, we have associativity and commutativity of the algebra V.

We also check that $D = T$ is a derivation of V. Indeed by Lemma 73

$$\begin{aligned} T(ab) &= TY(a,z)b|_{z=0} = [T, Y(a,z)]b|_{z=0} + Y(a,z)(Tb)|_{z=0} \\ &= Y(Ta,z)b|_{z=0} + Y(a,z)(Tb)|_{z=0} \\ &= (Ta)b + a(Tb). \end{aligned}$$

Finally we must confirm that the original vertex algebra structure is recovered from the commutative product as per the example above, i.e., we must confirm that

$$Y(a,z)b = (e^{zT}a)b.$$

For $b = \left|0\right\rangle$ this is already proved in Theorem 56. Returning to the calculations above, we have

$$Y(a,z)b = Y(a,z)Y(b,w)\left|0\right\rangle|_{w=0} = Y(b,w)Y(a,z)\left|0\right\rangle|_{w=0} = Y(b,w)(e^{zT}a)|_{w=0} = b(e^{zT}a) = (e^{zT}a)b.$$

□

Lemma 73. *In a vertex algebra* $(V, Y, \left|0\right\rangle, T)$ *we have*

$$\partial_z Y(a,z) = [T, Y(a,z)] = Y(Ta,z).$$

Proof. The first equality is of course one of the axioms. □

Lemma 74. *Let A be a vector space equipped with a product $*$. Suppose $*$ has a right unit 1 (i.e., $a * 1 = a$ for all $a \in A$), and the operations L_a of left multiplication by $a \in A$ mutually commute for all elements of A,*

$$a * (b * c) = b * (a * c) \quad \textit{for all } a, b, c \in A.$$

Then in fact $(A, *)$ is associative and commutative.

Proof. Firstly the product is commutative because

$$a * b = a * (b * 1) = b * (a * 1) = b * a.$$

It is associative because

$$(a * b) * c = c * (a * b) = a * (c * b) = a * (b * c).$$

Finally 1 is also a left unit because of commutativity. □

3.10 Universal affine vertex algebras

The definition of $V^k(\mathfrak{g})$ given in this section works for a general Lie algebra and invariant bilinear form $(,)$. But to avoid annoying notation issues, we will work with the most relevant case from the outset: Let $\mathfrak{g}$ be a finite dimensional simple Lie algebra, and let $(,) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ be the unique invariant bilinear form normalised so that $(\theta, \theta) = 2$ where $\theta \in \mathfrak{h}^*$ is a long root.

In fact $(,)$ is related to the Killing form by multiplication by some scalar factor, and the scalar factor is actually an important invariant of $\mathfrak{g}$. Indeed we define the number $h^\vee$ by the relation

$$(x, y) = \frac{1}{2h^\vee}\kappa(x, y),$$

where κ is the Killing form. Now $h^\vee$ is called the dual Coxeter number of $\mathfrak{g}$. For example the dual Coxeter number of sl_n is n and the dual Coxeter number of E_7 is 18, etc.

Now let

$$\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K$$

where

$$[at^m, bt^n] = [a, b]t^{m+n} + m\delta_{m,-n}(,)K, \quad \text{and } K \text{ central}$$

be the corresponding affine Lie algebra.

Let $\hat{\mathfrak{g}}_+ \subset \hat{\mathfrak{g}}$ denote the subalgebra

$$\hat{\mathfrak{g}}_+ = \mathfrak{g}[t] \oplus \mathbb{C}K.$$

We will consider the following:

Definition 51. Fix $k \in \mathbb{C}$, and let $\mathbb{C}v_k$ be the $\hat{\mathfrak{g}}_+$-module in which

$$\mathfrak{g}[t]v_k = 0, \quad \text{and} \quad Kv_k = kv_k.$$

The vacuum $\hat{\mathfrak{g}}$-module of level k is the induced module

$$V^k(\mathfrak{g}) = U(\hat{\mathfrak{g}}) \otimes_{U(\hat{\mathfrak{g}}_+)} \mathbb{C}v_k.$$

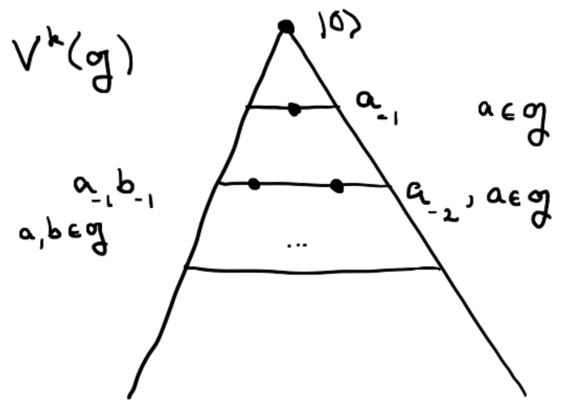
We typically write the operator at^m as a_m. By the PBW theorem $V^k(\mathfrak{g})$ has a basis of the form

$$a^{i_1}_{-m_1} a^{i_2}_{-m_2} \cdots a^{i_s}_{-m_s} v_k \tag{26}$$

where $\{a^1, \dots a^N\}$ is a basis of $\mathfrak{g}$ and the $a^{i_j}_{m_j}$ are arranged in lexicographic order, i.e., for example

$$m_1 \geq m_2 \geq \ldots \geq m_s$$

and if $m_j = m_{j+1}$ then $i_j \geq i_{j+1}$.



We also had the Kac-Moody algebra $\tilde{\mathfrak{g}} = \hat{\mathfrak{g}} \oplus \mathbb{C}d$ with

$$[d, at^m] = mat^m.$$

Most of the time we pay no attention to d, but sometimes it is useful. If we include d in $\hat{\mathfrak{g}}_+$, and choose to set $dv_k = 0$, then $V^k(\mathfrak{g})$ becomes a $\tilde{\mathfrak{g}}$-module, and on the monomial (26) above, d acts by the scalar

$$\sum m_j.$$

This is precisely because $[d, at^m] = mat^m$. If we denote by $V^k(\mathfrak{g})_n$ the subspace spanned by all monomials of degree n, then I ask:

Exercise 75. *Check that a_n has degree $-n$.*

We may write down a formula for the graded dimension

$$\mathrm{Ch}_{V^k(\mathfrak{g})} = \sum_{n=0}^{\infty} \dim(V^k(\mathfrak{g})_n) q^n.$$

By using what we know about the PBW theorem and symmetric algebras, we deduce

$$\mathrm{Ch}_{V^k(\mathfrak{g})} = \prod_{m=1}^{\infty} (1-q^m)^{-d}, \qquad d = \dim(\mathfrak{g}).$$

We are going to construct a vertex algebra structure on $V^k(\mathfrak{g})$. First we will invent some generating fields. For $a \in \mathfrak{g}$ let

$$a(z) = J^a(z) = \sum_{n \in \mathbb{Z}} at^n z^{-n-1}.$$

It is easy to see (especially using the grading introduced above) that each $a(z)$ is a quantum field. We shall now try to compute

$$[a(z), b(w)].$$

Indeed

$$\begin{aligned}
[a(z), b(w)] &= \sum_{m,n} [a_m, b_n] z^{-m-1} w^{-n-1} \\
&= \sum_{m,n} [a,b]_{m+n} z^{-m-1} w^{-n-1} + (a,b) \sum_{m,n} m \delta_{m,-n} z^{-m-1} w^{-n-1} K \\
&= \sum_{m,n} [a,b]_{m+n} w^{-(m+n)-1} z^{-m-1} w^m + (a,b) \sum_{m} m z^{-m-1} w^{m-1} k I_V \\
&= [a,b](w) \delta(z,w) + (a,b) \partial_w \delta(z,w) k I_V.
\end{aligned}$$

Success! The quantum fields $a(z)$ and $b(z)$ form a local pair. In λ-bracket notation we have

$$[a_\lambda b] = [a,b] + k(a,b)\lambda.$$

Since $V^k(\mathfrak{g})$ is spanned by the monomials (26), it is clearly a good idea to set $vac = v_k$, and we already guarantee the axiom that V be generated from $|0\rangle$ by the fields.

Next we need a candidate for T. The relation $[T, a(z)] = \partial_z a(z)$ directly implies

$$[T, at^n] = -nat^{n-1}.$$

This, together with the requirement that $T|0\rangle = 0$, determines the action of T on all monomials (26) recursively. Explicitly

$$\begin{aligned}
T\left(a^{i_1}_{-m_1} a^{i_2}_{-m_2} \cdots a^{i_s}_{-m_s} v_k\right) = m_1 &\left(a^{i_1}_{-m_1-1} a^{i_2}_{-m_2} \cdots a^{i_s}_{-m_s} v_k\right) \\
+ m_2 &\left(a^{i_1}_{-m_1} a^{i_2}_{-m_2-1} \cdots a^{i_s}_{-m_s} v_k\right) \\
+ &\cdots \\
+ m_s &\left(a^{i_1}_{-m_1} a^{i_2}_{-m_2} \cdots a^{i_s}_{-m_s-1} v_k\right)
\end{aligned}$$

A little more poetically, $T = -d/dt$.

Proposition 76. *The $\hat{\mathfrak{g}}$-module $V^k(\mathfrak{g})$, equipped with $|0\rangle = v_k$, $T = -d/dt$, and*

$$\mathcal{F} = \{a(z) \mid a \in \mathfrak{g}\}$$

is a vertex algebra.

3.11 Simple affine vertex algebras

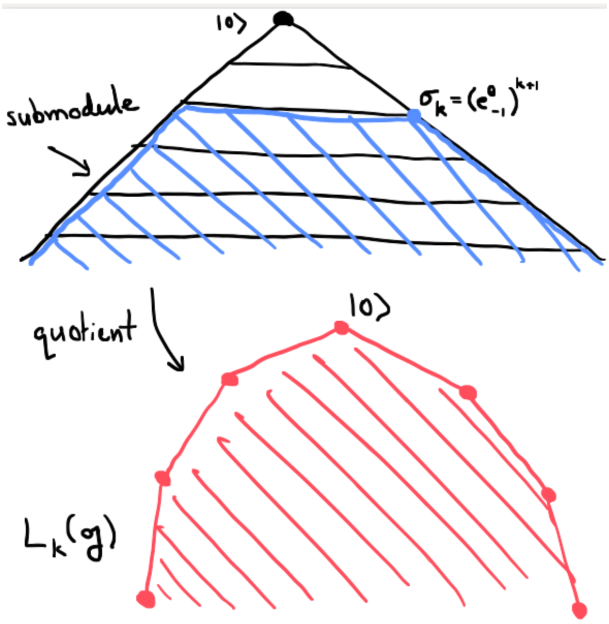
Now assume $k \in \mathbb{Z}_+$. Let θ be the highest root of $\mathfrak{g}$. (Example: in $\mathfrak{g} - \mathfrak{sl}_2$ we have $e = e^\theta$ and $f = e^{-\theta}$.) It turns out that

$$\sigma_k = (e^\theta_{(-1)})^{k+1}\,|0\rangle$$

is a singular vector in the $\hat{\mathfrak{g}}$-module $V^k(\mathfrak{g})$. That is, any lowering operator annihilates it, in fact it suffices to check that

$$e^{-\theta}_{(1)}\sigma_k = 0.$$

Exercise 77. *Explain why this check suffices, and do the calculation.*



Let

$$N_k = U(\hat{\mathfrak{g}})\sigma_k \subset V^k(\mathfrak{g})$$

denote the submodule generated by σ_k.

Proposition 78 (Kac, IDLA, (10.4.6))**.** *The quotient $\hat{\mathfrak{g}}$-module*

$$L_k(\mathfrak{g}) = V^k(\mathfrak{g})/N_k.$$

is irreducible

We now claim that $N_k \subset V^k(\mathfrak{g})$ is actually a vertex algebra ideal. In fact we claim in general

Proposition 79. *Any $\hat{\mathfrak{g}}$-submodule $I \subset V^k(\mathfrak{g})$ is a vertex algebra ideal.*

Proof. Firstly it is clear that

$$a_{(n)}i \in I$$

for all $i \in I$ and $a \in \mathfrak{g}$. This is the very definition of being a $\hat{\mathfrak{g}}$-submodule. Next we argue that the same is true for arbitrary $a \in V^k(\mathfrak{g})$, not just $a \in \mathfrak{g}$.

Remember that $V^k(\mathfrak{g})$ is in bijection with the completed space of quantum fields $\overline{\mathcal{F}}$, defined to be the span of all iterated n^{th}-products of fields from $\mathcal{F} = \{a(z) \mid a \in \mathfrak{g}\}$. Explicit formulas for these fields are complicated, but the point is that ultimately all their coefficients are linear combinations of compositions of coefficients of the $a(z)$. So I is stable under the action of all these coefficients.

Now we wish to show that

$$i_{(n)}a \in I$$

for all $i \in I$ and $a \in V^k(\mathfrak{g})$. It is not clear how to do this directly, but an idea emerges: let us use the skew-symmetry identity

$$a_{(n)}b = -\sum_{j \in \mathbb{Z}_+} (-1)^{n+j} \frac{1}{j!} T^j (b_{(n+j)}a).$$

If we could be sure that I were stable under T then we would easily conclude that

$$i_{(n)}a$$

is a linear combination of powers of T applied to elements of I, and hence lies in I. $\square$

So the question is... how can we be sure that $\hat{\mathfrak{g}}$-submodules of $V^k(\mathfrak{g})$ are stable under T? Since T is not an element of $\hat{\mathfrak{g}}$ (or even of $\tilde{\mathfrak{g}}$) it is not clear.

We will see that $\hat{\mathfrak{g}}$-submodules are indeed T-invariant. The way out of this conundrum, which we see in the next section, is surprisingly subtle.

3.12 Conformal gradings

Definition 52 (FBZ 1.2.1)**.** A conformal grading on a vertex algebra V is a $\mathbb{Z}$-grading

$$V = \bigoplus_{n \in \mathbb{Z}} V_n$$

such that for $a \in V_n$ and $m \in \mathbb{Z}$ the operator $a_{(m)}$ has degree $n - m - 1$. That is, if we write $\Delta(a) = n$ for $a \in V_n$, then

$$\Delta(a_{(m)}b) = \Delta(a) + \Delta(b) - m - 1. \tag{27}$$

To understand this peculiar definition, we observe the following example.

Recall that H is $\mathbb{Z}_+$-graded by

$$\deg(h_{-n_1}h_{-n_2}\cdots h_{-n_s}) = \sum_j n_j.$$

We claim this is a conformal grading. Certainly $\Delta(h) = 1$ and

$$h_{(n)} : H_k \to H_{k-n} = H_{k+\Delta(h)-n-1}$$

as required. But how are we to check that the same holds for more complicated elements of H?

Observe that, since

$$[L_\lambda h] = (T + \lambda)h$$

we have $L_{(0)}h = Th$ and $L_{(1)}h = h$, and so by the commutator formula

$$[L_{(m)}, h_{(n)}] = \sum_{j\geq 0} \binom{m}{j} (L_{(j)}h)_{(m+n-j)} = (Th)_{(m+n)} + mh_{(m+n-1)}.$$

Recall that

$$L(z) = \sum_n L_n z^{-n-1} = \sum_n L_n z^{-n-2},$$

so $L_{(n)} = L_{n-1}$. Thus

$$[L_0, h_{(n)}] = [L_{(1)}, h_{(n)}] = (Th)_{(n+1)} + h_{(n)} = -nh_{(n)}.$$

In the last equality here we have used the general relation

$$(Ta)_{(n)} = -na_{(n-1)}$$

which is equivalent to translation invariance.

Starting from $L_0 |0\rangle = 0$ it follows that

$$L_0(h_{(-n_1)}h_{(-n_2)}\cdots h_{(-n_s)}) = \sum_j n_j(h_{(-n_1)}h_{(-n_2)}\cdots h_{(-n_s)}),$$

so the grading deg is precisely the grading by eigenvalue of L_0. We also recall that $L_{-1} = T$. This will be useful as well.

Exercise 80. *If $L_0 a = \Delta(a)a$ and similarly for b, deduce from the commutator formula that*

$$\Delta(a_{(n)}b) = \Delta(a) + \Delta(b) - n - 1.$$

I did this calculation in class. Maybe include it here.

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3.13 Conformal structures

This calculation is pretty general, and leads to a definition.

Definition 53 (FBZ, 2.5.8)**.** The vertex algebra V is conformal, of central charge c if it contains a vector ω whose field

$$Y(\omega, z) = L(z) = \sum L_n z^{-n-2}$$

equips V with a representation of the Virasoro algebra of central charge c, such that

- $L_{-1} = T$,
- L_0 acts diagonalisably with eigenvalues in $\mathbb{Z}_+$ and finite dimensional eigenspaces.[3] Denote by V_n the $L_0 = n$ eigenspace.

Thus our first examples H and Vir^c are conformal vertex algebras. The eigenvalue grading on a conformal vertex algebra is a conformal grading. A homomorphism $\varphi : V_1 \to V_2$ of conformal vertex algebras is conformal if $\varphi(\omega_1) = \omega_2$. Notice that the set of conformal homomorphisms $\varphi : V_1 \to V_2$ is *not* a vector space.

The fact that, in a conformal vertex algebra, $T = L_{-1} = L_{(0)}$ is a coefficient of one of the quantum fields, simplifies many definitions.

Lemma 81. *An ideal I in a conformal vertex algebra V is a vector subspace such that*

$$a_{(n)}i \quad \textit{for all } a \in V \textit{ and } i \in I,$$

Lemma 82. *If (M, Y^M) is a module(1) for a conformal vertex algebra V with conformal vector ω then*

$$L^M(z) = Y^M(\omega, z)$$

turns M into a representation of the Virasoro Lie algebra at the same central charge as V such that

$$[L^M_{-1}, Y^M(a, z)] = \partial_z Y^M(a, z).$$

So M, with $T^M = L^M_{-1}$, automatically becomes a module(2).

3.14 The Sugawara construction

We have gotten to know the affine vertex algebras, and we have seen the notion of conformal structure in a vertex algebra. Now we will see that the affine vertex algebras are conformal.

Let $\mathfrak{g}$ be a finite dimensional simple Lie algebra and let $(,)$ be the nondegenerate invariant bilinear form on $\mathfrak{g}$ normalised so that $(\theta, \theta) = 2$ as usual.

[3] A common variant on this definition allows $V_n = 0$ for all $n < N$ for some fixed possibly negative $N \in \mathbb{Z}$.

Definition 54. Consider the universal affine vertex algebra $V^k(\mathfrak{g})$. Select bases $\{a^i\}$ and $\{b^i\}$ of $\mathfrak{g}$, dual with respect to $(,)$. Consider the element

$$L = \frac{1}{2(k+h^\vee)} \sum_i : a^i b^i :$$

is a conformal vector of central charge

$$c_k = \frac{k \dim(\mathfrak{g})}{k+h^\vee}$$

To prove that this is a conformal vector we must compute $[L_\lambda L]$, and we must verify that $L_{-1} = T$.

....

I did the calculation in detail in class, using λ-brackets

....

It turns out that if $k = 1$ and $\mathfrak{g}$ is simply laced then, in the simple quotient $L_1(\mathfrak{g})$, the Sugawara vector coincides [Kac, VA, Section 5.7] with the lattice vertex algebra Virasoro vector. Consequently

$$\dim(\mathfrak{g}) = (1 + h^\vee) \operatorname{rank}(\mathfrak{g}).$$

For $\mathfrak{g} = E_7$, for example, we have $\operatorname{rank}(\mathfrak{g}) = 7$, $h^\vee = 18$ and $\dim(\mathfrak{g}) = 133$. Sure enough

$$133 = 7 \times 19.$$

Let us remark that the relation above does not hold for non-simply laced types, but

$$\dim(\mathfrak{g}) = (1 + h) \operatorname{rank}(\mathfrak{g})$$

does always hold. Here h is the Coxeter number, defined to be the order of the product of all simple reflections.

Exercise 83. *Does this relation have a vertex algebraic interpretation?*

3.15 Lattice vertex algebras

Now let us examine an important construction: that of lattice vertex algebras. Eventually we will associate a vertex algebra V_L to any integral lattice L. But to warm up we start with the simplest case in which L is just $\mathbb{Z}$.

Recall the $\mathfrak{a}$-modules H^μ (recall that $h_0 = 0$ in H, while in H^μ we have $h_0 = \mu$). We shall attempt to construct a vertex algebra structure on the direct sum

$$V = \bigoplus_{\mu \in \mathbb{Z}} H^\mu.$$

We denoted by $|\mu\rangle$ the highest weight vector of H^μ. First let's try to figure out

$$\Gamma^\mu(z) = Y(|\mu\rangle, z).$$

We will build the vertex algebra in such a way that $T = L_{-1}$. It's easy to check that

$$L_{-1}|\mu\rangle = \mu h_{(-1)}|\mu\rangle,$$

so we ought to have

$$\partial_z \Gamma^\mu(z) = \mu : h(z)\Gamma^\mu(z) :$$

This, very intuitively, suggests

$$\begin{aligned}\Gamma^\mu(z) &= C : \exp \mu \int h(z)\, dz : \\ &= C \exp\left(\mu \sum_{n<0} h_n \frac{z^{-n}}{-n}\right) \exp\left(\mu \sum_{n>0} h_n \frac{z^{-n}}{-n}\right) z^{\mu h_0}.\end{aligned}$$

This is surprisingly close to the truth. Or rather...

We start from the relations

$$h_{(0)}|\mu\rangle = \mu|\mu\rangle, \quad \text{and} \quad h_{(>0)}|\mu\rangle = 0$$

which hold by definition. In other words

$$[h_\lambda |\mu\rangle] = \mu|\mu\rangle.$$

Consequently

$$[h(z), \Gamma^\mu(w)] = \mu\Gamma^\mu(w)\delta(z, w).$$

In particular

$$[h_n, \Gamma^\mu(w)] = \mu w^n \Gamma^\mu(w).$$

and

$$[h_0, \Gamma^\mu(w)] = \mu\Gamma^\mu(w).$$

Thus

$$\Gamma^\mu(w) : H^\beta \to H^{\beta+\mu}[[z, z^{-1}]].$$

Let's introduce the map $u : V \to V$, characterised by

$$u|\beta\rangle = |\beta + 1\rangle, \qquad uh_n = h_n u \text{ for all } n \neq 0.$$

Now we consider

$$u^{-\mu} \exp\left(\mu \sum_{n<0} h_n \frac{z^{-n}}{+n}\right) \Gamma^\mu(z) \exp\left(\mu \sum_{n>0} h_n \frac{z^{-n}}{+n}\right)$$

as a map

$$H^\beta \to H^\beta[[z, z^{-1}]].$$

By explicit calculation this commutes with each h_n, $n \neq 0$. Therefore it is some polynomial (which can depend on μ and β).

Above we saw intuitively that this polynomial should be of the form

$$C z^{\mu h_0}|_{H^\beta} = C_{\mu,\beta} z^{\mu\beta}.$$

We can see this more rigorously by considering actions of L_0. Indeed

$$L_0 \left|\mu\right\rangle = \left(\frac{1}{2}h_0^2 + \sum_{n>0} h_{-n}h_n\right)\left|\mu\right\rangle = \frac{\mu^2}{2}\left|\mu\right\rangle.$$

By thinking about conformal weights, it is clear that the unique term of

$$\Gamma^\mu(z) = \sum_{n\in\mathbb{Z}} \Gamma^\mu_{(n)} z^{-n-1}$$

that maps $\mathbb{C}\left|\beta\right\rangle$ to $\mathbb{C}\left|\beta+\mu\right\rangle$, corresponding to $n = -\mu\beta - 1$, is

$$\Gamma^\mu_{(-\mu\beta-1)} z^{\mu\beta}.$$

We are trying to construct a vertex algebra structure on the direct sum

$$V = \bigoplus_{\beta\in\mathbb{Z}} H^\beta,$$

and we discovered that

$$\Gamma^\mu(z) = Y(\left|\mu\right\rangle, z)$$

is given by

$$\Gamma^\mu(z)|_{H^\beta} = C_{\mu,\beta} \exp\left(\mu \sum_{n<0} h_n \frac{z^{-n}}{-n}\right) \exp\left(\mu \sum_{n>0} h_n \frac{z^{-n}}{-n}\right) z^{\mu\beta} u^\mu.$$

Let us rewrite this in the following way:

(1) We rewrite

$$V = \bigoplus_{\beta\in\mathbb{Z}} H^\beta = \bigoplus_{\beta\in\mathbb{Z}} H^\beta \otimes \mathbb{C}e^\beta,$$

(2) we rewrite u^μ as e^μ which is really an abuse of notation, by this we mean the operator

$$e^\mu : v \otimes e^\beta \mapsto v \otimes e^\mu e^\beta,$$

so that we may write

$$\Gamma^\mu(z) = \exp\left(\mu \sum_{n<0} h_n \frac{z^{-n}}{-n}\right) \exp\left(\mu \sum_{n>0} h_n \frac{z^{-n}}{-n}\right) z^{\mu h_0} e^\mu,$$

and (3) we absorb $C_{\mu,\beta}$ into the definition of an algebra $\mathbb{C}^C[\mathbb{Z}]$ via

$$e^\mu e^\beta = C_{\mu,\beta} e^{\mu+\beta}.$$

The final step is to use locality to determine the structure of the algebra.

Using the computation

$$\left[\sum_{n>0} h_n \frac{z^{-n}}{n}, \sum_{m<0} h_m \frac{w^{-m}}{m}\right] = -\sum_{n>0} \frac{1}{n}\left(\frac{w}{z}\right)^n = \log(1-w/z)$$

we are able to prove

$$\Gamma^\mu(z)\Gamma^\beta(w) = i_{z,w}(z-w)^{\mu\beta} : \Gamma^\mu(z)\Gamma^\beta(w) :$$

One finds

$$\begin{aligned} &: \Gamma^\mu(z)\Gamma^\beta(w) : = e^\mu e^\beta (\text{junk}) \\ &: \Gamma^\mu(z)\Gamma^\beta(w) : = e^\beta e^\mu (\text{same junk}) \end{aligned}$$

So

$$\Gamma^\mu(z)\Gamma^\beta(w) - \Gamma^\beta(w)\Gamma^\mu(z)$$

only has a chance of being a local distribution if

$$e^\mu e^\beta = (-1)^{\mu\beta} e^\beta e^\mu$$

for all μ, β.

This is actually impossible, as we see by taking $\mu = \beta = 1$. So our construction of a vertex algebra structure on V has failed.

But if we take, instead of V, rather

$$V_0 = \bigoplus_{\beta \in 2\mathbb{Z}} H^\beta$$

then we get a vertex algebra, by taking $C_{\mu,\beta} = 1$ for all μ, β.

There is a way to save V though. We should regard V as a *vector superspace*, with the parity of H^μ given by the parity of μ. In this case locality means, by definition, that

$$\Gamma^\mu(z)\Gamma^\beta(w) - (-1)^{p(\mu)p(\beta)} \Gamma^\beta(w)\Gamma^\mu(z)$$

be a local distribution. So now we can just choose $C_{\mu,\beta} = 1$ for all μ, β, and we are done because we have defined

$$(-1)^{p(\mu)p(\beta)} = (-1)^{\mu\beta}.$$

Definition 55. We call V the lattice vertex superalgebra associated with the lattice $\mathbb{Z}$. Similarly V_0 is the lattice vertex algebra associated with the lattice $2\mathbb{Z}$.

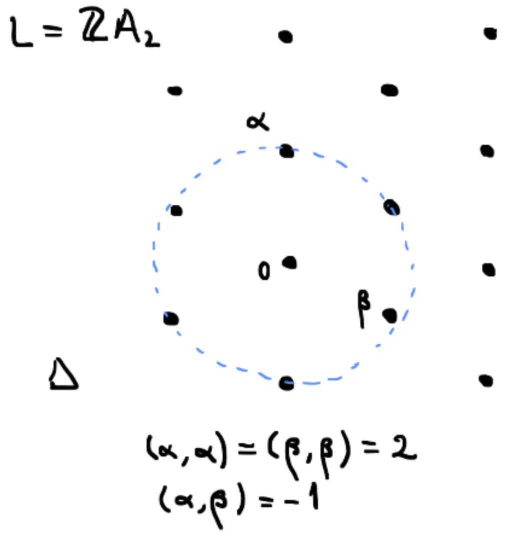
As the name suggests, there is actually a *lattice vertex (super)algebra* V_L for every integral lattice L.

Definition 56. An integral lattice is a free abelian group L together with a bilinear form

$$(\cdot,\cdot) : L \times L \to \mathbb{Z}.$$

The rank of L is its rank as a free abelian group. The lattice is called even if $|\alpha|$ is even for all $\alpha \in L$.

For example the A_2 root lattice.



The ambient space $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$ inherits the bilinear form $(\cdot,\cdot)$. From this we can construct

$$H(\mathfrak{h}) = U(\hat{\mathfrak{h}}) \otimes_{U(\mathbb{C}K+\mathfrak{h}[t])} \mathbb{C}\,|0\rangle$$

as in the first lecture. This is a vertex algebra, with

$$[h_\lambda h'] = (h,h')\lambda$$

(where $h = h_{(-1)}\,|0\rangle = ht^{-1}\,|0\rangle$).

Now put

$$V_L = \bigoplus_{\alpha \in L} H(\mathfrak{h})^\alpha \otimes e^\alpha,$$

and define $Y(|\alpha\rangle\,, z)$ similarly as before. We just need to choose

$$C : L \times L \to \mathbb{C}^\times$$

such that

$$C_{\alpha,\beta} = (-1)^{p(\alpha)p(\beta)+(\alpha,\beta)} C_{\beta,\alpha}$$
$$C_{0,\alpha} = C_{\alpha,0} = 1.$$

Exercise 84. *Deduce the cocycle condition*

$$C_{\alpha,\beta}C_{\alpha+\beta,\gamma} = C_{\alpha,\beta+\gamma}C_{\beta,\gamma}$$

from the Borcherds identity with appropriate choices of $n, m, k \in \mathbb{Z}$. This apparently necessary property of $C : L \times L \to \mathbb{C}^\times$ is not very apparent directly from the condition of locality. Discuss.

The vertex algebra $H(\mathfrak{h})$ has a Virasoro element:

$$L = \frac{1}{2}\sum_i :h^i h^i:$$

where $\{h^i\}$ is a basis of $\mathfrak{h}$ orthonormal with respect to $(\cdot,\cdot)$.

Exercise 85. *Check that*

$$[L_\lambda L] = (T + 2\lambda)L + \frac{\dim \mathfrak{h}}{12}\lambda^3.$$

So the lattice vertex algebra has central charge equal to the rank of L.

Let V be a vertex algebra with a Virasoro vector L, and consider the grading

$$V = \bigoplus_n V_n$$

by eigenvalues of L_0, known as *conformal weight*. For V_L the conformal weight of

$$h^1_{-n_1}h^2_{-n_2}\cdots h^s_{-n_s} \otimes e^\alpha$$

is $|\alpha|^2/2 + \sum_i n_i$ Recall also

$$\Delta(a_{(n)}b) = \Delta(a) + \Delta(b) - n - 1.$$

So if $a, b \in V_1$ then

$$a_{(0)}b \in V_1$$

as well.

Theorem 86. *If $TV_0 = 0$, in particular if $V_0 = \mathbb{C}\,|0\rangle$, then $[a,b] = a_{(0)}b$ defines a Lie algebra structure on V_1.*

Proof. Firstly put $n = m = k = 0$ in the Borcherds identity to obtain the Jacobi identity directly.

Now we need $b_{(0)}a = -a_{(0)}b$. But recall skew-symmetry

$$[b_\lambda a] = -[a_{-\lambda-T}b].$$

Rather

$$Y(b,z)a = e^{zT}Y(a,-z)b.$$

Expanding out completely gives

$$b_{(n)}a = -(-1)^n \sum_{j\in\mathbb{Z}_+} \frac{(-1)^j}{j!} T^j(a_{(n+j)}b).$$

$\square$

If we take $V = V_L$, then $V_0 = \mathbb{C}\,|0\rangle$ and

$$V_1 = \mathfrak{h} \oplus \bigoplus_{\alpha\Delta} \mathbb{C}e^{\alpha}$$

where

$$\Delta = \{\alpha \in L \mid (\alpha,\alpha) = 2\}.$$

PICTURE OF PARABOLOID

Indeed if L is the root lattice of a simply laced simple Lie algebra $\mathfrak{g}$, then the Lie algebra $(V_L)_1$ actually recovers $\mathfrak{g}$. Incidentally, Richard Borcherds has commented that this is probably the most uniform known construction of the finite dimensional simple Lie algebras.

TALK ABOUT EXPLICIT COCYCLES, AND CONSTRUCTION OF FINITE DIMENSIONAL SIMPLE LIE ALGEBRAS

Suppose $\mathfrak{g}$ is simple and simply laced with root lattice L. So we have $V = V_L$ with its weight-one space $V_1 \cong \mathfrak{g}$. We also have an infinite family $V = V^k(\mathfrak{g})$ with $V_1 \cong \mathfrak{g}$ for all of these. But V_L is clearly not isomorphic to any of the $V^k(\mathfrak{g})$ because its "shape" is different.

Exercise 87. *"Shape"? What am I talking about?*

Theorem 88 (Frenkel-Kac-Segal)**.** *If $\mathfrak{g}$ is simple and simply laced with root lattice, then there is an isomorphism*

$$V_L \cong V_1(\mathfrak{g}).$$

3.16 Restricted $\hat{\mathfrak{g}}$-modules and $V^k(\mathfrak{g})$-modules

The Zhu algebra construction gives us a powerful means to construct and classify modules over a vertex algebra. But the proof of Zhu's theorem requires some technical results about vertex algebra modules, which must be proved first to get the theory off the ground.

Let $\mathfrak{g}$ be a Lie algebra and $\hat{\mathfrak{g}}$ its affinisation, as in the first lecture. A $\hat{\mathfrak{g}}$-module M is said to be *smooth* if, for each $x \in M$, and $a \in \mathfrak{g}$, there exists N such that

$$(at^p) \cdot x = 0 \quad \text{for all } p \geq N.$$

More generally let U be an associative algebra equipped with a system of (not necessarily unital) subalgebras

$$U_0 \supset U_1 \supset U_2 \supset \dots$$

satisfying

- $\bigcap_p U_p = 0$,
- For any $u \in U$, and any $p \in \mathbb{Z}_{\geq 0}$ now matter how big, there is an ℓ (possibly much bigger) such that
$$U_\ell u \subset U U_p.$$

A U-module M is said to be smooth if, for each $x \in M$, there exists N such that

$$U_p \cdot x = 0 \quad \text{for all } p \geq N.$$

We explain the proof of the following theorem

Theorem 89. *A $V^k(\mathfrak{g})$-module is the same thing as a smooth $\hat{\mathfrak{g}}$-module of level k.*

Returning to the more general context, we consider the completion (formally it is a categorical limit)

$$\begin{tikzcd}
 & & U^c \arrow[dll] \arrow[dl] \arrow[d] & \\
U/UU_0 & U/UU_1 \arrow[l, two heads] & U/UU_2 \arrow[l, two heads] & \cdots \arrow[l, two heads]
\end{tikzcd}$$

Equivalently

$$U^c = \{\sum_i s_i \mid \sum_i s_i \bmod UU_p \text{ is finite for all } p\}.$$

Let us explain the relation between the two descriptions of U^c. By definition of limit, U^c is the subset of infinite tuples

$$(u_0, u_1, u_2, \dots) \in \prod_{i \in \mathbb{Z}_{\geq 0}} U/UU_i$$

such that, for every p, the image of u_i in the projection $U/UU_i \to U/UU_{i-1}$ coincides with u_{i-1}. If we now choose a representative $\overline{u}_i$ of $u_i \in U/UU_i$ in U, and set $s_i = \overline{u}_i - \overline{u}_{i-1}$ then our element of U^c becomes

$$(s_0, s_0 + s_1, s_0 + s_1 + s_2, \ldots),$$

which we identify with $\sum_i s_i$.

We note that the action of U on a smooth U-module extends to a well-defined action of the completion U^c. Now let

$$a(z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1} \in U^c[[z^{\pm 1}]]$$

(such objects we call formal distributions). We call $a(z)$ *continuous* if

$$a_{(n)} \in UU_p$$

for all $n \geq N$ for some N, possibly dependent on $a(z)$ and on p. Denote by $U^c[[z^{\pm 1}]]^{\text{cont}}$ the space of continuous formal distributions.

Proposition 90. *If $a(z)$ and $b(z)$ lie in $U^c[[z^{\pm 1}]]^{cont}$ then their n^{th} products are well defined and also lie in $U^c[[z^{\pm 1}]]^{cont}$, for all $n \in \mathbb{Z}$.*

Now let $\mathcal{F} \subset U^c[[z^{\pm 1}]]^{\text{cont}}$ such that

- $I_{U^c} \in \mathcal{F}$,
- All pairs of elements of $\mathcal{F}$ are local.

Let $\overline{\mathcal{F}}$ be the closure of $\mathcal{F}$, i.e., the span of iterated n^{th} products of elements of $\mathcal{F}$, as we have seen for quantum fields. Then

Theorem 91. *The vector space $\overline{\mathcal{F}}$, equipped with*

- $|0\rangle = I_{U^c}$,
- $Ta(z) = \partial_z a(z)$
- $Y(a(z), x) = \sum_{n\in\mathbb{Z}} a(z)_{(n)}(-)x^{-n-1}$,

is a vertex algebra.

The main thing to verify is locality.

Now let M be a restricted U-module, and $\overline{\mathcal{F}}$ a vertex algebra as above. Then by assigning

$$Y^M(a(z), x) = a(x) = \sum_{n\in\mathbb{Z}} a_{(n)}|_M x^{-n-1},$$

we obtain an $\overline{\mathcal{F}}$-module, essentially tautologically.

Now we come to the example relevant for Theorem 89. Consider the Lie subalgebra $\mathfrak{g}t^{\geq p} \subset \hat{\mathfrak{g}}$. In general we denote by $U(\mathfrak{g})_+ \subset U(\mathfrak{g})$ the subalgebra (ideal in fact) generated by $i(\mathfrak{g}) \subset U(\mathfrak{g})$. This is called the augmentation ideal, and it is clear that $U(\mathfrak{g})/U(\mathfrak{g})_+ = \mathbb{C}1$. Anyway, we take $U = U(\hat{\mathfrak{g}})/(K - k1)$ and for U_p we take $U(\mathfrak{g}t^{\geq p})_+$, or rather its image in $U(\hat{\mathfrak{g}})/(K - k1)$.

Now we take the following set of mutually local continuous formal distributions

$$\mathcal{F} = I_{U^c} \cup \{a(z) = \sum at^n z^{-n-1} \mid a \in \mathfrak{g}\},$$

and we obtain a vertex algebra $\overline{\mathcal{F}}$ spanned by products

$$a^1(z)_{(n_1)} a^2(z)_{(n_2)} \cdots a^s(z)_{(n_s)} |0\rangle .$$

Now we observe that $V^k(\mathfrak{g})$ itself is a smooth $\hat{\mathfrak{g}}$-module of level k, hence a smooth U-module, and thus a module over the vertex algebra $\overline{\mathcal{F}}$, in which (for generating fields $a(z)$, $a \in \mathfrak{g}$)

$$Y(a(z), x) = a(x) = \sum_n (at^n) x^{-n-1}.$$

Composing the map from $\overline{\mathcal{F}}$ to the set of quantum fields on $V^k(\mathfrak{g})$, with the state-field correspondence, we obtain a map

$$\phi : \overline{\mathcal{F}} \to V^k(\mathfrak{g})$$

sending $|0\rangle$ to $|0\rangle$ and n^{th}-products to n^{th}-products, i.e., a homomorphism of vertex algebras. It is easy to see that ϕ is surjective, we wish to show it is an isomorphism.

To see this, we would like to argue that $\overline{\mathcal{F}}$ is a highest weight $\hat{\mathfrak{g}}$-module, and then invoke the universal property of the vacuum $\hat{\mathfrak{g}}$-module $V^k(\mathfrak{g})$. The action will be

$$\rho(at^m)c(z) = a(z)_{(m)}c(z), \qquad \rho(K) = kI.$$

Why is this a $\hat{\mathfrak{g}}$-action? Well, since $\overline{\mathcal{F}}$ is a vertex algebra, we may use the commutator formula to see that

$$[\rho(at^m), \rho(bt^n)] = [a(z)_{(m)}, b(z)_{(n)}] = \sum_{j \in \mathbb{Z}_+} \binom{m}{j} \left(a(z)_{(j)} b(z)\right)_{(m+n-j)} .$$

Now the terms $a(z)_{(j)}b(z)$ are computed directly from the definition of the series $a(z) = \sum_n (at^n) z^{-n-1}$ and the commutation relations of $\hat{\mathfrak{g}}$, we have

$$a(z)_{(0)}b(z) = [a,b](z), \qquad a(z)_{(1)}b(z) = (a,b)kI_U, \qquad a(z)_{(\geq 2)}b(z) = 0.$$

Thus

$$\begin{aligned}[\rho(at^m), \rho(bt^n)] &= [a,b](z)_{(m+n)} + m(a,b)kI(z)_{(m+n-1)} \\ &= [a,b](z)_{(m+n)} + m\delta_{m,-n}(a,b)kI = \rho([at^m, bt^n]).\end{aligned}$$

Because $I \in \overline{\mathcal{F}}$ is annihilated by $a(z)_{(\geq 0)}$, it follows that there is a map of $\hat{\mathfrak{g}}$-modules $\psi : V^k(\mathfrak{g}) \to \overline{\mathcal{F}}$. Examining the constructions, we see that the composition $\phi \circ \psi$ is the identity on $V^k(\mathfrak{g})$. Thus both ϕ and ψ are linear isomorphisms, and so $V^k(\mathfrak{g}) \cong \overline{\mathcal{F}}$ as vertex algebras.

3.17 Review of limits and colimits

The notions of limit and colimit in category theory are very general. Let J be a "diagram category". This more or less just means a nonempty category, but the purpose of the terminology is to suggest that J should be something small and discrete, like

$$\bullet \longrightarrow \bullet$$

or

$$\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \\ \bullet & \end{tikzcd}$$

or

$$\mathbb{N}^{\downarrow} \qquad \bullet \longleftarrow \bullet \longleftarrow \bullet \longleftarrow \cdots$$

or

$$\mathbb{N}^{\uparrow} \qquad \bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \cdots$$

as opposed to some oceanically large category like, say, **Top**.

Now let $\mathcal{C}$ be a category. By definition a J-diagram in $\mathcal{C}$ is just a functor $D : J \to \mathcal{C}$. In the case of the first example above, then, a J-diagram is precisely a morphism in $\mathcal{C}$.

Consider an object $X \in \mathcal{C}$ and a J-diagram D in $\mathcal{C}$. If we have a morphism $f_j : X \to D(j)$ for each $j \in J$ such that the triangle

$$\begin{tikzcd} & X \arrow[dl] \arrow[dr] & \\ D(i) \arrow[rr, "D(a)"] & & D(j) \end{tikzcd}$$

commutes for all morphism $a : i \to j$ in J, then we refer to such a thing as a morphism from X to the J-diagram D.

Given $X \in \mathcal{C}$ with a morphism $f : X \to D$, and $X' \to X$ an honest morphism in $\mathcal{C}$, we can obviously compose to obtain a morphism from $f' : X' \to D$. So we have a collection of pairs (X, f).

Definition 57. The limit of the diagram D is an object X and morphism $X \to D$ such that any $X' \to D$ is of the form

$$X' \to X \to D$$

for a unique $h : X' \to X$.

The colimit of a diagram D is defined in a similar way; as an "initial object" among objects in $\mathcal{C}$ that receive a system of morphisms from D.

Definition 58. A filtered diagram category J is one in which

- For all $j, j' \in J$ there exists $k \in J$ and morphisms

$$\begin{tikzcd} & k & \\ j \arrow[ur] & & j' \arrow[ul] \end{tikzcd}$$

- For any two morphisms $u, v : i \to j$ there exists $w : j \to k$ such that $wu = wv$.

A filtered diagram in $\mathcal{C}$ is then a functor from a filtered category to $\mathcal{C}$, and a (co)limit of a filtered diagram is called a filtered (co)limit.

One could take a limit over an $\mathbb{N}^{\uparrow}$-diagram, but it would be a silly thing to do, the first object in the chain satisfies the defining property. More interesting is to take a limit of an $\mathbb{N}^{\downarrow}$-diagram. Such an object may or may not exist, depending on the category, etc. Similarly one could take a colimit of an $\mathbb{N}^{\downarrow}$-diagram, but it would be silly. More interesting is to take a colimit of an $\mathbb{N}^{\uparrow}$-diagram.

A number of confusing pieces of terminology:

- "Inverse limit" and "Projective limit" are both synonyms for limit, typically taken over an $\mathbb{N}^{\downarrow}$-diagram.
- "Direct limit" and "Inductive limit" are both synonyms for colimit, typically taken over an $\mathbb{N}^{\uparrow}$-diagram.

Some examples of ....

3.18 Representations and the Zhu algebra

Now we introduce a very powerful tool in the representation theory of vertex algebras: the Zhu algebra. The downside is that we have to give up some generality. The Zhu algebra can be defined for any vertex algebra, but it only really tells us something about representations of V when V is conformal.

Definition 59. Let V be a conformal vertex algebra. A weak V-module is a V-module.

Definition 60. Let V be a conformal vertex algebra. A positive energy V-module (also known as an admissible V-module [DLM]) is a V-module M with a grading

$$M = \bigoplus_{n \in \mathbb{Z}_+} M_n$$

in which

$$a^M_{(n)} : M_k \to M_{k+\Delta-n-1}$$

for all $a \in V_\Delta$.

Returning again to the relation

$$\Delta(a_{(n)}x) = \Delta(x) + \Delta(a) - n - 1$$

we observe that

$$a_{(\Delta(a)-1)}$$

is distinguished by the fact that it preserves degree. It is thus convenient to write

$$Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1} = \sum_{n\in\mathbb{Z}} a_n z^{-n-\Delta(a)}$$

so

$$a_n : M_k \to M_{k-n}.$$

Definition 61. Let V be a conformal vertex algebra. An ordinary V-module is a V-module M with a grading

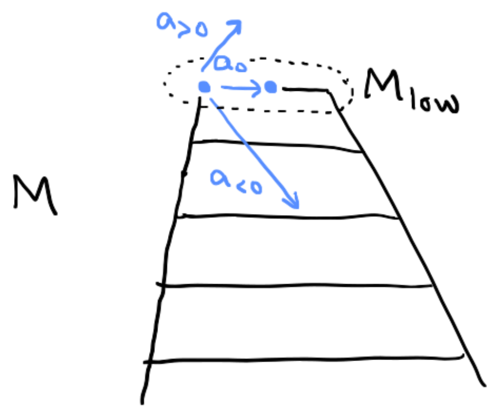
$$M = \bigoplus_{\lambda\in\mathbb{C}} M_\lambda$$

by finite dimensional eigenspaces of L_0, and such that for any λ we have $M_{\lambda+n} = 0$ for sufficiently negative n.

CLAIM:

$$\{\text{ordinary } V\text{-modules}\} \subset \{\text{admissible } V\text{-modules}\} \subset \{\text{weak } V\text{-modules}\}.$$

Let us now think about the lowest nonzero graded piece M_{low} of a positive energy module M. Specifically Zhu pondered what sort of algebra acts on M_{low} by virtue of its being the lowest piece of a V-module.



First we rewrite the Borcherds identity in terms of the new indexing:

$$\sum_{j\in\mathbb{Z}_+} \binom{m+\Delta(a)-1}{j} (a_{(n+j)}b)_{n+m+k}c = \sum_{j\in\mathbb{Z}_+} (-1)^j \binom{n}{j} \left(a_{m+n-j}b_{k+j}c - (-1)^n b_{n+k-j}a_{m+j}c\right).$$

(Having put $m + \Delta(a) - 1$ in place of m, $k + \Delta(b) - 1$ in place of k, and left n alone.)

Now let $c \in M_{\text{low}}$ and $k = 0$, $m = 1$ and $n = -1$, so

$$\sum_{j \in \mathbb{Z}_+} \binom{\Delta(a)}{j} (a_{(j-1)}b)_0 c = a_0 b_0 c.$$

In other words, if we define

$$a * b = \sum_{j \in \mathbb{Z}_+} \binom{\Delta(a)}{j} a_{(j-1)} b$$

then we obtain, in M_{low},

$$(a * b)_0 = a_0 b_0.$$

On the other hand, if we choose $m = k = 1$ and $n = -2$, then we find

$$\sum_{j \in \mathbb{Z}_+} \binom{\Delta(a)}{j} (a_{(j-2)}b)_0 c = 0.$$

So the idea is that V/J, where J is the span of all elements

$$a \circ b = \sum_{j \in \mathbb{Z}_+} \binom{\Delta(a)}{j} a_{(j-2)} b,$$

... equipped with the product $*$, might be some sort of algebra. And M_{low} should become a module.

Rather amazingly this is exactly what happens, and more.

Theorem 92. *The subspace $J \subset V$ is a 2-sided ideal for the product $*$. Furthermore the quotient $\text{Zhu}(V) = V/J$ with the induced product, is associative.*

If M is a positive energy V-module, then $a \mapsto a_0$ turns M_{low} into a left $\text{Zhu}(V)$-module. The module M is irreducible if and only if M_{low} is, and this establishes a bijection between isomorphism classes of irreducible modules.

Here is a similar, but slightly different, construction.

Definition 62. Let M be a weak V-module, and define

$$\Omega(M) = \{x \in M | a_n x = 0 \text{ for all } a \in V \text{ and } n > 0\}.$$

Then a left $\text{Zhu}(V)$-module structure on $\Omega(M)$ is defined by $(a + J)x = a_0 x$.

Example 7. We sketch a proof that

$$\text{Zhu}(V^k(\mathfrak{g})) \cong U(\mathfrak{g}).$$

Let $a \in V_1 = \mathfrak{g}$, and let $b \in V$ arbitrary. Then

$$a * b = a_{-1}b + a_0 b.$$

Let us define a linear map

$$F : U(\mathfrak{g}) \to V^k(\mathfrak{g})/J$$

by

$$a^1 a^2 \cdots a^s \mapsto a^1 * a^2 * \cdots * a^s = (a^1_{-1} + a^1_0)(a^2_{-1} + a^2_0) \cdots (a^s_{-1} + a^s_0) \left|0\right\rangle .$$

To show it is well-defined, simply check that

$$(a_{-1} + a_0)(b_{-1} + b_0) - (b_{-1} + b_0)(a_{-1} + a_0) - ([a, b]_{-1} + [a, b]_0)$$

reduces to

$$a_0(b_{-2} + b_{-1}) = [a, b]_{-2} + [a, b]_{-1} = [a, b] \circ \left|0\right\rangle \in J.$$

By the way, it is easy to see that all

$$a_{-n-1}b + a_{-n}b$$

also lie in J. So it is plausible that all of $V^k(\mathfrak{g})$ can be reduced to the image of F modulo J. So F is an isomorphism.

Things start to get interesting when we pass to the quotient

$$L_k(\mathfrak{g}) = V^k(\mathfrak{g})/(\sigma_k).$$

Correspondingly we find that

$$\mathrm{Zhu}(L_k(\mathfrak{g})) \cong U(\mathfrak{g})/((e^\theta)^{k+1}).$$

So let's say M is an irreducible positive energy $L_k(\mathfrak{g})$-module with finite dimensional graded pieces. Then we know that M_{low} is a finite dimensional $U(\mathfrak{g})$-module on which $(e^\theta)^{k+1} = 0$.

We have an $\mathfrak{sl}_2$-triple $e^\theta, h^\theta, f^\theta$. Now let $M_{\mathrm{low}} \cong L(\lambda)$, and consider a (maximal) weight string relative to the triple:

$$v_\mu, f^\theta v_\mu, (f^\theta)^2 v_\mu, \ldots, (f^\theta)^s v_\mu = v_{\mu - s\theta}.$$

Then $s = \langle \mu, \theta^\vee \rangle$ by $\mathfrak{sl}_2$-theory. So we have $(e^\theta)^{k+1} = 0$ in $L(\lambda)$ if and only if $\langle \mu, \theta^\vee \rangle \leq k$.

Theorem 93. *The irreducible positive energy $L_k(\mathfrak{g})$-modules are precisely the integrable highest weight $\hat{\mathfrak{g}}$-modules $L_k(\lambda)$ of highest weight $k\Lambda_0 + \lambda$.*

Lemma 94 (Kac, IDLA, Theorem 10.7)**.** *There are no $\hat{\mathfrak{g}}$-module extensions between $L_k(\lambda)$ and $L_k(\mu)$ for $\lambda, \mu \in P^k_+$.*

Definition 63. A vertex algebra is said to be rational if all positive energy modules are completely reducible.

Proposition 95. *The vertex algebra $L_k(\mathfrak{g})$ is rational.*

Categories of representations of rational vertex algebras have very rich structure. In particular there is a braided tensor structure, and we obtain modular tensor categories.

For V_L the category is pointed, with Grothendieck group $\mathbb{C}[L^\vee/L]$, there is a way to construct a braided tensor category on the category whose objects are $[h]$ for $h \in H$, with

$$[h_1] \otimes [h_2] = [h_1 h_2],$$

In fact [EGNO Section 8.4] any pointed braided tensor category is of this form, the only thing to prove is that the 3-cocycle in this case *is* the one coming from $L^\vee/L$.

For $L_k(\mathfrak{g})$ the representation category is a quotient category of the category of tilting modules over the quantum group $U_q(\mathfrak{g})$ at an appropriate root of unity (see Bakalov-Kirillov, Section 3.3).

3.19 Integrable modules in category $\mathcal{O}$

Our next goal will be to prove rationality of $L_k(\mathfrak{g})$ for $k \in \mathbb{Z}_+$. As preparation we must do a little more representation theory of Kac-Moody algebras, beyond what we needed for the Kac-Weyl character formula.

Definition 64 (IDLA, 3.6)**.** A $\mathfrak{g}(A)$-module is said to be integrable if

- It is a weight module with respect to $\mathfrak{h}$,
- e_i and f_i are locally nilpotent for $i = 1, \dots n$.

Lemma 96. *If the $\mathfrak{g}(A)$-module V is generated by weight vectors v on which e_1, etc. act nilpotently, then V is integrable.*

Proof. It suffices to prove that the same holds for any monomial

$$x_{i_1} x_{i_2} \cdots x_{i_s} v$$

where the x_i are e_i and f_i in any order. Because of the Serre and other relations in $\mathfrak{g}(A)$, it holds. □

Definition 65. Let V be a weight module for $\mathfrak{g}(A)$. A vector $v \in V_\lambda$ is called primitive if there exists a submodule $U \subset V$ such that

- $v \notin U$,
- $E_i v \in U$ for all simple roots E_i.

Lemma 97 ((3.2.4))**.** *A primitive weight λ of an integrable $\mathfrak{g}(A)$-module satisfies $< \lambda, \alpha_i^\vee > \in \mathbb{Z}_+$ for $i = 1, \ldots, n$.*

Proof. Indeed if $v \in V_\lambda$ is a primitive vector, with $E_i v \in U$ then in V/U we have $E_i v = 0$ and $F_i^{N+1} v = 0$ for some $N \in \mathbb{Z}_+$. Then by sl_2-theory, $< \lambda, \alpha_i^\vee >= N$. □

Lemma 98 (IDLA, Lemma 9.8(b))**.** *A primitive vector $v \in M_\lambda$ satisfies*

$$\Omega v - (\lambda, \lambda + 2\rho) v \in U$$

for some submodule U not containing v.

Proof. Clear from the explicit formula for Ω and the definition of primitive vector. □

Proposition 99 (IDLA 9.5)**.** *Let $V \in \mathcal{O}$ for $\mathfrak{g}(A)$ satisfying the following property: Whenever λ and $\mu \leq \lambda$ are primitive weights of V then $\mu = \lambda$. Then V is completely reducible.*

Proof. Let

$$V^0 = \{v \in V \mid \mathfrak{n}_+ v = 0\}.$$

Since V^0 is $\mathfrak{h}$-invariant it is a sum of weight spaces V_λ^0. Let L denote the set of weights. These are primitive (even singular) weights.

Let $\lambda \in L$ and $v \in V_\lambda^0$ and consider

$$U(\mathfrak{g}) v \subset V.$$

We claim this submodule is irreducible. Indeed since it is a highest weight module, if it is not irreducible then it contains a singular vector of weight $\mu \leq \lambda$, contradicting our hypothesis.

So $V' = U(\mathfrak{g}) V^0 \subset V$ is completely reducible. It remains to prove that $V = V'$. Suppose otherwise, and consider $V/V' \neq 0$. Then there is some weight vector $v \in V_\mu$ such that $v \notin V'$ and some $0 \neq E_i v \in V'$. But then some highest weight vector $\lambda \geq \mu + \alpha_i$ exists in $V' \subset V$ and the pair $\lambda > \mu$ of primitive weights again contradicts the hypotheses. □

Theorem 100 (IDLA Thm 10.7)**.** *Let M be an integrable $\mathfrak{g}(A)$-module in category $\mathcal{O}$. Then M is isomorphic to a direct sum of irreducible highest weight modules $L(\Lambda)$ for $\Lambda \in P_+$.*

Proof. Lemma 97 asserts that, since M is integrable, any primitive weight λ of M satisfies $< \lambda, \alpha_i^\vee > \in \mathbb{Z}_+$.

Suppose M had two primitive weights λ and $\mu = \lambda - \beta$ for some $\beta \in \mathbb{Q}_+ \backslash 0$. Then we calculate

$$2(\lambda + \rho, \beta) - (\beta, \beta) = (\lambda + \mu + 2\rho, \beta) > 0$$

(for ρ the inequality follows from its construction, for λ and μ the inequality follows from the previous paragraph).

Now let us assume, without loss of generality, that M is indecomposable. Since $M \in \mathcal{O}$ we know that Ω is locally finite in M. It follows that $\Omega - a$ is locally nilpotent in M for some $a \in \mathbb{C}$. Indeed any vector $v \in M$ is contained in some finite dimensional subspace, on which Ω has an eigenvector, then we look at generalised eigenspaces (which are $\mathfrak{g}$-invariant) and invoke indecomposability to reach the desired conclusion.

Lemma 98 asserts that any primitive vector $v \in M_\lambda$ satisfies

$$\Omega v - (\lambda, \lambda + 2\rho)v \in U$$

for some submodule U not containing v.

It follows that two primitive weights λ and μ of M satisfy

$$(\lambda, \lambda + 2\rho) = (\mu, \mu + 2\rho).$$

If $\mu = \lambda - \beta$ for some $\beta \in Q_+\backslash 0$ then it follows that

$$2(\lambda + \rho, \beta) - (\beta, \beta) = 0,$$

which contradicts the inequality above.

Therefore if λ and $\mu = \lambda - \beta$, for $\beta \in Q_+$, are primitive weights then $\mu = \lambda$ and, by Proposition 99, M is completely reducible. □

3.20 Rationality of $L_k(\mathfrak{g})$

Lemma 101. *Let M be a positive energy $L_k(\mathfrak{g})$-module. Then M is integrable as a $\tilde{\mathfrak{g}}$-module.*

TO AVOID A HOPELESS CLASH OF NOTATION, I WILL USE CAPITAL LETTERS WHEN I AM WORKING WITH KAC-MOODY ALGEBRA GENERATORS. SO FOR THEM SUBSCRIPTS HAVE NO RELATION TO VERTEX-ALGEBRA PRODUCTS.

Proof. It is clear that E_0 is locally nilpotent on M.

By Lemma **??** $E_1, \dots E_n$ and $F_1, \dots F_n$ are nilpotent on M_{low}. From this it follows that M_{low} is a direct sum of finite dimensional irreducible $\mathfrak{g}$-modules.

It is also easy to see that F_0 is locally nilpotent on M_{low}. Indeed

$$: e_{-\theta}(z) :^{k+1} = 0,$$

and if we apply this relation to a lowest weight vector v in the $\mathfrak{g}$-module M_{low} we obtain

$$F_0^{k+1} v = (e_{-\theta})_{-1}^{k+1} v = 0.$$

Local nilpotence follows by Lemma 96.

If M were generated by M_{low} as a $\tilde{\mathfrak{g}}$-module then we would be done. In general we proceed by induction. Let M' be the submodule generated, and let $M_1 = M/M'$. With its induced grading M_1 is $\mathbb{Z}_{\geq 1}$-graded, in particular it is positive energy. We repeat the process, defining $M_0 = M$, $M_1 = M/M'$, M_2, M_3, etc.

For any $v \in M$ there exists s such that $v = 0$ in M_{s+1}. By induction each E_i and F_i acts nilpotently on v. $\square$

3.21 Example: Wakimoto modules

We will consider a vertex algebra V which contains, beyond the vacuum vector $|0\rangle$, two vectors β and γ which satisfy

$$\begin{aligned} &[\beta_\lambda \beta] = [\gamma_\lambda \gamma] = 0 \\ &[\beta_\lambda \gamma] = 1 \quad \text{(and consequently } [\gamma_\lambda \beta] = -1). \end{aligned}$$

Remembering that, by the definition of the λ-bracket notation, this means that

$$[Y(\beta, z), Y(\gamma, w)] = \delta(z, w).$$

Actually we can easily construct such a vertex algebra, as follows. Let

$$\mathcal{A} = U \otimes \mathbb{C}[t, t^{-1}] \oplus \mathbb{C}K$$

where U is a two-dimensional vector space with nondegenerate antisymmetric bilinear form $\langle \cdot, \cdot \rangle$, and bracket

$$[xt^m, yt^n] = \delta_{m,-n} \langle x, y \rangle K, \qquad [K, \mathfrak{g}] = 0.$$

As always we write u_n for ut^n. Put $\mathcal{A}_+ = U \otimes \mathbb{C}[t] \oplus \mathbb{C}K$ and

$$V = U(\mathcal{A}) \otimes_{U(\mathcal{A}_+)} \mathbb{C} |0\rangle$$

where $U \otimes \mathbb{C}[t] |0\rangle = 0$ and $K |0\rangle = |0\rangle$ as usual.

Let $\{\beta, \gamma\}$ be a basis of U satisfying $\langle \beta, \gamma \rangle = 1$ and $\langle \beta, \beta \rangle = \langle \gamma, \gamma \rangle = 0$. It is not hard to see that the fields

$$\beta(z) = \sum_{n \in \mathbb{Z}} \beta_n z^{-n-1} \quad \text{and} \quad \gamma(z) = \sum_{n \in \mathbb{Z}} \gamma_n z^{-n} \quad \text{(no, it's not a misprint)}$$

satisfy

$$[\beta(z), \gamma(w)] = \delta(z, w)$$

as required. So in V our desired vectors are

$$\beta = \beta_{-1} |0\rangle \quad \text{and} \quad \gamma = \gamma_0 |0\rangle .$$

But anyway... Let us now define

$$\begin{aligned}E &= \beta\\ H &= -2 :\gamma\beta:\\ F &= - :\gamma(\gamma\beta): -2T\gamma.\end{aligned}$$

Exercise 102. *Calculate the λ-brackets between E, H and F.*

After seeing the definition of the affine vertex algebra $V^k(\mathfrak{g})$ associated to the Lie algebra $\mathfrak{g}$ at level k, do the following exercise.

Exercise 103. *Conclude that E, H and F satisfy the relations of generating fields of $V^{-2}(\mathfrak{sl}_2)$.*

This example can be thought of an *affinisation* of the following representation of $\mathfrak{sl}_2$:

$$\begin{aligned}e &\mapsto \frac{\partial}{\partial u}\\ h &\mapsto -2u\frac{\partial}{\partial u}\\ f &\mapsto -u^2\frac{\partial}{\partial u}.\end{aligned}$$

in $\mathbb{C}[u]$.

Exercise 104. *Verify that this does indeed define a representation of $\mathfrak{sl}_2$.*

But there are two interesting new features in the affine case:

- It was necessary to introduce a "quantum correction" $-2T\gamma$ for the construction to work,
- The construction yields a representation of a specific level k, in this case $k = -2$.

The $\widehat{\mathfrak{sl}}_2$-module constructed this way is called a *Wakimoto module*.

Let G be a simple Lie group. There is a general construction of a representation of its Lie algebra $\mathfrak{g}$ associated with the "action" of G on the big cell of the flag variety

$$B_-\backslash G.$$

The representation of $\mathfrak{sl}_2$ that appeared above is just the $G = SL_2$ case of this more general construction.

Exercise 105. *See if you can spell out the action of $G = SL_2$ on $B_-\backslash G$ and differentiate it to recover the representation of $\mathfrak{sl}_2$ given above. Try doing $G = SL_3$.*

The Wakimoto construction was generalised by Feigin and Frenkel from $G = SL_2$ to general G, to obtain a realisation of the affine vertex algebra

$$V^{-h^\vee}(\mathfrak{g}),$$

where $h^\vee$ is the *dual Coxeter number* of $\mathfrak{g}$. The level $k = -h^\vee$ is the so-called *critical level* and is very important in the theory of affine Lie algebras. For $\mathfrak{sl}_2$ we have $h^\vee = 2$.